

# AN EXPONENT OF 1.04273 FOR THE UNIT DISTANCE PROBLEM

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**ABSTRACT.** Let  $u(U)$  count the unordered pairs at distance one in a finite planar set  $U$ . We construct finite sets  $U_j$  with  $|U_j| \rightarrow \infty$  and  $u(U_j)/|U_j|^{1.04273} \rightarrow \infty$ . The largest exponent previously claimed, 1.0358, is in the author's unpublished manuscript. The method is the number-field construction of OpenAI and Sawin: unit distances come from elements of relative norm one in quadratic extensions, and the fields come from an infinite pro-2 class field tower. The new ingredients are quadratic extensions of mixed signature, with an exact average over their norm-one units, and a tower over the real quadratic field  $\mathbb{Q}(\sqrt{241})$ , in which 2, 3 and 5 split. Its Golod-Shafarevich polynomial contains two copies of the local conditions at these primes but only one constant term, and the extra room lets the tower be ramified only above 2, 3 and 5; the root discriminant of its fields is about 286. The relative zeta value is bounded through the 255 quadratic Hecke  $L$ -functions of the degree-512 field generated over  $\mathbb{Q}(\sqrt{241})$  by the square roots of its  $\{2, 3, 5\}$ -units. Finite facts and numerical inequalities are certified by exact computation and interval arithmetic. A significant portion of this work was verified in Lean.

\* **AI Methodology.** This work was done with extensive AI use, with agents developing the mathematics and writing the paper. The listed author is the human responsible for having supervised and prompted the AI agents that produced this work. The construction over  $\mathbb{Q}(\sqrt{241})$ , the computations that certify it, the Lean formalization of the reduction of Theorem 1.1 to one explicit zeta inequality (Remark 4.28), and this paper were produced with a combination of Claude Opus 5.5, ChatGPT Sol 5.6, ChatGPT Astra 6, working as agents in Claude Code and Codex. The construction adapts the author's previous manuscript [22], which was also written with AI tools.

## 1. INTRODUCTION

For a finite set  $U \subset \mathbb{R}^2$ , let

$$u(U) = \#\{\{x, y\} \subset U : |x - y| = 1\}, \quad u(n) = \max_{|U|=n} u(U),$$

where the pairs are unordered and the distance is Euclidean. The unit distance problem of Erdős [6] asks for the order of growth of  $u(n)$ . Our main result is the following.

**Theorem 1.1.** *There are finite sets  $U_j \subset \mathbb{R}^2$ , with  $|U_j| \rightarrow \infty$ , such that*

$$\frac{u(U_j)}{|U_j|^{1.04273}} \rightarrow \infty.$$

*In particular,  $u(n) \geq n^{1.04273}$  for arbitrarily large integers  $n$ .*

The largest exponent previously claimed, 1.0358324, is in the author's unpublished manuscript [22]. Theorem 1.1 uses the same number-field method with several new ingredients: fields of mixed signature, averaged over their norm-one units; a pro-2 class field tower over the real quadratic field  $\mathbb{Q}(\sqrt{241})$  with a nonabelian dyadic local group; an explicit bound for the relative zeta value from the quadratic  $L$ -functions of a Kummer field of degree 512; and optimized archimedean profiles and weighted windows. Section 1.3 describes them. The theorem concerns an unbounded sequence of cardinalities. Its proof uses no unproved hypothesis, such as the generalized Riemann hypothesis,

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but it is computer-assisted: finite facts about the tower and several numerical inequalities are certified by exact computation and interval arithmetic, as described in Section 1.6.

**1.1. Background.** Erdős [6] showed that a suitable section of a scaled integer lattice gives

$$u(n) \geq n^{1+c/\log \log n}$$

for some  $c > 0$  and arbitrarily large  $n$ , and he conjectured that this is essentially optimal [6, 7]. The best upper bound,

$$u(n) = O(n^{4/3}),$$

is due to Spencer, Szemerédi and Trotter [34]; Székely [36] gave a proof by crossing numbers.

In 2026 a team at OpenAI [27] disproved Erdős’s conjecture. They constructed sets with  $u(U) \geq |U|^{1+\delta}$  for a fixed  $\delta > 0$  and arbitrarily large  $|U|$ , replacing the integer lattice by ideal lattices in number fields of large degree and small root discriminant. Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimerman, Wang and Wood [1] simplified the argument and obtained  $\delta \approx 6 \cdot 10^{-38}$ . Sawin [32] made every step explicit and sharpened it, and reached the exponent 1.014114.

Improvements followed within days in online discussions. On the Erdős Problems forum, mlewko [20] reached 1.031849 in Sawin’s criterion with finite data found by ChatGPT, and on MathOverflow spiderduckpig [35] adjusted these data to reach 1.031883. In a comment the next day, spiderduckpig proposed a structural change: in the Golod–Shafarevich step of Sawin’s argument [32, Lemma 11], replace the unweighted count of relations by a count weighted by degree in the Zassenhaus filtration. This was reported to give  $\delta \geq 0.0333487$  [35]. Weighted counts of this kind underlie the towers of [22] and of the present paper. The author’s MathOverflow answer [21] then combined this count with covolume counting, dyadic ramification and Euler products over a fixed subfield. The manuscript [22] is a formal write-up of that construction, and it records  $\delta = 0.0358324$ . Table 1 lists the explicit exponents, and the repository [28] records further bounds posted online.

TABLE 1. Explicit lower bounds  $u(n) \geq n^{1+\delta}$  for arbitrarily large  $n$ , in increasing order. The exponents are rounded down and constant factors are omitted. The last column gives the kind of source; apart from Erdős’s, none of these bounds had appeared in a refereed journal when this paper was written.

| Exponent                 | Source                 | Main new input                                                        | Kind of source          |
|--------------------------|------------------------|-----------------------------------------------------------------------|-------------------------|
| $1 + c/\log \log n$      | Erdős [6]              | integer lattice                                                       | journal                 |
| $1 + \delta, \delta > 0$ | OpenAI [27]            | ideal lattices from class field towers                                | manuscript              |
| $1 + 6 \cdot 10^{-38}$   | Alon et al. [1]        | simplified argument                                                   | preprint                |
| 1.014114                 | Sawin [32]             | explicit and sharpened criterion                                      | preprint                |
| 1.015263                 | Emmerich [4]           | optimized parameters                                                  | preprint                |
| 1.031185                 | Emmerich, Cordella [5] | extended range of primes                                              | certificate             |
| 1.031849                 | mlewko [20]            | finite data found by ChatGPT                                          | forum post              |
| 1.0333487                | spiderduckpig [35]     | weighted Zassenhaus count                                             | online post             |
| 1.0358324                | Naslund [22, 21]       | covolume counting, dyadic ramification, fixed-subfield Euler products | manuscript, online post |
| 1.04273                  | this paper             | see Section 1.3                                                       |                         |

**1.2. The construction in outline.** The construction carries Erdős’s lattice argument to number fields of large degree. Let  $K$  be such a field, of degree  $2d$ , with an involution  $\iota$  whose fixed field is  $F$ . We take the points of a translated ideal lattice of  $K$  that lie in a bounded region of  $K \otimes \mathbb{R}$ . A complex embedding in which  $\iota$  acts as complex conjugation maps them injectively to the plane. It sends every difference of relative norm one over  $F$  to a unit vector. The exponent is then set by a balance, per unit of  $d$ . The gain is the number of equal-norm choices: a prime of  $F$  that splits in  $K$

and is used with exponent  $k$  gives  $k + 1$  ideals of the same relative norm. Selecting a common ideal class loses part of this gain. Through the class-number formula, that loss is governed by the root discriminant  $\lambda$  of  $K$  and the value at  $s = 1$  of  $L_F(s) = \zeta_K(s)/\zeta_F(s)$ . The geometric window decides how many of the resulting displacements keep both endpoints in the region, and its volume fixes the number of points. Using a prime  $\mathfrak{p}$  with exponent  $k$  also makes the ideal lattice denser by the factor  $N\mathfrak{p}^k$ , and the exponent  $\delta$  charges the gain for the points this adds.

Quantitatively, suppose that such fields exist with  $d \rightarrow \infty$ , a fixed root discriminant  $\lambda$ , fixed local types at the selected primes, and  $\theta = c/d$  bounded below, where  $(b, c)$  is the signature of  $F$  and  $d = b + 2c$ ; thus  $2\theta$  is the proportion of non-real embeddings of  $F$ . With hard windows at the selected primes, the exponent  $1 + \delta$  is attained when

$$(1.2) \quad \sum_q \frac{\log(k_q + 1) - \delta k_q f_q \log q}{e_q f_q} - \left(\frac{1}{2} - \delta\right) \log \lambda - C + \mathcal{A}_\delta(\theta) > 0.$$

Here  $q$  runs over the rational primes below the selected primes,  $e_q$  and  $f_q$  are their absolute ramification indices and residue degrees in  $F$ ,  $k_q$  is the exponent used at the primes above  $q$ ,  $C$  is an upper bound for  $d^{-1} \log L_F(1)$ , and

$$\mathcal{A}_\delta(\theta) = (1 - \delta) \log 2 + (1 - \theta) \log \pi + (1 - 2\theta) J_{\mathbb{R}} + \theta J_{\mathbb{C}}$$

is the archimedean contribution, in which  $J_{\mathbb{R}}$  and  $J_{\mathbb{C}}$  measure the overlap of the archimedean profiles at the real and at the complex places of  $F$  under unit displacements (Proposition 5.18). Section 6 replaces the hard windows by weighted shells (Proposition 6.15), and Section 8 verifies the resulting inequality at  $\delta = 0.04273$ .

The fields must form an infinite family of growing degree with fixed  $\lambda$  and fixed local types. Such families come from infinite pro-2 class field towers with prescribed local Galois groups. The Golod–Shafarevich criterion [10] proves that a tower is infinite when a certain power series in a variable  $t$  takes a negative value; the use of Frobenius conditions to control splitting in such towers goes back to Hajir, Maire and Ramakrishna [11].

**1.3. New ingredients.** The construction differs from those of Sawin [32] and of [22] in four ways.

*Mixed signature.* In Sawin’s construction, and in [22], the field  $K$  is a CM field: complex conjugation is central in  $\text{Gal}(K/\mathbb{Q})$ ,  $F$  is totally real, every norm-one element has modulus one at every archimedean place, and the norm-one units are finite. Here  $K$  is totally imaginary and Galois over  $B = \mathbb{Q}(\sqrt{241})$ , and  $F$  is the fixed field of one complex conjugation  $\iota_1$ , which is not central: its conjugacy class has at least  $2^{15}$  elements, so  $\theta \geq 65535/131072$  and almost every place of  $F$  is complex. Above a complex place of  $F$  the two coordinates of a norm-one element can have moduli  $e^u$  and  $e^{-u}$ , and the norm-one units, of rank  $c$ , move  $u$  along a lattice. Averaging over a fundamental domain of that lattice counts the resulting displacements exactly, and the class-number formula cancels the regulator of these units together with the capitulation kernel (Proposition 5.2). Mixed signature therefore costs only a factor  $\pi$  per complex place of  $F$ , and a profile that couples the two coordinates at that place more than repays it.

*A tower over  $\mathbb{Q}(\sqrt{241})$ .* Over a totally real base field in which every prime used by the construction splits completely, the Golod–Shafarevich series contains a copy of each local condition for every prime of the base, but only one constant term. The field  $B = \mathbb{Q}(\sqrt{241})$  is the real quadratic field of smallest discriminant in which 2, 3 and 5 all split. Over  $B$  we prescribe a nonabelian dyadic local Galois group of order 32 at each of the two primes above 2, a tame local group  $C_2 \times C_2$  at each of the four primes above 3 and 5, and conditions that make the Frobenius elements have order dividing 4 at the two primes above 29 and at the prime 7, which is inert in  $B$ . The relations are counted through their images in the Zassenhaus filtration, and one dyadic relation is implied by the others through Hilbert reciprocity. The resulting tower is infinite, it is ramified only above 2, 3 and

5, and its Galois group has 8 generators. The root discriminant of its fields is

$$\lambda = 2^{9/4}\sqrt{3615} \approx 286.0, \quad 3615 = 3 \cdot 5 \cdot 241.$$

Sawin's fields come from a pro-2 tower of unramified extensions of  $\mathbb{Q}(\sqrt{3 \cdot 5 \cdots 43})$ , so they are ramified over  $\mathbb{Q}$  at 2 and at the thirteen odd primes up to 43, and their root discriminant is about  $1.6 \cdot 10^8$ . Theorem 2.24 states the family of fields.

*The zeta value.* Every field  $K$  of Theorem 2.24 contains the Kummer field  $E = B(\sqrt{V})$ , of degree 512 over  $\mathbb{Q}$ , where  $V$  is the group of  $\{2, 3, 5\}$ -units of  $B$  modulo squares. Its Dedekind zeta function is the product of  $\zeta_B$  and 255 quadratic Hecke  $L$ -functions of  $B$ , which we evaluate with certified approximate functional equations (Sections 3 and 4). Together with refinements from the exact local types of the tower, this gives the bound  $C = 0.04871285$  for  $d^{-1} \log L_F(1)$  (Theorem 4.1). It replaces Louboutin's bound [16], which Sawin uses, and the Euler products over a fixed multiquadratic subfield of [22].

*Profiles and shells.* Sawin counts points in a sup-norm window by packing, and [22] counts them by covolume in a Euclidean ball. We use a Gaussian profile at the real places of  $F$ , a coupled Student–Bernstein profile at the complex places, and weighted valuation shells at the selected primes, and we count points by Poisson summation (Sections 5–7). Table 2 compares Sawin's example with the present construction.

TABLE 2. Sawin's example and the present construction. The pairs  $(e, f)$  are absolute ramification indices and residue degrees of the selected primes of  $F$ ; Sawin's residue degrees are at most 2.

|                              | Sawin [32]                               | this paper                                        |
|------------------------------|------------------------------------------|---------------------------------------------------|
| base field of the tower      | $\mathbb{Q}(\sqrt{3 \cdot 5 \cdots 43})$ | $\mathbb{Q}(\sqrt{241})$                          |
| primes ramified in the tower | 2 and 3, 5, ..., 43                      | those above 2, 3, 5                               |
| fields $K$                   | CM, $\theta = 0$                         | mixed signature, $\theta \geq 65535/131072$       |
| selected primes, $(e, f)$    | 22 primes up to 179                      | 2: (8, 4); 3, 5: (2, 2);<br>7: (1, 8); 29: (1, 4) |
| root discriminant $\lambda$  | about $1.6 \cdot 10^8$                   | $2^{9/4}\sqrt{3615} \approx 286.0$                |
| $\ell = \log \lambda$        | 18.90                                    | 5.656                                             |
| zeta term                    | $1 + \log \log \lambda \approx 3.94$     | $C = 0.04871285$                                  |
| exponent $1 + \delta$        | 1.014114                                 | 1.04273                                           |

1.4. **Why the exponent improves.** Write  $\ell = \log \lambda$ ,

$$J = \sum_q \frac{\log(k_q + 1)}{e_q f_q}, \quad H = \sum_q \frac{k_q \log q}{e_q},$$

so that the sum in (1.2) is  $J - \delta H$ . Sawin's criterion [32, Proposition 10] applies to Galois CM fields of unbounded degree with relative root discriminant  $\lambda$ , and uses a sup-norm window with parameter  $R > 1$ . Rearranged, it gives arbitrarily large sets with  $u(U) \gg |U|^{1+\delta}$  whenever (1.3) holds; inequality (1.4) is (1.2) with  $\mathcal{A}_\delta(\theta)$  written out and regrouped:

$$(1.3) \quad (J - \delta H) + \left(\log 2\pi - \frac{1}{2}\ell - 1 - \log \ell\right) + \left(2\log(1 - R^{-1}) - 2\delta \log(2R + e^{-H/2})\right) \geq 0,$$

$$(1.4) \quad (J - \delta H) + \left(\log 2 + (1 - \theta) \log \pi - \frac{1}{2}\ell - C\right) + \left(\delta \ell - \delta \log 2 + (1 - 2\theta)J_{\mathbb{R}} + \theta J_{\mathbb{C}}\right) > 0.$$

In both, the three groups are the gain from equal-norm choices, the loss in selecting a common ideal class, and the count of vertices together with the window.

*The fields.* The dominant change is in  $\ell$ . Sawin's example has  $\ell > 18.9$ ; here  $\ell < 5.657$ , so the term  $-\frac{1}{2}\ell$  improves by more than 6.6 per unit of  $d$ . The price is paid in the first group: Sawin

selects 22 primes with residue degree at most 2, and we select only the primes above 2, 3, 5, 7 and 29, several with residue degree 4 or 8.

*Ideal-class selection.* Both criteria use the analytic class-number formula, which is exact in either signature; they differ in how they bound the zeta value. At  $\theta = 0$  the constants agree, and the groups differ only in  $C$  against  $1 + \log \ell$ . In Sawin's criterion,  $1 + \log \ell$  is the sum of  $\log(e/2) + \log \ell$ , which is Louboutin's bound [16, Corollary 3] for  $d^{-1} \log L_F(1)$  read through the class-number formula, and  $\log 2$ , for the factor  $2^d$  allowed for units in Sawin's modified class group [32, Lemma 6]. The manuscript [22] removed the unit factor and sharpened Louboutin's bound with Euler products over a fixed subfield. Here Proposition 5.2 accounts for the unit index exactly, and the Kummer field gives  $C = 0.04871285$ , whereas  $1 + \log \ell$  would exceed 2.73 even at our value of  $\lambda$ . Mixed signature costs  $\theta \log \pi$  in this group, since each place of  $F$  carries one factor  $\pi$  in the class-number formula.

*Vertices and window.* Sawin's packing count gives  $\delta(H + 2 \log(2R + e^{-H/2}))$ , which does not involve  $\lambda$ . Counting by covolume, as in [22] and here, returns  $\delta \ell$ , because a larger discriminant makes the ideal lattice sparser. The overlap  $2 \log(1 - R^{-1})$  of the sup-norm window and its volume become the profile functionals  $J_{\mathbb{R}}$  and  $J_{\mathbb{C}}$ .

*Signature.* CM fields have  $\theta = 0$ . With the other data fixed, mixed signature adds  $\theta(J_{\mathbb{C}} - 2J_{\mathbb{R}} - \log \pi)$  to the left side of (1.4). The certificate gives  $J_{\mathbb{C}} - 2J_{\mathbb{R}} - \log \pi > 0.632$ , so with  $\theta \geq 65535/131072$  this term exceeds 0.316 per unit of  $d$ ; at  $\theta = 0$  the certified margin would be negative (Remark 8.6).

Table 8 lists the certified terms of the margin at  $\delta = 0.04273$ . Their sum leaves a margin of about  $1.77 \cdot 10^{-4}$  per unit of  $d$ . With the same profiles, and shell weights optimized in the same way, the margin at  $\delta = 0.0428$  is negative.

**1.5. Organization.** Section 2 constructs the tower over  $B$  and proves Theorem 2.24, which supplies the fields. Section 3 describes the Kummer field and its quadratic  $L$ -functions, and Section 4 proves the ceiling  $d^{-1} \log L_F(1) < C$  (Theorem 4.1). Sections 5–7 carry out the geometric construction: the norm-one mass formula and the transfer to planar sets, the weighted windows at the selected primes, and the archimedean profiles. These sections hold for any quadratic extension with the stated properties. Section 8 evaluates the final inequality at  $\delta = 0.04273$  and completes the proof of Theorem 1.1.

**1.6. Computations and data.** The proof uses finite computations: linear algebra over  $\mathbb{F}_2$  for the tower, exact rational arithmetic for the Golod–Shafarevich series, certified evaluations of  $L$ -functions in the ball arithmetic of Arb, now part of FLINT [15, 38], and directed interval bounds for the geometric margin [40]. PARI/GP [41] is used for the arithmetic of number fields and for independent checks. Each finite fact is verified by a supplementary program, named in the text where the fact is used. These certificates concern exact arithmetic and rigorous inequalities, not the statistical accuracy of floating-point approximations.

The programs and their data are in the directory `papers/0.04273/certificates` of the public repository [24]. Some certified routines, for the degree-two approximate functional equations, the archimedean profiles and the finite windows, are imported unchanged from an earlier supplementary archive of the author, in the directory `papers/0.0418235/certificates` of the same repository, together with a manifest of the SHA-256 hashes of its members. A replay program runs every step and records its results. The computations were run with PARI/GP 2.17.2, python-flint 0.9.0 (FLINT 3.6.0) and mpmath 1.3.0.

A separate Lean 4 development [23], built on Mathlib [2, 39], proves the planar conclusion with the slightly smaller exponent 1.0427 from one explicit numerical inequality for the Dedekind zeta function of  $E$  (Remark 4.28). That inequality has not been proved in Lean, so the formalization is conditional, and it is not used as a substitute for the arguments here.

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2. THE TOWER OVER  $\mathbb{Q}(\sqrt{241})$ 

This section constructs the number fields of Theorem 2.24 as finite layers of an infinite pro-2 extension of  $B = \mathbb{Q}(\sqrt{241})$  with prescribed local Galois groups. Subsection 2.1 describes  $B$  and its  $\{2, 3, 5\}$ -units. Subsection 2.2 recalls the local Galois groups involved and constructs the dyadic local field. Subsection 2.3 shows that the Galois group  $\mathcal{G}_B$  of the maximal pro-2 extension of  $B$  unramified outside  $2, 3, 5$  has a presentation with eight generators and seven local relators. Subsection 2.4 imposes the local conditions and defines the quotient  $G_B$ . Subsection 2.5 computes the first three layers of the Zassenhaus filtration of  $G_B$ : they retain the prescribed local groups, and complex conjugation has many conjugates. Subsections 2.6 and 2.7 prove that  $G_B$  is infinite, by a filtered Fox calculus and the Golod–Shafarevich method [10]. Subsection 2.8 proves Theorem 2.24.

We use the following conventions. Subgroups of profinite groups are closed, generation is topological, and the normal closure of a set is the smallest closed normal subgroup containing it. The commutator is  $[g, h] = g^{-1}h^{-1}gh$ . For a finite 2-group  $\Delta$  with augmentation ideal  $\mathfrak{I} \subset \mathbb{F}_2[\Delta]$ , the Zassenhaus filtration is  $D_n\Delta = \{g : g - 1 \in \mathfrak{I}^n\}$ ; for a pro-2 group it is defined through finite quotients. Then  $[D_m, D_n] \subset D_{m+n}$ ,  $g^2 \in D_{2n}$  for  $g \in D_n$ , a surjection maps  $D_n$  onto  $D_n$ , and  $\text{gr } \Delta = \bigoplus_{n \geq 1} D_n\Delta / D_{n+1}\Delta$  is a restricted Lie algebra over  $\mathbb{F}_2$  whose bracket is induced by commutators and whose restricted square  $v \mapsto v^{[2]}$  is induced by squaring [13, 31, 8]. By Lazard’s formula [8, Proposition 3.2],  $D_n = \prod_{i, 2^j \geq n} \gamma_i^{2^j}$ , where  $\gamma_i$  is the lower central series. In particular  $D_1$  is the whole group,  $D_2$  is the Frattini subgroup, and  $D_3 = D_2^2[D_2, D_1]$ . For a free pro-2 group  $\mathcal{F}$  of rank  $n$ ,  $\text{gr } \mathcal{F}$  is the free restricted Lie algebra on  $\mathcal{F}/D_2\mathcal{F} \cong \mathbb{F}_2^n$  [31, 8]. We realize it inside the free associative algebra  $\mathbb{F}_2\langle X_1, \dots, X_n \rangle$ , with bracket  $[a, b] = ab - ba$  and restricted square  $a^{[2]} = a^2$ ; its layers of degrees one, two and three have dimensions  $n$ ,  $n(n+1)/2$  and  $(n^3 - n)/3$ .

**2.1. The Base Field.** Let  $B = \mathbb{Q}(\sqrt{241})$ , with real places  $v_1$  ( $\sqrt{241} \mapsto +\sqrt{241}$ ) and  $v_2$  ( $\sqrt{241} \mapsto -\sqrt{241}$ ). Put

$$\begin{aligned}
 (2.1) \quad & \alpha_0 = -1, \quad \alpha_1 = \varepsilon = -71011068 + 4574225\sqrt{241}, \\
 & \alpha_2 = \frac{-6101-393\sqrt{241}}{2}, \quad \alpha_3 = \frac{6101-393\sqrt{241}}{2}, \\
 & \alpha_4 = 31 - 2\sqrt{241}, \quad \alpha_5 = 31 + 2\sqrt{241}, \quad \alpha_6 = 326 - 21\sqrt{241}, \quad \alpha_7 = 326 + 21\sqrt{241}.
 \end{aligned}$$

Their norms  $N_{B/\mathbb{Q}}(\alpha_i)$  are  $1, -1, -2, -2, -3, -3, -5, -5$ , and  $\alpha_2\alpha_3 = 2$ ,  $\alpha_4\alpha_5 = -3$ ,  $\alpha_6\alpha_7 = -5$ . For a nonzero ideal  $\mathfrak{a}$  of  $\mathcal{O}_B$  we write  $N\mathfrak{a} = |\mathcal{O}_B/\mathfrak{a}|$  for its absolute norm.

**Lemma 2.2.** (a)  $\mathcal{O}_B = \mathbb{Z}[(1 + \sqrt{241})/2]$ , the discriminant of  $B$  is 241, and  $B$  has class number one.

(b) The unit  $\varepsilon$  has norm  $N_{B/\mathbb{Q}}(\varepsilon) = -1$ , and the classes of  $-1$  and  $\varepsilon$  form a basis of  $\mathcal{O}_B^\times / \mathcal{O}_B^{\times 2}$ .

(c) In  $\mathcal{O}_B$  we have  $2 = \mathfrak{p}_1\mathfrak{p}_2$ ,  $3 = \mathfrak{q}_1\mathfrak{q}_2$ ,  $5 = \mathfrak{r}_1\mathfrak{r}_2$  and  $29 = \mathfrak{t}_1\mathfrak{t}_2$ , where

$$\mathfrak{p}_j = \alpha_{1+j}\mathcal{O}_B, \quad \mathfrak{q}_j = \alpha_{3+j}\mathcal{O}_B, \quad \mathfrak{r}_j = \alpha_{5+j}\mathcal{O}_B, \quad \mathfrak{t}_j = (-14127 \pm 910\sqrt{241})\mathcal{O}_B,$$

with the sign  $+$  for  $j = 1$ . These eight primes are distinct and of degree one. The prime 7 is inert, and we write  $\mathfrak{t}_0 = 7\mathcal{O}_B$ , so that  $N\mathfrak{t}_0 = 49$ . The prime 241 ramifies. At each of the eight primes of degree one, above  $p \in \{2, 3, 5, 29\}$ , the completion of  $B$  is  $\mathbb{Q}_p$ , and  $\sqrt{241}$  maps to the square root of 241 in  $\mathbb{Z}_p$  given in Table 3.

(d) Let  $S = \{\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2\}$  and  $V = \mathcal{O}_B[1/30]^\times / \mathcal{O}_B[1/30]^{\times 2}$ . The classes of  $\alpha_0, \dots, \alpha_7$  form a basis of  $V$ , and  $V$  is the subgroup of  $B^\times / B^{\times 2}$  of classes with even valuation at every prime outside  $S$ . Moreover  $\text{Pic}(\mathcal{O}_B[1/30]) = 0$ .

*Proof.* (a) Since  $241 \equiv 1 \pmod{4}$ , the ring of integers and the discriminant are as stated. The Minkowski bound is  $\sqrt{241}/2 < 8$ , so every ideal class contains an integral ideal of norm at most 7, which is a product of primes of norm at most 7. By (c), these are the six primes above 2, 3, 5, and they are principal.

(b) A direct computation gives  $71011068^2 - 241 \cdot 4574225^2 = -1$ . By Dirichlet's theorem  $\mathcal{O}_B^\times = \{\pm 1\} \times \eta^\mathbb{Z}$  for a fundamental unit  $\eta$ , and  $\varepsilon = \pm \eta^k$ . Since  $N_{B/\mathbb{Q}}(\varepsilon) = -1$ , the exponent  $k$  is odd, so  $-1$  and  $\varepsilon$  generate  $\mathcal{O}_B^\times$  modulo squares. Neither  $-1$  nor  $\pm \varepsilon$  is a square, because  $B$  is real and  $N_{B/\mathbb{Q}}(\pm \varepsilon) = -1$ .

(c) We have  $241 \equiv 1 \pmod{8}$ ,  $241 \equiv 1 \pmod{3}$ ,  $241 \equiv 1 \pmod{5}$ ,  $241 \equiv 9 \pmod{29}$  and  $241 \equiv 3 \pmod{7}$ , and 3 is not a square modulo 7. Hence 2, 3, 5, 29 split, 7 is inert and 241 ramifies. The listed generators have norms  $\pm 2, \pm 3, \pm 5, 29$ , so they generate primes of degree one, and the displayed products show that the two generators above each  $p$  generate the two distinct primes above  $p$ ; for 29 the product of the two generators is 29. A prime of degree one above  $p$  is the preimage of  $p\mathbb{Z}_p$  under an embedding  $B \rightarrow \mathbb{Q}_p$ , which sends  $\sqrt{241}$  to one of the two square roots of 241 in  $\mathbb{Z}_p$ ; the completion there is  $\mathbb{Q}_p$ , and the prime contains the stated generator exactly for the square root in Table 3.

(d) Every  $S$ -unit  $u$  satisfies  $u\mathcal{O}_B = \prod_{i \geq 2} (\alpha_i \mathcal{O}_B)^{a_i}$ , so  $u \prod_{i \geq 2} \alpha_i^{-a_i}$  is a unit, and by (b) the classes of  $\alpha_0, \dots, \alpha_7$  generate  $V$ . If  $\prod_i \alpha_i^{a_i}$  with  $a_i \in \{0, 1\}$  is a square, the valuations at  $S$  give  $a_i = 0$  for  $i \geq 2$ , and then (b) gives  $a_0 = a_1 = 0$ . If  $b \in B^\times$  has even valuation outside  $S$ , then  $b\mathcal{O}_B = \mathfrak{a}^2 \prod_{\mathfrak{p} \in S} \mathfrak{p}^{a_{\mathfrak{p}}}$  with  $\mathfrak{a}$  principal by (a), so  $b$  is an  $S$ -unit times a square. Finally  $\text{Pic}(\mathcal{O}_B[1/30])$  is a quotient of  $\text{Pic}(\mathcal{O}_B) = 0$ .  $\square$

The classes in Table 3 follow from the residues of  $\sqrt{241}$  given there; at  $p = 2$  they need its image modulo 32. For example, at  $\mathfrak{p}_1$  we have  $4574225 \equiv 1$  and  $-71011068 \equiv 4$  modulo 8, so  $\varepsilon \equiv 4 + 7 \equiv 3 \equiv -5 \pmod{8}$ . At the inert prime  $\mathfrak{t}_0$ , an element  $a \in \mathcal{O}_B$  prime to  $\mathfrak{t}_0$  is a square modulo  $\mathfrak{t}_0$  exactly when  $a^{24} \equiv 1$ , that is, when the reduction of  $N_{B/\mathbb{Q}}(a) \equiv a^8$  is a square in  $\mathbb{F}_7$ . Hence  $\alpha_i$  is a square modulo  $\mathfrak{t}_0$  exactly for  $i = 0, 4, 5, 6, 7$ .

**2.2. Local Galois Groups.** For a place  $\nu$  of  $B$  let  $\mathcal{G}_\nu$  be the Galois group of the maximal pro-2 extension of the completion  $B_\nu$ ; for a real place,  $\mathcal{G}_\nu = \text{Gal}(\mathbb{C}/\mathbb{R})$ . We write  $(\cdot, \cdot)$  for the Hilbert symbol of order two of a local field [19, Chapter III, Section 4]. For odd  $p$  and  $p$ -adic units  $u, w$ , and for 2-adic units  $u, w$ ,

$$(2.3) \quad (p^a u, p^b w)_p = (-1)^{ab(p-1)/2} \left(\frac{u}{p}\right)^b \left(\frac{w}{p}\right)^a, \quad (2^a u, 2^b w)_2 = (-1)^{\frac{u-1}{2} \frac{w-1}{2} + a \frac{w^2-1}{8} + b \frac{u^2-1}{8}}$$

[33, Chapter III, Theorem 1], [19, Chapter VIII, Section 5]. A defining relator of a pro-2 group with respect to generators  $g_1, \dots, g_k$  is an element of the free pro-2 group on  $k$  letters whose normal closure is the kernel of the map to the group; its quadratic initial is its class in the degree-two layer of the Zassenhaus filtration of the free group.

TABLE 3. The image of  $\sqrt{241}$  in  $\mathbb{Z}_p$  at each prime of degree one, given by its residue, and in  $\mathbb{R}$  at the real places; the classes of  $\alpha_0, \dots, \alpha_7$  in  $\mathbb{Q}_p^\times/\mathbb{Q}_p^{\times 2}$  at these primes, and their signs at the real places. At  $p = 2$  the classes are written as  $\pm 1, \pm 5, \pm 2, \pm 10$ ; at odd  $p$  as  $1, u, p, up$ , where  $u = -1, 2, 2$  is a nonsquare unit for  $p = 3, 5, 29$ .

| place            | $\sqrt{241} \mapsto$ | $\alpha_0$ | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ | $\alpha_5$ | $\alpha_6$ | $\alpha_7$ |
|------------------|----------------------|------------|------------|------------|------------|------------|------------|------------|------------|
| $\mathfrak{p}_1$ | 7 mod 32             | -1         | -5         | -10        | -5         | 1          | 5          | -5         | 1          |
| $\mathfrak{p}_2$ | 25 mod 32            | -1         | 5          | 5          | 10         | 5          | 1          | 1          | -5         |
| $\mathfrak{q}_1$ | 2 mod 3              | -1         | 1          | -1         | 1          | 3          | -1         | -1         | -1         |
| $\mathfrak{q}_2$ | 1 mod 3              | -1         | -1         | -1         | 1          | -1         | 3          | -1         | -1         |
| $\mathfrak{r}_1$ | 1 mod 5              | 1          | 2          | 2          | 1          | 1          | 2          | 10         | 2          |
| $\mathfrak{r}_2$ | 4 mod 5              | 1          | 2          | 1          | 2          | 2          | 1          | 2          | 10         |
| $\mathfrak{t}_1$ | 3 mod 29             | 1          | 1          | 2          | 1          | 1          | 2          | 2          | 2          |
| $\mathfrak{t}_2$ | 26 mod 29            | 1          | 1          | 1          | 2          | 2          | 1          | 2          | 2          |
| $v_1$            | $+\sqrt{241}$        | -          | +          | -          | -          | -          | +          | -          | +          |
| $v_2$            | $-\sqrt{241}$        | -          | -          | +          | +          | +          | -          | +          | -          |

- Lemma 2.4.** (a) If  $\nu$  is real, then  $\mathcal{G}_\nu = \langle \iota \rangle \cong C_2$  with the single defining relator  $\iota^2$ , and  $\iota(\sqrt{b}) = -\sqrt{b}$  exactly when  $b < 0$ .
- (b) Let  $p$  be an odd prime and  $\varpi$  a uniformizer of  $\mathbb{Q}_p$ . Then  $\mathcal{G}_{\mathbb{Q}_p}$  is generated by a generator  $\tau$  of its inertia group and a Frobenius lift  $\varphi$  with  $\varphi(\sqrt{\varpi}) = \sqrt{\varpi}$ , with the single defining relator  $\varphi\tau\varphi^{-1}\tau^{-p}$ . For  $b \in \mathbb{Q}_p^\times$ ,  $\tau(\sqrt{b}) = -\sqrt{b}$  exactly when  $v_p(b)$  is odd; for a unit  $b$ ,  $\varphi(\sqrt{b}) = -\sqrt{b}$  exactly when  $b$  is not a square modulo  $p$ . The relator has quadratic initial  $[\tau, \varphi] + \frac{p-1}{2}\tau^{[2]}$ .
- (c) Let  $E_2 = \mathbb{Q}_2(\sqrt{-1}, \sqrt{2}, \sqrt{5})$ , and let  $x, y, z \in \mathcal{G}_{\mathbb{Q}_2}$  multiply the triple  $(\sqrt{2}, \sqrt{-1}, \sqrt{5})$  by the signs  $(-, +, +)$ , by  $(+, -, +)$  and by  $(+, -, -)$ . Then  $x, y, z$  generate  $\mathcal{G}_{\mathbb{Q}_2}$ . If  $b \equiv (-1)^{a_1}2^{a_2}5^{a_3}$  modulo squares, they multiply  $\sqrt{b}$  by  $(-1)^{a_2}$ ,  $(-1)^{a_1}$  and  $(-1)^{a_1+a_3}$ , that is, by the Hilbert symbols  $(5, b)_2$ ,  $(-1, b)_2$  and  $(-2, b)_2$ . The group  $\mathcal{G}_{\mathbb{Q}_2}$  has a single defining relator  $r$  with respect to  $x, y, z$ , and every such  $r$  has quadratic initial  $y^{[2]} + [x, y] + [x, z]$ .

*Proof.* (a) is clear.

(b) By Iwasawa's theorem [26, Theorem 7.5.3], the Galois group of the maximal tamely ramified extension of  $\mathbb{Q}_p$  is generated by a Frobenius lift  $\varphi'$  and a generator  $\tau$  of tame inertia, with the single relation  $\varphi'\tau\varphi'^{-1} = \tau^p$ . Since  $p$  is odd, every 2-extension of  $\mathbb{Q}_p$  is tamely ramified, so  $\mathcal{G}_{\mathbb{Q}_p}$  is the pro-2 group with the same presentation. The extension  $\mathbb{Q}_p(\sqrt{\varpi})$  is ramified, so  $\tau$  moves  $\sqrt{\varpi}$ ; replacing  $\varphi'$  by  $\varphi'\tau$  if necessary gives  $\varphi$ . This substitution does not change the relator, since  $(\varphi'\tau)\tau(\varphi'\tau)^{-1} = \varphi'\tau\varphi'^{-1}$ . The actions on square roots hold because  $\mathbb{Q}_p(\sqrt{b})$  is ramified exactly when  $v_p(b)$  is odd, and because  $\varphi$  acts on unramified extensions as the Frobenius. Modulo  $D_3$ , the element  $\varphi\tau\varphi^{-1}\tau^{-1}$  has class  $[\varphi, \tau] = [\tau, \varphi]$ , and  $\tau^{1-p}$  has class  $\frac{p-1}{2}\tau^{[2]}$ .

(c) By Kummer theory the maximal elementary abelian quotient of  $\mathcal{G}_{\mathbb{Q}_2}$  is  $\text{Gal}(E_2/\mathbb{Q}_2)$ , since  $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$  has basis  $-1, 2, 5$ . The images of  $x, y, z$  form a basis of it, so  $x, y, z$  generate [26, Proposition 3.9.1]. The action on  $\sqrt{b}$  is read off from the actions on  $\sqrt{-1}, \sqrt{2}, \sqrt{5}$ , and it agrees with the stated Hilbert symbols by (2.3). By [26, Theorem 7.5.11],  $\mathcal{G}_{\mathbb{Q}_2}$  is a Demuškin group of rank 3: it has a single defining relator  $r$ , and  $H^2(\mathcal{G}_{\mathbb{Q}_2}, \mathbb{F}_2) \cong \mathbb{F}_2$ . For  $b \in \mathbb{Q}_2^\times$  let  $\chi_b$  be the character with  $g(\sqrt{b}) = (-1)^{\chi_b(g)}\sqrt{b}$ . The characters dual to  $x, y, z$  are  $\chi_2, \chi_{-5}, \chi_5$ . Inflation identifies  $H^2(\mathcal{G}_{\mathbb{Q}_2}, \mathbb{F}_2)$  with  $H^2$  of the absolute Galois group of  $\mathbb{Q}_2$  [26, Proposition 7.5.8], and there the invariant of  $\chi_a \cup \chi_b$  is given by the Hilbert symbol  $(a, b)_2$  [19, Chapter III, Section 4]. By (2.3), the matrix of the

indicator of  $(a, b)_2 = -1$  on  $2, -5, 5$  is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

The evaluation  $\text{tr}_r$  of the transgression at  $r$  is an isomorphism  $H^2(\mathcal{G}_{\mathbb{Q}_2}, \mathbb{F}_2) \rightarrow \mathbb{F}_2$  [26, Proposition 3.9.12(iii)], and so is the invariant, so the two coincide. We now apply the relation–cup identity [26, Proposition 3.9.13(ii)], see also [30, Proposition 3.2]. Write  $g_1, g_2, g_3$  for  $x, y, z$ . The identity writes the relator as  $r = \prod_j g_j^{q a_j} \prod_{k < l} [g_k, g_l]^{a_{kl}} r'$ , with  $r'$  in the third term of the descending  $q$ -central series, and states that  $\text{tr}_r$  of the cup product of the characters dual to  $g_k$  and  $g_l$  is  $-a_{kl}$  for  $k < l$  and  $-\binom{q}{2} a_k$  for  $k = l$ . Here  $q$ , the smallest elementary divisor of the abelianization of  $\mathcal{G}_{\mathbb{Q}_2}$ , is 2: by local class field theory this abelianization is the pro-2 completion of  $\mathbb{Q}_2^\times$ , which is  $\mathbb{Z}_2^2 \times \mathbb{Z}/2$ . The third term of the 2-central series is  $(D_2)^2[D_2, D_1] = D_3$  by Lazard's formula, the class of  $g_j^2$  in  $\text{gr}_2$  is  $g_j^{[2]}$ , and the signs disappear modulo 2, with  $\binom{2}{2} = 1$ . Hence the off-diagonal entries of the matrix are the coefficients of  $[x, y], [x, z], [y, z]$  in the quadratic initial of  $r$ , and the diagonal entries, which are the values of the cup squares  $\chi \cup \chi$ , are the coefficients of  $x^{[2]}, y^{[2]}, z^{[2]}$ . For  $p = 2$  the cup square of a character is its Bockstein, so the diagonal coefficients are also given by [26, Proposition 3.9.14].  $\square$

We now construct the dyadic local field. Let  $U_4$  be the unramified extension of  $\mathbb{Q}_2$  of degree four, with Frobenius  $\text{Fr}$ ; let  $i = \sqrt{-1}$ ,  $k = \mathbb{Q}_2(i)$ ,  $\alpha = 1 + 2i$ , and

$$M_8 = \mathbb{Q}_2(i, \sqrt{5}, \sqrt{\alpha}), \quad M_{16} = M_8(\sqrt{2}), \quad L_2 = M_{16}U_4.$$

Since  $\alpha\bar{\alpha} = 5$  with  $\bar{\alpha} = 1 - 2i$ , the field  $M_8$  contains  $\sqrt{\alpha} = \sqrt{5}/\sqrt{\alpha}$ . Let  $\sigma_y$  and  $\sigma_z$  be the automorphisms of  $M_8$  given by

$$\sigma_y : i \mapsto -i, \sqrt{5} \mapsto \sqrt{5}, \sqrt{\alpha} \mapsto \sqrt{5}/\sqrt{\alpha}; \quad \sigma_z : i \mapsto -i, \sqrt{5} \mapsto -\sqrt{5}, \sqrt{\alpha} \mapsto \sqrt{5}/\sqrt{\alpha}.$$

**Lemma 2.5.** (a) *The extension  $M_8/\mathbb{Q}_2$  is Galois of degree 8, and its Galois group  $\langle \sigma_y, \sigma_z \rangle$  is dihedral.*

(b) *The extension  $L_2/\mathbb{Q}_2$  is Galois of degree 32, with ramification index 8 and residue degree 4. There are unique  $x, y, z \in \text{Gal}(L_2/\mathbb{Q}_2)$  such that  $x$  is trivial on  $M_8$  and  $U_4$  and negates  $\sqrt{2}$ ;  $y$  restricts to  $\sigma_y$  on  $M_8$ , fixes  $\sqrt{2}$  and is trivial on  $U_4$ ; and  $z$  restricts to  $\sigma_z$  on  $M_8$ , fixes  $\sqrt{2}$  and restricts to  $\text{Fr}$  on  $U_4$ . They act on  $E_2 \subset L_2$  as in Lemma 2.4(c), and the map sending the generators of*

$$D = \langle x, y, z \mid x^2, y^2, z^4, [y, z]^2, [x, y], [x, z], [[y, z], z] \rangle$$

*to  $x, y, z$  is an isomorphism  $D \cong \text{Gal}(L_2/\mathbb{Q}_2)$ . The inertia group is  $\langle x, y, [y, z] \rangle \cong C_2^3$ .*

(c) *The Zassenhaus layers of  $D$  are  $\text{gr}_1 D = \langle \bar{x}, \bar{y}, \bar{z} \rangle \cong \mathbb{F}_2^3$  and  $\text{gr}_2 D = \langle z^2, [y, z] \rangle \cong \mathbb{F}_2^2$ , and  $D_3(D) = 1$ , so the augmentation Hilbert polynomial of  $\mathbb{F}_2[D]$  is  $P_D(t) = (1+t)^3(1+t^2)^2$ .*

(d) *The different of  $L_2/\mathbb{Q}_2$  has valuation 18 in  $L_2$ ; equivalently, its normalized exponent is  $18/8 = 9/4$ .*

*Proof. Quadratic extensions of  $k$ .* Put  $\varpi = 1 + i$ , a uniformizer of  $k$ , so that  $v_k(2) = 2$ , and for  $b \in k^\times$  let  $f(b)$  be the exponent of the conductor of  $k(\sqrt{b})/k$ , which for a quadratic extension equals the exponent of its discriminant. Then  $f(5) = 0$ , since  $\mathbb{Q}_2(\sqrt{5})/\mathbb{Q}_2$  is unramified. Since  $\alpha = 1 + \varpi^2$ , the element  $\varpi_\alpha = (\sqrt{\alpha} - 1 - \varpi)/\varpi$  is a root of the Eisenstein polynomial  $X^2 + \frac{2(1+\varpi)}{\varpi}X + \frac{2}{\varpi}$ ; its derivative at  $\varpi_\alpha$  is the sum of terms of valuations 5 and 2 in  $k(\sqrt{\alpha})$ , so  $f(\alpha) = 2$ . Next  $2\alpha = \varpi^2\alpha'$  with  $\alpha' = 2 - i$ , and  $\sqrt{\alpha'} - 1$  is a root of  $X^2 + 2X + 1 - \alpha'$ , which is Eisenstein since  $v_k(1 - \alpha') = 1$ ; its derivative  $2\sqrt{\alpha'}$  has valuation 4, so  $f(2\alpha) = 4$ . Similarly  $2 = -i\varpi^2$ ,  $-i \equiv i$  modulo squares, and  $\sqrt{i} - 1$  is a root of  $X^2 + 2X + 1 - i$ , so  $f(2) = 4$ . Finally  $f(5b) = f(b)$  when  $b \notin k^{\times 2} \cup 5k^{\times 2}$ :

the field  $k(\sqrt{5}, \sqrt{b})$  is unramified over both  $k(\sqrt{b})$  and  $k(\sqrt{5b})$ , so the two ways of computing its discriminant over  $k$  give  $2f(b) = 2f(5b)$ .

(a) Since  $k(\sqrt{5})/k$  is unramified and  $k(\sqrt{\alpha})/k$  is ramified,  $\alpha$  is not a square in  $k(\sqrt{5})$ , so  $[M_8 : \mathbb{Q}_2] = 8$ . The field  $M_8$  contains the conjugate  $\sqrt{\alpha}$  of  $\sqrt{\alpha}$ , so it is Galois over  $\mathbb{Q}_2$ , and an automorphism is determined by the images  $\pm i, \pm\sqrt{5}$  and a square root of the image of  $\alpha$ ; in particular  $\sigma_y$  and  $\sigma_z$  exist. Now  $\sigma_y^2 = 1$ ,  $\sigma_z^2$  fixes  $i, \sqrt{5}$  and negates  $\sqrt{\alpha}$ ,  $\sigma_z$  has order four, and  $\sigma_y\sigma_z\sigma_y = \sigma_z^{-1}$ . So  $\text{Gal}(M_8/\mathbb{Q}_2) = \langle \sigma_y, \sigma_z \rangle$  is dihedral of order eight, with center  $\langle \sigma_z^2 \rangle$  and maximal abelian subextension  $\mathbb{Q}_2(i, \sqrt{5})$ .

(b) The quadratic subfields of  $M_8$  are  $\mathbb{Q}_2(\sqrt{b})$  for  $b = -1, 5, -5$ , so  $\sqrt{2} \notin M_8$ , and  $M_{16}$  has degree 16 and group  $\text{Gal}(M_8/\mathbb{Q}_2) \times C_2$ . This group has elementary abelian abelianization, so the maximal unramified subextension of  $M_{16}$ , which is cyclic over  $\mathbb{Q}_2$  and contains  $\mathbb{Q}_2(\sqrt{5})$ , is  $\mathbb{Q}_2(\sqrt{5})$ . Hence  $M_{16}$  has ramification index 8,  $M_{16} \cap U_4 = \mathbb{Q}_2(\sqrt{5})$ , and  $[L_2 : \mathbb{Q}_2] = 16 \cdot 4/2 = 32$ . Since  $L_2$  contains  $U_4$  and has ramification index at least 8, its residue degree is 4 and its ramification index is 8. Restriction identifies  $\text{Gal}(L_2/\mathbb{Q}_2)$  with the set of triples  $(\sigma, \epsilon, \mu)$ , where  $\sigma \in \text{Gal}(M_8/\mathbb{Q}_2)$ ,  $\epsilon = \pm 1$  is the sign by which the element multiplies  $\sqrt{2}$ , and  $\mu \in \text{Gal}(U_4/\mathbb{Q}_2)$  agrees with  $\sigma$  on  $\sqrt{5}$ ; both sets have 32 elements. Since  $\text{Fr}$  negates  $\sqrt{5}$ , the triples

$$x = (1, -1, 1), \quad y = (\sigma_y, 1, 1), \quad z = (\sigma_z, 1, \text{Fr}), \quad w = [y, z] = (\sigma_z^2, 1, 1)$$

exist and are the unique elements described in the statement. They act on  $E_2$  as required. The elements  $x$  and  $w$  are central involutions,  $y$  is an involution, and  $z$  has order four. Hence the relations of  $D$  hold. Conversely, in  $D$  these relations make  $w = [y, z]$  central:  $w^y = w^{-1} = w$ , and  $[w, z] = 1$  is imposed. Since  $zy = yzw$ , every element of  $D$  is  $x^{a_1}y^{a_2}z^{a_3}w^{a_4}$  with  $a_1, a_2, a_4 \in \mathbb{F}_2$  and  $a_3 \in \mathbb{Z}/4$ , and

$$(a_1, a_2, a_3, a_4)(a'_1, a'_2, a'_3, a'_4) = (a_1 + a'_1, a_2 + a'_2, a_3 + a'_3, a_4 + a'_4 + a_3a'_2).$$

These 32 normal forms have distinct images in  $\text{Gal}(L_2/\mathbb{Q}_2)$ : the third component determines  $a_3$ , the second  $a_1$ , and  $\sigma_y^{a_2}\sigma_z^{a_3+2a_4}$  determines  $a_2$  and  $a_4$ . So  $D \cong \text{Gal}(L_2/\mathbb{Q}_2)$ . The inertia group consists of the elements trivial on  $U_4$ , namely  $\langle x, y, w \rangle$ .

(c) The Frattini subgroup of  $D$  is  $\langle z^2, w \rangle$ , which is central of exponent two, and  $D$  has exponent four. So Lazard's formula gives  $D_3(D) = [[D, D], D][D, D]^2D^4 = 1$ , and the Hilbert polynomial follows from Jennings' theorem [13].

(d) The extension  $M_{16}/k$  is abelian of type  $(2, 2, 2)$ , generated by  $\sqrt{5}, \sqrt{\alpha}, \sqrt{2}$ . We apply the conductor–discriminant formula for abelian extensions [19, Chapter V, Theorem 3.27] to the extension  $\mathbb{Q}(i)(\sqrt{5}, \sqrt{1+2i}, \sqrt{2})$  of  $\mathbb{Q}(i)$ , whose completion at the unique prime above  $1+i$  is  $M_{16}/k$ . Taking  $(1+i)$ -adic valuations gives

$$v_k(\mathfrak{d}_{M_{16}/k}) = \sum_b f(b) = 0 + 0 + 2 + 2 + 4 + 4 + 4 + 4 = 20,$$

the sum over the classes  $b = 1, 5, \alpha, 5\alpha, 2, 10, 2\alpha, 10\alpha$ . Hence

$$v_2(\mathfrak{d}_{M_{16}/\mathbb{Q}_2}) = [M_{16} : k] v_2(\mathfrak{d}_{k/\mathbb{Q}_2}) + 20 = 8 \cdot 2 + 20 = 36.$$

Since  $M_{16}$  has residue degree two over  $\mathbb{Q}_2$ , its different has valuation  $36/2 = 18$ . The extension  $L_2/M_{16}$  is unramified, so the different of  $L_2/\mathbb{Q}_2$  also has valuation 18, and  $18/8 = 9/4$ .  $\square$

The field  $E_2$  is the maximal elementary abelian subextension of  $L_2$ , so  $\text{Gal}(L_2/E_2)$  is the Frattini subgroup  $\langle z^2, [y, z] \rangle$ , which is central of exponent two. Hence any triple in  $\text{Gal}(L_2/\mathbb{Q}_2)$  that acts on  $E_2$  as  $x, y, z$  do differs from  $x, y, z$  by central elements of order at most two; it satisfies the relations of  $D$  and generates, so it also defines an isomorphism  $D \cong \text{Gal}(L_2/\mathbb{Q}_2)$ . The description of the inertia group, however, refers to the triple of Lemma 2.5(b).

**2.3. The Global Group and Its Complete Presentation.** Let  $B_S$  be the maximal pro-2 extension of  $B$ , in a fixed algebraic closure, that is unramified at every finite prime outside  $S$ ; no condition is imposed at  $v_1, v_2$ . Let  $\mathcal{G}_B = \text{Gal}(B_S/B)$ , and let  $E = B(\sqrt{\alpha_0}, \dots, \sqrt{\alpha_7}) = B(\sqrt{V})$  be the Kummer field. Every prime above 2 lies in  $S$ , so a quadratic extension  $B(\sqrt{b})$  lies in  $B_S$  exactly when  $b$  has even valuation outside  $S$ . By Lemma 2.2(d),  $E$  is the maximal elementary abelian extension of  $B$  in  $B_S$ , and  $[E : B] = 256$ . For  $g \in \mathcal{G}_B$  define its elementary image  $\bar{g} \in \mathbb{F}_2^8$  by  $g(\sqrt{\alpha_i}) = (-1)^{\bar{g}_i} \sqrt{\alpha_i}$ . This identifies  $\mathcal{G}_B/D_2\mathcal{G}_B = \text{Gal}(E/B)$  with  $\mathbb{F}_2^8$ . We write vectors as strings  $v_0 v_1 \dots v_7$ . For  $a \in V$  let  $\chi_a$  be the Kummer character,  $g(\sqrt{a}) = (-1)^{\chi_a(g)} \sqrt{a}$ ; then  $a \mapsto \chi_a$  identifies  $V$  with the dual of  $\mathbb{F}_2^8$ , and  $\chi_{\alpha_i}(g) = \bar{g}_i$ .

For a place  $\nu$  of  $B$ , a choice of a place of  $B_S$  above  $\nu$  gives a homomorphism  $\mathcal{G}_\nu \rightarrow \mathcal{G}_B$ , the local map at  $\nu$ ; another choice changes it by an inner automorphism of  $\mathcal{G}_B$ . For a finite place  $\nu \notin S$ , the local map kills inertia, and its image is generated by a Frobenius element  $\text{Frob}_\nu$ . Let  $\Sigma = S \cup \{v_1, v_2\}$ . For  $\nu \in \Sigma$  we fix generators of  $\mathcal{G}_\nu$  as follows, and use the same names for their images in  $\mathcal{G}_B$ :  $\iota_j$  at  $v_j$  (Lemma 2.4(a)), so that  $\iota_1, \iota_2$  are complex conjugations;  $\tau_\nu, \varphi_\nu$  at  $\nu \in \{\mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2\}$ , with the generator  $\alpha_i$  of  $\nu$  as uniformizer  $\varpi$  in Lemma 2.4(b); and  $x_j, y_j, z_j$  at  $\mathfrak{p}_j$ , the images of one fixed triple  $x, y, z \in \mathcal{G}_{\mathbb{Q}_2} = \mathcal{G}_{\mathfrak{p}_j}$  lifting the elements  $x, y, z$  of Lemma 2.5(b). This triple satisfies the hypothesis of Lemma 2.4(c). We also write  $\text{Frob}_{t_k}$  for  $k = 0, 1, 2$ .

TABLE 4. Elementary images in  $\mathbb{F}_2^8$  of the local generators. Entry  $i$  is 1 exactly when the element negates  $\sqrt{\alpha_i}$ . The third and sixth columns give the action on  $\sqrt{b}$  for  $b$  in the completion.

| element                 | image    | action                     | element                    | image    | action                            |
|-------------------------|----------|----------------------------|----------------------------|----------|-----------------------------------|
| $\iota_1$               | 10111010 | sign at $v_1$              | $\iota_2$                  | 11000101 | sign at $v_2$                     |
| $x_1$                   | 00100000 | $(5, b)_{\mathfrak{p}_1}$  | $x_2$                      | 00010000 | $(5, b)_{\mathfrak{p}_2}$         |
| $y_1$                   | 11110010 | $(-1, b)_{\mathfrak{p}_1}$ | $y_2$                      | 10000001 | $(-1, b)_{\mathfrak{p}_2}$        |
| $z_1$                   | 10000100 | $(-2, b)_{\mathfrak{p}_1}$ | $z_2$                      | 11111000 | $(-2, b)_{\mathfrak{p}_2}$        |
| $\tau_{\mathfrak{q}_1}$ | 00001000 | $(-1, b)_{\mathfrak{q}_1}$ | $\varphi_{\mathfrak{q}_1}$ | 10100111 | $(-\alpha_4, b)_{\mathfrak{q}_1}$ |
| $\tau_{\mathfrak{q}_2}$ | 00000100 | $(-1, b)_{\mathfrak{q}_2}$ | $\varphi_{\mathfrak{q}_2}$ | 11101011 | $(-\alpha_5, b)_{\mathfrak{q}_2}$ |
| $\tau_{\mathfrak{r}_1}$ | 00000010 | $(2, b)_{\mathfrak{r}_1}$  | $\varphi_{\mathfrak{r}_1}$ | 01100101 | $(-\alpha_6, b)_{\mathfrak{r}_1}$ |
| $\tau_{\mathfrak{r}_2}$ | 00000001 | $(2, b)_{\mathfrak{r}_2}$  | $\varphi_{\mathfrak{r}_2}$ | 01011010 | $(-\alpha_7, b)_{\mathfrak{r}_2}$ |
| $\text{Frob}_{t_1}$     | 00100111 | $(29, b)_{t_1}$            | $\text{Frob}_{t_2}$        | 00011011 | $(29, b)_{t_2}$                   |
| $\text{Frob}_{t_0}$     | 01110000 | $(7, b)_{t_0}$             |                            |          |                                   |

Table 4 lists the elementary images. They follow from Lemma 2.4 and Table 3. For example, at the places above 3 and 5 the vector of  $\tau_\nu$  marks the odd valuations, and that of  $\varphi_\nu$  marks the units that are not squares modulo  $\nu$ , with a zero at the generator of  $\nu$ . The columns headed “action” express the same actions as Hilbert symbols, that is, as local Artin symbols [19, Chapter III, Remark 4.5]. The supplementary program `kummer241.gp` checks the norms of the  $\alpha_i$ , the generators of the primes in Lemma 2.2(c) and the residues of  $\sqrt{241}$  in Table 3, and recomputes Table 4 from these Hilbert symbols.

Fix a minimal presentation  $1 \rightarrow R \rightarrow \mathcal{F} \rightarrow \mathcal{G}_B \rightarrow 1$ , where  $\mathcal{F}$  is a free pro-2 group of rank 8. Minimality means  $R \subset D_2\mathcal{F}$ , so  $\mathcal{F}/D_2\mathcal{F} = \mathbb{F}_2^8$  with the coordinates above. Put  $\mathfrak{L} = \text{gr } \mathcal{F}$ , so that  $\dim \mathfrak{L}_1 = 8$ ,  $\dim \mathfrak{L}_2 = 36$  and  $\dim \mathfrak{L}_3 = 168$ . For  $v, w \in \mathfrak{L}_1$  we have  $(v + w)^{[2]} = v^{[2]} + w^{[2]} + [v, w]$ , and the square of an element of  $\mathcal{F}$  with image  $v$  has class  $v^{[2]}$ . For each local generator  $g$  above choose a lift  $\hat{g} \in \mathcal{F}$ , and define the local relators

$$(2.6) \quad \rho_{v_j} = \hat{\iota}_j^2, \quad \rho_\nu = \hat{\varphi}_\nu \hat{\tau}_\nu \hat{\varphi}_\nu^{-1} \hat{\tau}_\nu^{-p} \quad (\nu \in \{\mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2\}), \quad \rho_{\mathfrak{p}_j} = r(\hat{x}_j, \hat{y}_j, \hat{z}_j),$$

where  $p \in \{3, 5\}$  is the rational prime below  $\nu$ , and  $r$  is a defining relator of  $\mathcal{G}_{\mathbb{Q}_2}$  with respect to the fixed triple  $x, y, z$ , the same for  $j = 1, 2$ . They lie in  $R$ , because each local relation holds in  $\mathcal{G}_\nu$ . Their classes in  $\mathfrak{L}_2$  are obtained by substituting elementary images into the quadratic initials of Lemma 2.4:

$$(2.7) \quad \bar{v}_j^{[2]}, \quad [\bar{\tau}_\nu, \bar{\varphi}_\nu] + \frac{p-1}{2} \bar{\tau}_\nu^{[2]}, \quad \bar{y}_j^{[2]} + [\bar{x}_j, \bar{y}_j] + [\bar{x}_j, \bar{z}_j].$$

The results below hold for every minimal presentation and every choice of lifts.

The classes (2.7) have an arithmetic meaning. The elements  $e_i^{[2]}$  and  $[e_i, e_j]$  ( $i < j$ ) of the standard basis form a basis of  $\mathfrak{L}_2$ , so there is a linear isomorphism  $\beta$  from  $\mathfrak{L}_2$  onto the space of symmetric bilinear forms on  $V$  with

$$\beta(v^{[2]})(a, b) = \chi_a(v)\chi_b(v), \quad \beta([v, w])(a, b) = \chi_a(v)\chi_b(w) + \chi_a(w)\chi_b(v) \quad (v, w \in \mathfrak{L}_1).$$

It is defined by these formulas for basis vectors  $v, w$ , which send the basis of  $\mathfrak{L}_2$  to a basis of the symmetric bilinear forms. The formulas then hold for all  $v, w$ , because the bracket is bilinear,  $(v + w)^{[2]} = v^{[2]} + w^{[2]} + [v, w]$ , and squaring is the identity on  $\mathbb{F}_2$ . Equivalently, if  $\xi \in \mathfrak{L}_2$  is written in the free algebra  $\mathbb{F}_2\langle X_0, \dots, X_7 \rangle$  as  $\sum_{i,j} n_{ij} X_i X_j$ , then  $\beta(\xi)(\alpha_i, \alpha_j) = n_{ij}$ . For  $\nu \in \Sigma$ , the local Hilbert form  $H_\nu$  on  $V$  is the symmetric bilinear form with  $H_\nu(a, b) = 1$  if  $(a, b)_\nu = -1$  and  $H_\nu(a, b) = 0$  otherwise.

**Lemma 2.8.** *For every  $\nu \in \Sigma$ ,  $\beta$  maps the class of  $\rho_\nu$  to  $H_\nu$ . The eight classes (2.7) sum to zero, and the seven classes with  $\nu \neq \mathfrak{p}_1$  are linearly independent.*

*Proof.* For a local generator  $g$  at  $\nu$ ,  $\chi_a(\bar{g})$  is the value at  $g$  of the local Kummer character of the image of  $a$  in  $B_\nu$ . Hence  $\beta$  of the class of  $\rho_\nu$  is the pullback to  $V$  of a symmetric bilinear form on  $B_\nu^\times/B_\nu^{\times 2}$ , as is  $H_\nu$ , and it suffices to compare the two on a basis of  $B_\nu^\times/B_\nu^{\times 2}$ . At a real place the only class is  $-1$ , and both forms take the value 1 on  $(-1, -1)$ . At  $\nu$  above  $p \in \{3, 5\}$ , take the basis  $\varpi, u$  with  $\varpi$  the generator of  $\nu$  and  $u$  a nonsquare unit. By Lemma 2.4(b),  $\chi_\varpi(\tau) = \chi_u(\varphi) = 1$  and  $\chi_\varpi(\varphi) = \chi_u(\tau) = 0$ , so the form of  $[\tau, \varphi] + \frac{p-1}{2} \tau^{[2]}$  takes the values  $\frac{p-1}{2}, 1, 0$  on  $(\varpi, \varpi), (\varpi, u), (u, u)$ ; by (2.3) these are the indicators of  $(\varpi, \varpi)_p = (-1)^{(p-1)/2}$ ,  $(\varpi, u)_p = -1$  and  $(u, u)_p = 1$ . At  $\mathfrak{p}_j$  the characters  $\chi_2, \chi_{-5}, \chi_5$  are dual to  $x, y, z$ , as in the proof of Lemma 2.4(c). So  $y^{[2]}$  contributes the entry at  $(-5, -5)$ ,  $[x, y]$  the entries at  $(2, -5)$  and  $(-5, 2)$ , and  $[x, z]$  those at  $(2, 5)$  and  $(5, 2)$ : the form of  $y^{[2]} + [x, y] + [x, z]$  on the basis  $2, -5, 5$  is the matrix of the Hilbert symbol displayed in that proof, from which this initial was obtained.

For  $a, b \in V$  and a place  $\nu' \notin \Sigma$ , the place  $\nu'$  is finite and not above 2, and  $a, b$  are units at  $\nu'$ , so  $(a, b)_{\nu'} = 1$  [19, Chapter VIII, 5.8]. By the product formula for the Hilbert symbol [19, Chapter V, Example 5.4],  $\sum_{\nu \in \Sigma} H_\nu = 0$ , so the eight classes sum to zero.

Finally, by Table 3 and (2.3), the seven pairs

$$(-1, \alpha_2), (-1, \varepsilon), (-1, \alpha_7), (-1, \alpha_4), (-1, \alpha_5), (\alpha_2, \alpha_6), (\varepsilon, \alpha_7)$$

have Hilbert symbol  $-1$  exactly at  $\{\mathfrak{p}_1, v_1\}$ ,  $\{\mathfrak{p}_1, v_2\}$ ,  $\{\mathfrak{p}_2, v_2\}$ ,  $\{\mathfrak{q}_1, v_1\}$ ,  $\{\mathfrak{q}_2, v_2\}$ ,  $\{\mathfrak{r}_1, v_1\}$  and  $\{\mathfrak{r}_2, v_2\}$ ; the supplementary program `cup241.gp` recomputes these symbols. Evaluating a relation  $\sum_{\nu \neq \mathfrak{p}_1} n_\nu H_\nu = 0$ , with  $n_\nu \in \mathbb{F}_2$ , at the first two pairs gives  $n_{v_1} = n_{v_2} = 0$ , and then at the other five pairs gives  $n_{\mathfrak{p}_2} = n_{\mathfrak{q}_1} = n_{\mathfrak{q}_2} = n_{\mathfrak{r}_1} = n_{\mathfrak{r}_2} = 0$ .  $\square$

**Lemma 2.9.** *The group  $\mathcal{G}_B$  has generator rank 8 and relation rank 7. The seven relators  $\rho_\nu$  with  $\nu \in \Sigma \setminus \{\mathfrak{p}_1\}$  form a minimal system of defining relations: their normal closure in  $\mathcal{F}$  is  $R$ . In particular the dyadic local relator  $\rho_{\mathfrak{p}_1}$  lies in the normal closure of the other seven.*

*Proof.* The generator rank is  $\dim H^1(\mathcal{G}_B, \mathbb{F}_2) = 8$  [26, Proposition 3.9.1].

Next we bound  $H^2$ . Let  $X = \text{Spec } \mathcal{O}_B[1/30]$ . Since 2 is invertible on  $X$ , the sheaf  $\mu_2$  is the constant sheaf  $\mathbb{F}_2$ . The finite subextensions of  $B_S/B$  correspond to the connected finite étale Galois

2-covers  $Y \rightarrow X$ , and the Hochschild–Serre spectral sequence of this tower [17, Theorem 14.9 and Remark 14.10] gives an exact sequence

$$0 \rightarrow H^1(\mathcal{G}_B, \mathbb{F}_2) \rightarrow H^1(X, \mathbb{F}_2) \rightarrow \left( \varinjlim_Y H^1(Y, \mathbb{F}_2) \right)^{\mathcal{G}_B} \rightarrow H^2(\mathcal{G}_B, \mathbb{F}_2) \rightarrow H^2(X, \mathbb{F}_2).$$

A class in  $H^1(Y, \mathbb{F}_2)$  is an étale double cover of  $Y$ . Its Galois closure over  $X$  is a finite étale Galois cover whose group embeds in the wreath product of  $\text{Gal}(Y/X)$  with  $C_2$ , hence a 2-cover in the tower, over which the class vanishes. So the direct limit is zero, and  $H^2(\mathcal{G}_B, \mathbb{F}_2)$  injects into  $H^2(X, \mu_2)$ . The Kummer sequence and  $\text{Pic}(X) = 0$  identify  $H^2(X, \mu_2)$  with  $\text{Br}(X)[2]$ , where  $\text{Br}(X) = H^2(X, \mathbb{G}_m)$ . By [29, Theorem 6.8.3 and Example 6.9.4],  $\text{Br}(X)$  is the kernel of the sum of the local invariants  $\bigoplus_{\nu \in \Sigma} \text{Br}(B_\nu) \rightarrow \mathbb{Q}/\mathbb{Z}$ . Each of the eight groups  $\text{Br}(B_\nu)[2]$  has order two, and the sum of the invariants maps their direct sum onto  $\frac{1}{2}\mathbb{Z}/\mathbb{Z}$ , so  $\dim \text{Br}(X)[2] = 7$ . We use only the resulting bound  $\dim H^2(\mathcal{G}_B, \mathbb{F}_2) \leq \dim \text{Br}(X)[2] \leq 7$ .

By Lemma 2.8, the classes in  $\mathfrak{L}_2$  of the seven relators  $\rho_\nu$ ,  $\nu \neq \mathfrak{p}_1$ , are linearly independent. Since  $R \subset D_2\mathcal{F}$ , we have  $R^2[R, \mathcal{F}] \subset D_3\mathcal{F}$ , so the classes of these seven relators in  $R/R^2[R, \mathcal{F}]$  are independent. On the other hand  $\dim R/R^2[R, \mathcal{F}] = \dim H^2(\mathcal{G}_B, \mathbb{F}_2) \leq 7$  by the five-term exact sequence of the presentation [26, Proposition 3.9.5], [30, (3.3)–(3.4)]. Hence the seven classes form a basis, the relation rank is 7, and by the pro-2 Nakayama lemma [26, Corollary 3.9.3] the seven relators generate  $R$  as a normal subgroup. Finally  $\rho_{\mathfrak{p}_1} \in R$ .  $\square$

By Lemma 2.8, the only linear relation among the eight classes (2.7) is that their sum vanishes, and this relation is Hilbert reciprocity. Hence any seven of the eight classes are linearly independent, and the proof of Lemma 2.9 shows that any seven of the eight local relators generate  $R$  as a normal subgroup. We omit the dyadic relator at  $\mathfrak{p}_1$ .

*Remark 2.10.* By Lemma 2.9, the bound  $\dim H^2(\mathcal{G}_B, \mathbb{F}_2) \leq \dim \text{Br}(X)[2] = 7$  is attained, so  $H^2(\mathcal{G}_B, \mathbb{F}_2) \rightarrow \text{Br}(X)[2]$  is an isomorphism. This can also be seen directly: the cup product  $\chi_a \cup \chi_b$  maps to the class of the quaternion algebra  $(a, b)$ , whose invariant at  $\nu$  is given by  $(a, b)_\nu$  [19, Chapter III, Remark 4.7], and the seven pairs in the proof of Lemma 2.8 give seven classes with independent invariant vectors. The supplementary program `cup241.gp` computes the invariant vectors at the eight places of  $\Sigma$  of all 36 algebras  $(\alpha_i, \alpha_j)$ .

**2.4. The Prescribed Local Groups.** We now cut  $\mathcal{G}_B$  down to prescribed local groups. At the dyadic places we use the restriction  $\mathcal{G}_{\mathfrak{p}_j} = \mathcal{G}_{\mathbb{Q}_2} \rightarrow \text{Gal}(L_2/\mathbb{Q}_2) = D$  of Lemma 2.5. At a place  $\nu$  above  $p \in \{3, 5\}$ , the maximal elementary abelian quotient of  $\mathcal{G}_\nu$  is  $\text{Gal}(\mathbb{Q}_p(\sqrt{u}, \sqrt{p})/\mathbb{Q}_p) \cong C_2 \times C_2$  for a nonsquare unit  $u$ ; this quotient has ramification index and residue degree two. At  $\mathfrak{t}_1, \mathfrak{t}_2$ , whose completions are  $\mathbb{Q}_{29}$ , and at  $\mathfrak{t}_0$ , whose residue field is  $\mathbb{F}_{49}$ , we bound the order of the Frobenius by four.

**Definition 2.11.** Let  $N$  be the normal closure in  $\mathcal{G}_B$  of the images of  $\ker(\mathcal{G}_{\mathfrak{p}_j} \rightarrow D)$  for  $j = 1, 2$ , of the Frattini subgroups  $D_2\mathcal{G}_\nu$  for  $\nu \in \{\mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2\}$ , and of  $\text{Frob}_{\mathfrak{t}_k}^4$  for  $k = 0, 1, 2$ . Put  $G_B = \mathcal{G}_B/N$  and  $\overline{G}_B = G_B/D_4G_B$ , and let  $B_G \subset B_S$  be the fixed field of  $N$ , so that  $\text{Gal}(B_G/B) = G_B$ .

This does not depend on the choice of places of  $B_S$  above each  $\nu$ . The image of  $\mathcal{G}_{\mathfrak{p}_j}$  in  $G_B$  is a quotient of  $D$ , the image of  $\mathcal{G}_\nu$  for  $\nu$  above 3 or 5 is a quotient of  $C_2 \times C_2$  in which inertia maps to the image of  $\tau_\nu$ , and  $\text{Frob}_{\mathfrak{t}_k}$  has order dividing four in  $G_B$ . Subsection 2.5 shows that these are equalities. Since  $D$  and  $\mathcal{G}_{\mathbb{Q}_2}$  both have generator rank three,  $\ker(\mathcal{G}_{\mathfrak{p}_j} \rightarrow D)$  lies in the Frattini subgroup of  $\mathcal{G}_{\mathfrak{p}_j}$ . Hence  $N \subset D_2\mathcal{G}_B$ ,  $G_B/D_2G_B = \text{Gal}(E/B)$ , and  $E \subset B_G$  is the fixed field of  $D_2G_B$ .

**Lemma 2.12.** *The kernel  $N_B$  of  $\mathcal{F} \rightarrow \mathcal{G}_B \rightarrow G_B$  is the normal closure of the following thirty elements:*

- (i) *the seven relators  $\rho_\nu$ ,  $\nu \in \Sigma \setminus \{\mathfrak{p}_1\}$ ;*

- (ii) for  $j = 1, 2$ :  $\hat{x}_j^2, [\hat{x}_j, \hat{y}_j], [\hat{x}_j, \hat{z}_j], [[\hat{y}_j, \hat{z}_j], \hat{z}_j], \hat{z}_j^4, [\hat{y}_j, \hat{z}_j]^2$ ;
- (iii) for  $\nu \in \{\mathbf{q}_1, \mathbf{q}_2, \mathbf{r}_1, \mathbf{r}_2\}$ :  $\hat{\tau}_\nu^2$  and  $\hat{\varphi}_\nu^2$ ;
- (iv) for  $k = 0, 1, 2$ :  $\hat{F}_k^4$ , where  $\hat{F}_k$  is the chosen lift of  $\text{Frob}_{t_k}$ .

Moreover  $N_B \subset D_2\mathcal{F}$ , and  $\rho_{p_1} \in N_B$ .

*Proof.* By Lemma 2.9, it suffices to show that the images in  $\mathcal{G}_B$  of the elements (ii)–(iv) generate  $N$  as a normal subgroup. Let  $\mathcal{F}_3$  be free on  $x, y, z$ , and let  $N_D$  be the normal closure of the seven relators of  $D$  in Lemma 2.5, so that  $\mathcal{F}_3/N_D = D$ . The relator  $r$  maps to 1 in  $D$ , so  $r \in N_D$ , and  $\ker(\mathcal{G}_{\mathbb{Q}_2} \rightarrow D)$  is the image of  $N_D$ . Since  $N_D \subset D_2\mathcal{F}_3$ , we have  $N_D^2[N_D, \mathcal{F}_3] \subset D_3\mathcal{F}_3$ . The seven relators of  $D$  span  $N_D/N_D^2[N_D, \mathcal{F}_3]$  [26, Corollary 3.9.3]; write the class of  $r$  as a combination of them. The classes in  $\text{gr}_2 \mathcal{F}_3$  of  $x^2, y^2, [x, y], [x, z]$  are independent, the relators  $z^4, [y, z]^2$  and  $[[y, z], z]$  lie in  $D_3\mathcal{F}_3$ , and  $r$  has class  $y^{[2]} + [x, y] + [x, z]$ ; so  $y^2$  has coefficient one. Replacing  $y^2$  by  $r$  therefore still gives a spanning set, and by [26, Corollary 3.9.3] the elements  $r, x^2, [x, y], [x, z], [[y, z], z], z^4, [y, z]^2$  generate  $N_D$  as a normal subgroup. Since  $r = 1$  in  $\mathcal{G}_{\mathbb{Q}_2}$ , the other six generate  $\ker(\mathcal{G}_{\mathbb{Q}_2} \rightarrow D)$ . For  $\nu$  above  $p \in \{3, 5\}$ , the quotient of  $\mathcal{G}_\nu$  by the normal closure of  $\tau^2, \varphi^2$  is generated by two involutions which commute, because  $\varphi\tau\varphi^{-1} = \tau^p = \tau$  there; so that normal closure is  $D_2\mathcal{G}_\nu$ . Finally  $N \subset D_2\mathcal{G}_B$  and  $R \subset D_2\mathcal{F}$  give  $N_B \subset D_2\mathcal{F}$ , and  $\rho_{p_1} \in R \subset N_B$ .  $\square$

**2.5. The Retained Quotient.** For  $n \geq 1$  let  $R_n \subset \mathfrak{L}_n$  be the image of  $N_B \cap D_n\mathcal{F}$ . Since  $N_B \subset D_2\mathcal{F}$  and  $D_n\mathcal{G}_B$  is the image of  $D_n\mathcal{F}$ ,

$$\text{gr}_1 G_B = \mathbb{F}_2^8, \quad \text{gr}_n G_B \cong \mathfrak{L}_n/R_n.$$

Consider the following 22 elements of  $\mathfrak{L}_2$ , written with the vectors of Table 4:

$$(2.13) \quad \begin{aligned} & \bar{\iota}_1^{[2]}, \bar{\iota}_2^{[2]}, \quad [\bar{\tau}_\nu, \bar{\varphi}_\nu] + \bar{\tau}_\nu^{[2]} \quad (\nu = \mathbf{q}_1, \mathbf{q}_2), \quad [\bar{\tau}_\nu, \bar{\varphi}_\nu] \quad (\nu = \mathbf{r}_1, \mathbf{r}_2); \\ & \bar{y}_j^{[2]} + [\bar{x}_j, \bar{y}_j] + [\bar{x}_j, \bar{z}_j], \quad \bar{x}_j^{[2]}, \quad [\bar{x}_j, \bar{y}_j], \quad [\bar{x}_j, \bar{z}_j] \quad (j = 1, 2); \\ & \bar{\tau}_\nu^{[2]}, \bar{\varphi}_\nu^{[2]} \quad (\nu = \mathbf{q}_1, \mathbf{q}_2, \mathbf{r}_1, \mathbf{r}_2). \end{aligned}$$

They are the classes of the eight local relators (2.7) and of the fourteen quadratic elements  $\hat{x}_j^2, [\hat{x}_j, \hat{y}_j], [\hat{x}_j, \hat{z}_j], \hat{\tau}_\nu^2, \hat{\varphi}_\nu^2$  of Lemma 2.12.

**Lemma 2.14.** *The space  $R_2$  is spanned by the 22 elements (2.13), and  $\dim R_2 = 21$ ; their only linear relation is that the eight classes of local relators sum to zero. Hence  $\dim \text{gr}_2 G_B = 15$ . Consequently, if  $g \in G_B$  has nonzero elementary image  $v$ , then  $g$  has order at least four in  $G_B/D_3G_B$  exactly when  $v^{[2]} \notin R_2$ ; otherwise  $g$  has order two there.*

*Proof.* Conjugates of an element of  $D_2\mathcal{F}$  have the same class modulo  $D_3\mathcal{F}$ , classes of products add, and  $\mathfrak{L}_2$  is finite. Hence, by Lemma 2.12,  $R_2$  is spanned by the classes of the thirty generators of  $N_B$ . Those in (i) have the classes (2.7), and the quadratic elements of (ii) and (iii) have the classes  $\bar{x}_j^{[2]}, [\bar{x}_j, \bar{y}_j], [\bar{x}_j, \bar{z}_j], \bar{\tau}_\nu^{[2]}, \bar{\varphi}_\nu^{[2]}$ . The remaining generators,  $[[\hat{y}_j, \hat{z}_j], \hat{z}_j], \hat{z}_j^4, [\hat{y}_j, \hat{z}_j]^2$  and the fourth powers in (iv), lie in  $D_3\mathcal{F}$ . The class of  $\rho_{p_1}$  also lies in  $R_2$ , since  $\rho_{p_1} \in N_B$ . The supplementary program `lie241.py` computes the rank of the 22 elements in the 36-dimensional space  $\mathfrak{L}_2$ : it is 21. The sum of the eight local classes vanishes by Lemma 2.8, that is, by Hilbert reciprocity, so this is the only relation; none of the fourteen quadratic elements of (ii) and (iii) is involved in it. For the last statement,  $g^2 \in D_2G_B$  has class  $v^{[2]}$  in  $\text{gr}_2 G_B = \mathfrak{L}_2/R_2$ , so  $g^2 \in D_3G_B$  exactly when  $v^{[2]} \in R_2$ .  $\square$

**Lemma 2.15.** (a)  $\dim \text{gr}_3 G_B = 26$ , so  $|\overline{G}_B| = 2^{8+15+26} = 2^{49}$ .

(b) For  $j = 1, 2$ , the local map induces an injection  $D \rightarrow G_B/D_3G_B$ , and  $\text{gr}_n D \rightarrow \text{gr}_n G_B$  is injective for  $n = 1, 2$ . For  $\nu$  above 3 or 5, the image of  $\mathcal{G}_\nu$  in  $G_B$  is  $\langle \tau_\nu \rangle \times \langle \varphi_\nu \rangle \cong C_2 \times C_2$ , and it injects into  $G_B/D_2G_B$ . For  $k = 0, 1, 2$ ,  $\text{Frob}_{t_k}$  has order exactly four in  $G_B$  and in  $G_B/D_3G_B$ . For  $j = 1, 2$ ,  $\iota_j$  has order two in  $G_B/D_2G_B$ .

- (c) The conjugacy class of  $\iota_1$  in  $\overline{G}_B$  has  $2^{15}$  elements.
- (d) The elementary images  $\bar{\iota}_1 \neq \bar{\iota}_2$  lie outside the span of the elementary images of the local generators at each of  $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2, \mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$ .

*Proof.* (a) Work in  $\mathcal{F}/D_4\mathcal{F}$ , in which  $D_2\mathcal{F}/D_4\mathcal{F}$  is elementary abelian and  $D_3\mathcal{F}/D_4\mathcal{F}$  is central. Let  $\rho_1, \dots, \rho_{21}$  be the generators of  $N_B$  of degree two, namely (i), the first three elements of (ii) for each  $j$ , and (iii). Their classes are the elements of (2.13) other than the class of  $\rho_{\mathfrak{p}_1}$ , and these are independent by Lemma 2.14. The cubic generators are  $\omega_j = [[\hat{y}_j, \hat{z}_j], \hat{z}_j]$ , and the remaining generators lie in  $D_4\mathcal{F}$ . For  $\rho \in D_2\mathcal{F}$  and  $g \in \mathcal{F}$  we have  $g^{-1}\rho g = \rho[\rho, g]$ , and modulo  $D_4\mathcal{F}$  the commutator  $[\rho, g]$  is central and depends only, and linearly, on the classes of  $\rho$  in  $\mathfrak{L}_2$  and of  $g$  in  $\mathfrak{L}_1$ . Hence the image of  $N_B$  in  $\mathcal{F}/D_4\mathcal{F}$  is the subgroup generated by the  $\rho_i$ , the  $[\rho_i, f]$  for  $f$  in a basis of  $\mathcal{F}$ , and the  $\omega_j$ . An element  $\prod_i \rho_i^{a_i}$  times an element of  $D_3\mathcal{F}$  lies in  $D_3\mathcal{F}$  only if all  $a_i = 0$ . Therefore  $R_3$  is spanned by the brackets  $[\bar{\rho}_i, \xi]$ , for  $\xi$  in a basis of  $\mathfrak{L}_1$ , and by  $[[\bar{y}_j, \bar{z}_j], \bar{z}_j]$ . The supplementary program `lie241c.py` computes these elements in the free associative algebra, where  $[a, \xi] = a\xi + \xi a$ : they span a space of dimension 142 in the 168-dimensional space  $\mathfrak{L}_3$ . So  $\dim \text{gr}_3 G_B = 26$ .

(b) The local map  $\mathcal{G}_{\mathfrak{p}_j} \rightarrow G_B$  factors through  $D$  by Definition 2.11, and the induced map  $D \rightarrow G_B$  preserves the Zassenhaus filtrations. It is injective on  $\text{gr}_1$  because  $\bar{x}_j, \bar{y}_j, \bar{z}_j$  are independent, and on  $\text{gr}_2$  because  $\bar{z}_j^{[2]}$  and  $[\bar{y}_j, \bar{z}_j]$ , the images of  $\bar{z}^2$  and  $[\bar{y}, \bar{z}]$ , are independent modulo  $R_2$ ; the program `lie241.py` checks both facts. If  $1 \neq g \in D$ , let  $n \leq 2$  be maximal with  $g \in D_n(D)$ ; then the image of  $g$  is nonzero in  $\text{gr}_n G_B$ , so it is not in  $D_3 G_B$ . For  $\nu$  above 3 or 5, the vectors  $\bar{\tau}_\nu$  and  $\bar{\varphi}_\nu$  are independent, and the image of  $\mathcal{G}_\nu$  is a quotient of  $C_2 \times C_2$ . For the Frobenius elements, `lie241.py` checks that  $\overline{\text{Frob}}_{\mathfrak{t}_k}^{[2]} \notin R_2$ , so Lemma 2.14 gives order at least four in  $G_B/D_3 G_B$ , while Definition 2.11 gives order at most four in  $G_B$ . Finally  $\bar{\iota}_j \neq 0$ .

(c) Write  $\overline{G} = \overline{G}_B$  and  $\iota = \iota_1$ . The map  $\mathfrak{L}_1 \rightarrow \text{gr}_2 \overline{G}$ ,  $v \mapsto [\bar{\iota}, v]$ , has rank 7 by `lie241.py`, so its kernel is the line through  $\bar{\iota}$ . The map  $\text{gr}_2 \overline{G} \rightarrow \text{gr}_3 \overline{G}$ ,  $u \mapsto [u, \bar{\iota}]$ , has rank 8 by `lie241c.py`. Since  $D_2 \overline{G}$  is abelian and  $D_3 \overline{G}$  is central in  $\overline{G}$ , the map  $h \mapsto [h, \bar{\iota}]$  is a homomorphism  $D_2 \overline{G} \rightarrow D_3 \overline{G}$ ; it factors through  $\text{gr}_2 \overline{G}$  and equals the second map. Its kernel  $C_{\overline{G}}(\iota) \cap D_2 \overline{G}$  therefore has order  $2^{15+26-8} = 2^{33}$ . If  $g \in C_{\overline{G}}(\iota)$ , then  $[\bar{\iota}, \bar{g}] = 0$  in  $\text{gr}_2 \overline{G}$ , so  $\bar{g} \in \{0, \bar{\iota}\}$ ; conversely  $\iota \in C_{\overline{G}}(\iota)$ . Hence  $|C_{\overline{G}}(\iota)| = 2^{34}$ , and the class of  $\iota$  has  $2^{49-34} = 2^{15}$  elements.

(d) This is a finite check on Table 4, carried out by `lie241.py`.  $\square$

**2.6. Filtered Fox Images.** We now describe the filtered Fox calculus. Let  $\Delta$  be a finite 2-group,  $\Lambda = \mathbb{F}_2[\Delta]$ , and  $\mathfrak{I}$  its augmentation ideal. Filtrations are decreasing, with  $F^0 M = M$  and  $F^n M = 0$  for large  $n$ . For a finite-dimensional filtered space put  $h_M(t) = \sum_n \dim(F^n M / F^{n+1} M) t^n$ . We filter  $\Lambda$  by  $F^n \Lambda = \mathfrak{I}^n$ , and  $\Lambda^m(-1)$  denotes  $\Lambda^m$  with  $F^n = (\mathfrak{I}^{n-1})^m$  for  $n \geq 1$ , so that  $h_{\Lambda^m(-1)} = m t h_\Lambda$ . Subspaces carry the induced filtration unless stated otherwise. For  $0 < t < 1$ ,

$$(2.16) \quad h_M(t) = (1-t) \sum_{n \geq 1} \dim(M / F^n M) t^{n-1}.$$

Hence, for  $0 < t < 1$ : (h1) if  $f: M \rightarrow M'$  is surjective with  $f(F^n M) \subset F^n M'$ , then  $h_{M'}(t) \leq h_M(t)$ ; (h2) if  $U_1 \subset U_2$  are subspaces of the same filtered space, then  $h_{U_1}(t) \leq h_{U_2}(t)$ ; (h3) if  $f$  is surjective and strict,  $f(F^n M) = F^n M'$ , then  $h_M = h_{\ker f} + h_{M'}$ ; and (h4) for subspaces  $U_1, U_2$  of a filtered space,

$$(2.17) \quad h_{U_1+U_2}(t) \leq h_{U_1}(t) + h_{U_2}(t) - h_{U_1 \cap U_2}(t).$$

For (h4), apply (h3) to  $U_1 \oplus U_2 \rightarrow U_1 + U_2$ ,  $(u_1, u_2) \mapsto u_1 + u_2$ , whose kernel is  $U_1 \cap U_2$ , with the image filtration  $F^n U_1 + F^n U_2$  on  $U_1 + U_2$ ; this filtration is contained in the induced one, and (h1) applies to the identity map.

Let  $\mathcal{F}$  be a free pro-2 group with basis  $f_1, \dots, f_n$  and  $\pi: \mathcal{F} \rightarrow \Delta$  a surjection. Since  $\Lambda^n \rtimes \Delta$ , with  $\Delta$  acting by left multiplication, is a finite 2-group, there is a unique homomorphism  $\mathcal{F} \rightarrow \Lambda^n \rtimes \Delta$

sending  $f_i$  to  $(e_i, \pi(f_i))$ , where  $e_1, \dots, e_n$  is the standard basis of  $\Lambda^n$ . We write it as  $w \mapsto (\partial w, \pi(w))$  and call  $\partial w$  the row of  $w$ . For  $w$  in the discrete free group on the  $f_i$ , the row  $\partial w \in \Lambda^n$  is the image of the vector of free derivatives of  $w$  [9, Section 2]. We use the following properties.

- (F1)  $\partial(ww') = \partial w + \pi(w)\partial w'$ .
- (F2)  $\sum_i \partial_i w (\pi(f_i) - 1) = \pi(w) - 1$ .
- (F3) If  $\pi(\rho) = 1$ , then  $\partial(u\rho u^{-1}) = \pi(u)\partial\rho$ ; and  $\partial$  restricts to a continuous homomorphism on  $\ker \pi$ .
- (F4) If  $\mathcal{F}'$  is free on  $f'_1, \dots, f'_m, u_1, \dots, u_m \in \mathcal{F}$ , and  $\partial'$  denotes the rows for the map  $\mathcal{F}' \rightarrow \Delta$ ,  $f'_l \mapsto \pi(u_l)$ , then  $\partial(W(u_1, \dots, u_m)) = \sum_l \partial'_l W \cdot \partial u_l$  for  $W \in \mathcal{F}'$ .

(F1) is the multiplication of the semidirect product. For (F2), the pairs  $(a, g)$  with  $\sum a_i(\pi(f_i) - 1) = g - 1$  form a subgroup containing the images of the  $f_i$ . (F3) follows from (F1). For (F4), both sides are the first components of homomorphisms  $\mathcal{F}' \rightarrow \Lambda^n \rtimes \Delta$  that agree on the basis.

**Lemma 2.18.** *Let  $\mathcal{M} = \{a \in \Lambda^n : \sum_i a_i(\pi(f_i) - 1) = 0\}$ , with the filtration induced from  $\Lambda^n(-1)$ . Then  $h_{\mathcal{M}} = 1 - (1 - nt)h_{\Lambda}$  and  $\mathcal{M} = \partial(\ker \pi)$ . If  $\rho$  lies in the normal closure of a set  $Y \subset \ker \pi$ , then  $\partial\rho$  lies in the  $\Lambda$ -span of  $\partial Y$ . In particular,  $\mathcal{M}$  is the  $\Lambda$ -span of the rows of any set of normal generators of  $\ker \pi$ .*

*Proof.* Since the  $\pi(f_i)$  generate  $\Delta$ , we have  $\mathfrak{I} = \sum_i \Lambda(\pi(f_i) - 1) \subset \sum_i \mathbb{F}_2(\pi(f_i) - 1) + \mathfrak{I}^2$ , hence  $\mathfrak{I}^m = \sum_i \mathfrak{I}^{m-1}(\pi(f_i) - 1)$  for all  $m \geq 1$ , by induction and because  $\mathfrak{I}$  is nilpotent. So  $\Lambda^n(-1) \rightarrow \mathfrak{I}$ ,  $a \mapsto \sum a_i(\pi(f_i) - 1)$ , is strict and surjective with kernel  $\mathcal{M}$ , and (h3) gives  $h_{\mathcal{M}} = nt h_{\Lambda} - (h_{\Lambda} - 1)$ .

By (F2),  $\partial(\ker \pi) \subset \mathcal{M}$ . Conversely, let  $\mathcal{F}^\circ \subset \mathcal{F}$  be the dense subgroup generated by the  $f_i$ , which is a free group on them. Then  $\pi(\mathcal{F}^\circ) = \Delta$ , and the kernel of  $\mathbb{F}_2[\mathcal{F}^\circ] \rightarrow \Lambda$  is  $\mathbb{F}_2[\mathcal{F}^\circ](\mathcal{F}^\circ \cap \ker \pi - 1)$ . Lift  $a \in \mathcal{M}$  to  $\tilde{a} \in \mathbb{F}_2[\mathcal{F}^\circ]^n$ . Then  $\sum_i \tilde{a}_i(f_i - 1) = \sum_l \zeta_l(n_l - 1)$  with  $n_l \in \mathcal{F}^\circ \cap \ker \pi$  and  $\zeta_l \in \mathbb{F}_2[\mathcal{F}^\circ]$ . By Fox's fundamental formula  $n_l - 1 = \sum_i (\partial n_l / \partial f_i)(f_i - 1)$ , where the  $\partial / \partial f_i$  are the free derivatives [9, Section 2], and the coefficients of the  $f_i - 1$  are unique, because  $\partial / \partial f_j$  maps  $\sum_i \zeta'_i(f_i - 1)$  to  $\zeta'_j$ . So  $\tilde{a}_i = \sum_l \zeta_l \partial n_l / \partial f_i$ . On  $\mathcal{F}^\circ$  the rows are the images of the vectors of free derivatives, so  $a = \sum_l \zeta_l \partial n_l$  lies in  $\Lambda \partial(\ker \pi)$ , which equals  $\partial(\ker \pi)$  by (F3). For the remaining statements, let  $M_Y$  be the  $\Lambda$ -span of  $\partial Y$ . By (F3), the elements  $n \in \ker \pi$  with  $\partial n \in M_Y$  form a closed normal subgroup of  $\mathcal{F}$  containing  $Y$ .  $\square$

The next lemma computes the cost of imposing a full local group. The groups  $\Gamma$  below are  $C_2$ ,  $C_2 \times C_2$ ,  $C_4$  and  $D$ , with standard generators  $\iota$ ;  $\tau, \varphi$ ; a generator; and  $x, y, z$ . Let  $k$  be the number of standard generators, so  $k = 1, 2, 1, 3$ . The augmentation Hilbert polynomials are  $P_\Gamma(t) = 1 + t$ ,  $(1 + t)^2$ ,  $(1 + t)(1 + t^2)$  and  $P_D(t)$ , and  $D_3(\Gamma) = 1$  in each case. Put

$$\psi_\Gamma(t) = kt - 1 + P_\Gamma(t)^{-1}.$$

**Lemma 2.19.** *Let  $\Gamma \leq \Delta$  be a subgroup isomorphic to one of these groups, with  $g_1, \dots, g_k \in \Gamma$  corresponding to its standard generators, and suppose that  $\text{gr}_n \Gamma \rightarrow \text{gr}_n \Delta$  is injective for  $n = 1, 2$ . Let  $\Lambda_\Gamma = \mathbb{F}_2[\Gamma] \subset \Lambda$ , with augmentation ideal  $\mathfrak{I}_\Gamma$ , and let  $\mathcal{M}_\Gamma \subset \Lambda_\Gamma^k(-1)$  be the kernel of  $b \mapsto \sum_l b_l(g_l - 1)$ .*

- (a) *There are elements  $\eta_j \in \Lambda$  with weights  $w_j$  such that  $\Lambda = \bigoplus_j \eta_j \Lambda_\Gamma$  and  $\mathfrak{I}^m = \bigoplus_j \eta_j \mathfrak{I}_\Gamma^{m-w_j}$  for every  $m$ , where  $\mathfrak{I}_\Gamma^{m'} = \Lambda_\Gamma$  for  $m' \leq 0$ ; moreover  $\sum_j t^{w_j} = h_\Lambda / P_\Gamma$ .*
- (b) *The  $\Lambda$ -module  $\Lambda \mathcal{M}_\Gamma \subset \Lambda^k(-1)$  has  $h_{\Lambda \mathcal{M}_\Gamma} = \psi_\Gamma h_\Lambda$ .*
- (c) *Let  $\mathcal{F}$  and  $\pi$  be as above, let  $u_1, \dots, u_k \in \mathcal{F}$  with  $\pi(u_l) = g_l$ , and let  $\mathcal{F}_k$  be free on  $k$  letters, mapping to  $\Gamma$  by the standard generators. Let  $U \subset \Lambda^n(-1)$  be the  $\Lambda$ -span of the rows  $\partial W(u_1, \dots, u_k)$  for all  $W$  in the kernel of  $\mathcal{F}_k \rightarrow \Gamma$ . Then  $h_U(t) \leq \psi_\Gamma(t) h_\Lambda(t)$  for  $0 < t < 1$ .*

*Proof.* (a) Since  $D_3(\Gamma) = 1$  and  $\text{gr } \Gamma \rightarrow \text{gr } \Delta$  is injective,  $\Gamma \cap D_m \Delta = D_m(\Gamma)$  for all  $m$ . Choose elements of  $\Gamma$  whose classes form a homogeneous basis of  $\text{gr } \Gamma$ , and extend them by elements of  $\Delta$  to a family whose classes form a homogeneous basis of  $\text{gr } \Delta$ . By the restricted Poincaré–Birkhoff–Witt form of Jennings' theorem [13, 31], the ordered products of the elements  $g - 1$  of this family, each

used at most once and with the elements of  $\Gamma$  last, form a basis of  $\Lambda$ , and those of weight at least  $m$  form a basis of  $\mathfrak{I}^m$ . The same holds for  $\Lambda_\Gamma$  and the elements of  $\Gamma$  alone. Let the  $\eta_j$  be the ordered products of the other elements. Comparing Hilbert series gives the last assertion.

(b) As in the proof of Lemma 2.18, the map  $\Lambda_\Gamma^k(-1) \rightarrow \mathfrak{I}_\Gamma$  is strict and surjective, so  $h_{\mathcal{M}_\Gamma} = ktP_\Gamma - (P_\Gamma - 1)$ . By (a),  $\Lambda^k(-1) = \bigoplus_j \eta_j \Lambda_\Gamma^k(-1)$  with  $F^m = \bigoplus_j \eta_j F^{m-w_j}$ , and  $\Lambda \mathcal{M}_\Gamma = \bigoplus_j \eta_j \mathcal{M}_\Gamma$ . Hence  $h_{\Lambda \mathcal{M}_\Gamma} = (h_\Lambda / P_\Gamma) h_{\mathcal{M}_\Gamma} = \psi_\Gamma h_\Lambda$ .

(c) For  $W$  in the kernel, the local row  $\partial' W \in \Lambda_\Gamma^k$  lies in  $\mathcal{M}_\Gamma$  by (F2), and by (F4)  $\partial W(u) = \sum_l \partial'_l W \partial u_l$ . Thus  $U$  is contained in the image of  $\Lambda \mathcal{M}_\Gamma$  under the map  $T_u: \Lambda^k(-1) \rightarrow \Lambda^n(-1)$ ,  $a \mapsto \sum_l a_l \partial u_l$ , which preserves filtrations because  $\partial u_l \in \Lambda^n = F^0$ . By (h2), (h1) and (b),  $h_U \leq h_{T_u(\Lambda \mathcal{M}_\Gamma)} \leq h_{\Lambda \mathcal{M}_\Gamma} = \psi_\Gamma h_\Lambda$ .  $\square$

For  $\Gamma = D$  we write  $d_D = \psi_D = 3t - 1 + P_D(t)^{-1}$ . In the proof of Proposition 2.23, the module of the dyadic place  $\mathfrak{p}_1$  and the modules of the other places share the row of one relator, and the next lemma computes the module generated by this row.

**Lemma 2.20.** *In the situation of Lemma 2.19 with  $\Gamma = D$ , suppose that  $u_1, u_2, u_3$  are the basis elements  $f_1, f_2, f_3$  of  $\mathcal{F}$ . Let  $r \in \mathcal{F}_3$  map to 1 in  $D$ , with class  $y^{[2]} + [x, y] + [x, z]$  in  $\text{gr}_2 \mathcal{F}_3$ , and let  $\rho = r(f_1, f_2, f_3)$ . Then*

$$h_{\Lambda \partial \rho} = s_D h_\Lambda, \quad s_D(t) = t^2 \left( 1 - \frac{t^7}{P_D(t)} \right).$$

*Proof.* By (F4),  $\partial \rho = (\partial' r, 0, \dots, 0)$ , where  $\partial' r \in \Lambda_D^3$  is the local row, so  $\Lambda \partial \rho$  is the module  $\Lambda \partial' r \subset \Lambda^3(-1)$  placed in the first three coordinates. By Lemma 2.19(a),  $\Lambda^3 = \bigoplus_j \eta_j \Lambda_D^3$  with  $F^n(\Lambda^3(-1)) = \bigoplus_j \eta_j F^{n-w_j}(\Lambda_D^3(-1))$ , and  $b \mapsto \eta_j b$  is injective on  $\Lambda_D^3$ . The submodule  $\Lambda \partial' r = \bigoplus_j \eta_j \Lambda_D \partial' r$  decomposes in the same way, so  $F^n(\Lambda \partial' r) = \bigoplus_j \eta_j F^{n-w_j}(\Lambda_D \partial' r)$  for the induced filtrations, and  $h_{\Lambda \partial \rho} = (h_\Lambda / P_D) h_{\Lambda_D \partial' r}$ . It therefore suffices to show that  $h_{\Lambda_D \partial' r} = t^2(P_D - t^7)$ , for the filtration induced from  $\Lambda_D^3(-1)$ . Write  $\mathfrak{I}_D$  for the augmentation ideal of  $\Lambda_D = \mathbb{F}_2[D]$ , and  $X_1, X_2, X_3, X_4$  for  $x-1, y-1, z-1, w-1$ , where  $w = [y, z]$ .

*The linear part of  $\partial' r$ .* Here  $\pi$  also denotes the map  $\mathcal{F}_3 \rightarrow D$ . For  $g, h \in \mathcal{F}_3$ , (F1) gives  $\partial'(g^2) = (1 + \pi(g))\partial' g$  and  $\partial'[g, h] = \pi(g)^{-1}(\pi(h)^{-1} - 1)\partial' g + \pi(g)^{-1}\pi(h)^{-1}(\pi(g) - 1)\partial' h$ . Moreover  $\partial' g$  is congruent modulo  $\mathfrak{I}_D$  to the image  $\bar{g} \in \mathbb{F}_2^3$  of  $g$  in  $\mathcal{F}_3/D_2\mathcal{F}_3$ , and  $\pi(g) - 1 \equiv \sum_l \bar{g}_l X_l$  modulo  $\mathfrak{I}_D^2$  by (F2). Hence, for  $q \in D_2\mathcal{F}_3$ , the row  $\partial' q$  has entries in  $\mathfrak{I}_D$ , and modulo  $\mathfrak{I}_D^2$  it is additive in  $q$ , because  $\pi(q) - 1 \in \mathfrak{I}_D^2$ . It vanishes modulo  $\mathfrak{I}_D^2$  on  $D_3\mathcal{F}_3 = (D_2\mathcal{F}_3)^2[D_2\mathcal{F}_3, \mathcal{F}_3]$  (Lazard's formula): for  $q \in D_2\mathcal{F}_3$  the formulas above give  $\partial'(q^2), \partial'[q, h] \in \mathfrak{I}_D^2$ , since  $\bar{q} = 0$  and  $\pi(q) - 1 \in \mathfrak{I}_D^2$ . So the row modulo  $\mathfrak{I}_D^2$  depends only on the class of  $q$  in  $\text{gr}_2 \mathcal{F}_3$ , the classes  $v^{[2]}$  and  $[v, v']$  giving the rows with  $l$ -th entries  $v_l \sum_m v_m X_m$  and  $v_l \sum_m v'_m X_m + v'_l \sum_m v_m X_m$ . For the class  $y^{[2]} + [x, y] + [x, z]$  this gives

$$\partial' r \equiv (X_2 + X_3, X_2 + X_1, X_1) \pmod{\mathfrak{I}_D^2}.$$

*Three facts about  $\Lambda_D$ .* Let  $A = \bigoplus_m \mathfrak{I}_D^m / \mathfrak{I}_D^{m+1}$ , and write  $X, Y, Z$  and  $W$  for the classes of  $X_1, X_2, X_3$  in degree one and of  $X_4$  in degree two. By Jennings' theorem [13] and Lemma 2.5(c), the products  $X_1^{a_1} X_2^{a_2} X_3^{a_3} X_4^{a_4}$  with  $a_1, a_2, a_3 \in \{0, 1\}$  and  $0 \leq a_4 \leq 3$ , of weight  $a_1 + a_2 + a_3 + 2a_4 \geq m$ , form a basis of  $\mathfrak{I}_D^m$ ; here  $X_3^2 = z^2 - 1$ . Hence the monomials  $X^{a_1} Y^{a_2} Z^{a_3} W^{a_4}$  form a basis of  $A$ , and, since the coefficients of  $P_D$  are 1, 3, 5, 7, 7, 5, 3, 1,

$$\dim \mathfrak{I}_D^m = 32, 31, 28, 23, 16, 9, 4, 1, 0 \quad (m = 0, \dots, 8).$$

Because  $x$  and  $w$  are central and  $x^2 = y^2 = w^2 = z^4 = 1$ , the elements  $X$  and  $W$  are central in  $A$  and  $X^2 = Y^2 = W^2 = Z^4 = 0$ . Moreover  $ZY = YZ + W$ , because  $X_3 X_2 + X_2 X_3 = zy + yz = zy(w - 1) \equiv X_4 \pmod{\mathfrak{I}_D^3}$ ; by induction  $Z^j Y = Y Z^j + j Z^{j-1} W$  for  $j \geq 1$ .

First,  $\mathfrak{I}_D^7 = \mathbb{F}_2 \Omega_D$ , where  $\Omega_D = \sum_{g \in D} g$ :  $\mathfrak{I}_D^7$  is one-dimensional and  $(g - 1)\mathfrak{I}_D^7 \subset \mathfrak{I}_D^8 = 0$ , so its nonzero element is invariant under left multiplication by  $D$ , and hence equal to  $\Omega_D$ .

Second, for  $m < 7$  the map  $A_m \rightarrow A_{m+1}^3$ ,  $\xi \mapsto (\xi X, \xi Y, \xi Z)$ , is injective. Let  $\xi = \sum_{\mu} b_{\mu} \mu$  over the basis monomials  $\mu = X^{a_1} Y^{a_2} Z^{a_3} W^{a_4}$  of degree  $m$ . Right multiplication by  $X$  maps the monomials with  $a_1 = 0$  to distinct basis monomials and kills the others, and right multiplication by  $Z$  maps those with  $a_3 \leq 2$  to distinct basis monomials and kills those with  $a_3 = 3$ . So  $\xi X = \xi Z = 0$  forces  $\xi = \sum_{a_2, a_4} b_{a_2 a_4} X Y^{a_2} Z^3 W^{a_4}$ . Then

$$\xi Y = b_{00}(XY Z^3 + X Z^2 W) + b_{01}XY Z^3 W + b_{10}XY Z^2 W,$$

a combination of distinct basis monomials, so  $\xi Y = 0$  forces  $\xi = b_{11}XY Z^3 W$ , which has degree 7. Hence  $\xi = 0$  when  $m < 7$ .

Third, the same holds for  $\xi \mapsto (\xi(Y + Z), \xi(Y + X), \xi X)$ , because this triple is obtained from  $(X, Y, Z)$  by the invertible matrix

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

This is the coefficient matrix of  $y^{[2]} + [x, y] + [x, z]$ , that is, the matrix of the 2-adic Hilbert symbol on  $2, -5, 5$  in the proof of Lemma 2.4(c). So the injectivity used here, and with it the saving  $s_D$ , comes from the nondegeneracy of the local Hilbert pairing, that is, from local duality. The supplementary program `dyadic241.py` confirms these three facts by direct computation in  $\Lambda_D$ .

*The filtration.* If  $\xi \in \mathfrak{I}_D^m \setminus \mathfrak{I}_D^{m+1}$  with  $m < 7$ , then by the third fact  $\xi \partial' r$  lies in  $F^{m+2}$  but not in  $F^{m+3}$ , whereas  $\Omega_D \partial' r = 0$  because  $\Omega_D \mathfrak{I}_D = 0$ . Now let  $\xi \in \Lambda_D$  with  $\xi \partial' r \in F^{m+2}$ , and suppose that  $\xi \notin \mathfrak{I}_D^m + \mathbb{F}_2 \Omega_D$ . Let  $m' < m$  be maximal with  $\xi \in \mathfrak{I}_D^{m'} + \mathbb{F}_2 \Omega_D$ , and write  $\xi = \xi' + a \Omega_D$  with  $\xi' \in \mathfrak{I}_D^{m'}$  and  $a \in \mathbb{F}_2$ . Then  $\xi' \notin \mathfrak{I}_D^{m'+1}$ , and  $m' < 7$  because  $\mathfrak{I}_D^7 = \mathbb{F}_2 \Omega_D$ ; so  $\xi \partial' r = \xi' \partial' r$  is not in  $F^{m'+3} \supset F^{m+2}$ , a contradiction. Therefore  $F^{m+2}(\Lambda_D \partial' r) = \mathfrak{I}_D^m \partial' r$  for every  $m \geq 0$ , the map  $\xi \mapsto \xi \partial' r$  has kernel  $\mathbb{F}_2 \Omega_D$ , and  $h_{\Lambda_D \partial' r} = \sum_{m < 7} \dim(\mathfrak{I}_D^m / \mathfrak{I}_D^{m+1}) t^{m+2} = t^2(P_D - t^7)$ .  $\square$

**2.7. Infinitude.** Put

(2.21)

$$\psi_1(t) = \frac{t^2}{1+t}, \quad \psi_2(t) = 2t - 1 + \frac{1}{(1+t)^2}, \quad \psi_4(t) = \frac{t^4}{(1+t)(1+t^2)}, \quad d_D(t) = 3t - 1 + \frac{1}{P_D(t)},$$

so that  $\psi_1, \psi_2, \psi_4, d_D$  are the functions  $\psi_{\Gamma}$  for  $\Gamma = C_2, C_2 \times C_2, C_4, D$ . With  $s_D$  as in Lemma 2.20, let

(2.22)

$$P_B(t) = 1 - 8t + 2\psi_1(t) + 4\psi_2(t) + 2d_D(t) - s_D(t) + 3\psi_4(t).$$

Exact rational arithmetic, carried out by the supplementary program `gs241.py`, gives

$$P_B\left(\frac{34}{117}\right) = -\frac{187433948535241}{88772225460489675} < 0.$$

**Proposition 2.23.** *The group  $G_B$  is infinite.*

*Proof.* Suppose that  $G_B$  is finite, and put  $\Delta = G_B$  and  $\Lambda = \mathbb{F}_2[\Delta]$ . By Table 4,  $\bar{x}_1, \bar{y}_1, \bar{z}_1$  are independent; choose  $g_4, \dots, g_8 \in \mathcal{G}_B$  such that  $\bar{x}_1, \bar{y}_1, \bar{z}_1, \bar{g}_4, \dots, \bar{g}_8$  is a basis of  $\mathbb{F}_2^8$ . Let  $\mathcal{F}$  be free on  $f_1, \dots, f_8$ , and send  $f_1, f_2, f_3$  to  $x_1, y_1, z_1$  and  $f_l$  to  $g_l$  for  $4 \leq l \leq 8$ . This is a minimal presentation of  $\mathcal{G}_B$ ; we take the lifts  $\hat{x}_1 = f_1$ ,  $\hat{y}_1 = f_2$ ,  $\hat{z}_1 = f_3$ , and arbitrary lifts of the other local generators. Let  $\pi: \mathcal{F} \rightarrow \Delta$  be the composite and  $\mathcal{M}$  as in Lemma 2.18, so that  $h_{\mathcal{M}} = 1 - (1 - 8t)h_{\Lambda}$ .

By Lemma 2.15(b), which concerns only  $\text{gr}_1$  and  $\text{gr}_2$ , the image in  $\Delta$  of each local group is the prescribed group  $\Gamma$  and satisfies the hypothesis of Lemma 2.19. We use eleven places: the real places  $v_1, v_2$ , with  $\Gamma = C_2$ ; the four places above 3 and 5, with  $\Gamma = C_2 \times C_2$ ; the dyadic places  $\mathfrak{p}_1, \mathfrak{p}_2$ , with  $\Gamma = D$ ; and  $t_0, t_1, t_2$ , with  $\Gamma = C_4$  generated by the Frobenius. For each of these places  $\nu$ , let  $U_{\nu}$  be the module  $U$  of Lemma 2.19(c) for this  $\Gamma$ , formed with the lifts of the local generators at  $\nu$ . Every generator of  $N_B$  in Lemma 2.12 is of the form  $W(u_1, \dots, u_k)$  for one of these places and some  $W$  in the kernel of  $\mathcal{F}_k \rightarrow \Gamma$ , and conversely every such element lies in  $N_B$ . By Lemma 2.18,  $\mathcal{M} = \sum_{\nu} U_{\nu}$ .

Let  $U'$  be the sum of the  $U_\nu$  over the ten places  $\nu \neq \mathfrak{p}_1$ . The row of  $\rho_{\mathfrak{p}_1}$  lies in  $U_{\mathfrak{p}_1}$ . Since  $\rho_{\mathfrak{p}_1}$  lies in the normal closure of the relators (i) of Lemma 2.12, its row also lies in the  $\Lambda$ -span of their rows (Lemma 2.18), which is contained in  $U'$ . Hence  $\Lambda \partial \rho_{\mathfrak{p}_1} \subset U_{\mathfrak{p}_1} \cap U'$ . By (2.17), (h2), Lemma 2.19(c) and Lemma 2.20, for  $0 < t < 1$ ,

$$1 - (1 - 8t)h_\Lambda(t) = h_{\mathcal{M}}(t) \leq h_{U_{\mathfrak{p}_1}}(t) + \sum_{\nu \neq \mathfrak{p}_1} h_{U_\nu}(t) - h_{\Lambda \partial \rho_{\mathfrak{p}_1}}(t) \leq (2\psi_1 + 4\psi_2 + 2d_D + 3\psi_4 - s_D)(t) h_\Lambda(t),$$

that is,  $1 \leq h_\Lambda(t)P_B(t)$ . Since  $h_\Lambda(t) > 0$ , this contradicts  $P_B(34/117) < 0$ .  $\square$

**2.8. The Fields.** Recall that  $\lambda = 2^{9/4}\sqrt{3615}$ , where  $3615 = 3 \cdot 5 \cdot 241$ , and note that  $65535/131072 = 1/2 - 2^{-17}$ .

**Theorem 2.24.** *For every power of two  $m \geq 2^{49}$  there is a finite Galois extension  $K/B$  of degree  $m$ , contained in  $B_G$ , such that  $G_B \rightarrow \overline{G}_B$  factors through  $\text{Gal}(K/B)$ . In particular  $K$  contains the fixed field of  $D_4G_B$ , hence the fixed field of  $D_3G_B$  and the Kummer field  $E$ . Every such  $K$  is totally imaginary. Let  $\iota_1$  be complex conjugation at a place of  $K$  above  $v_1$ , let  $F = K^{\langle \iota_1 \rangle}$  and  $d = [F : \mathbb{Q}] = m$ , and let  $(b, c)$  be the signature of  $F$  and  $\theta = c/d$ . Then  $F$  has a real place,*

$$\text{rd}(K) = \text{rd}(F) = \lambda, \quad 65535/131072 \leq \theta < \frac{1}{2}.$$

*The extension  $K/F$  is quadratic, unramified at every finite place, and split at every place of  $F$  above 2, 3, 5, 7 and 29. At these primes all places of  $K$ , and of  $F$ , have the absolute ramification indices and residue degrees of Table 5.*

TABLE 5. Absolute ramification indices and residue degrees of the places of  $K$  and  $F$  above the selected primes, the corresponding local extensions of the completions of  $B$ , and their contributions to  $\text{rd}(K)$ . The last row is the ramified prime of  $B$ , at which  $K/B$  is unramified.

| $p$ | $(e_p, f_p)$ | local extension over $B$                                                  | factor of $\text{rd}(K)$ |
|-----|--------------|---------------------------------------------------------------------------|--------------------------|
| 2   | (8, 4)       | $L_2/\mathbb{Q}_2$ , group $D$                                            | $2^{9/4}$                |
| 3   | (2, 2)       | $\mathbb{Q}_3(\sqrt{-1}, \sqrt{3})/\mathbb{Q}_3$ , group $C_2 \times C_2$ | $3^{1/2}$                |
| 5   | (2, 2)       | $\mathbb{Q}_5(\sqrt{2}, \sqrt{5})/\mathbb{Q}_5$ , group $C_2 \times C_2$  | $5^{1/2}$                |
| 7   | (1, 8)       | unramified of degree 4 over $B_{\mathfrak{t}_0}$                          | 1                        |
| 29  | (1, 4)       | unramified of degree 4 over $\mathbb{Q}_{29}$                             | 1                        |
| 241 | $e_p = 2$    | unramified                                                                | $241^{1/2}$              |

*Proof. Existence.* By Proposition 2.23,  $G_B$  has open normal subgroups of arbitrarily large index, and  $D_4G_B$  is open because  $G_B$  is finitely generated. Given  $m$ , choose an open normal subgroup  $U' \subset D_4G_B$  of  $G_B$  of index at least  $m$ , and put  $\Delta' = G_B/U'$  and  $N' = D_4G_B/U'$ . A nontrivial normal subgroup of a finite 2-group meets its center, so  $N'$  contains normal subgroups of  $\Delta'$  of every order dividing  $|N'|$ . Take one of order  $|\Delta'|/m$ , which divides  $|N'| = |\Delta'|/2^{49}$ . Its preimage  $U$  is open and normal in  $G_B$ , contained in  $D_4G_B$ , and of index  $m$ . Let  $K$  be its fixed field in  $B_G$ . Conversely, every  $K$  as in the statement is the fixed field of an open normal subgroup  $U \subset D_4G_B$ . We fix such a  $K$  and put  $H = \text{Gal}(K/B) = G_B/U$ .

*Complex places.* Since  $\sqrt{-1} = \sqrt{\alpha_0} \in E \subset K$ , the field  $K$  is totally imaginary, and  $[K : F] = 2$ . Let  $\sigma_1 : K \rightarrow \mathbb{C}$  induce the chosen place, so that  $\bar{\sigma}_1 = \sigma_1 \circ \iota_1$  and  $\sigma_1(F) \subset \mathbb{R}$ . The embeddings above  $v_1$  are  $\sigma_1 \circ h$  with  $h \in H$ , and two of them agree on  $F$  exactly when they differ by right multiplication by an element of  $\langle \iota_1 \rangle$ . The restriction of  $\sigma_1 \circ h$  to  $F$  is real exactly when  $\sigma_1 \circ \iota_1 h$  and  $\sigma_1 \circ h$  agree on  $F$ , that is, when  $h^{-1}\iota_1 h \in \langle \iota_1 \rangle$ , that is, when  $h \in C_H(\iota_1)$ . Similarly, if  $\iota'$  is

complex conjugation for an embedding  $\sigma_2$  above  $v_2$ , the restriction of  $\sigma_2 \circ h$  to  $F$  is real exactly when  $h^{-1}\iota'h = \iota_1$ . This never happens, because conjugation preserves elementary images and  $\iota' = \bar{\iota}_2 \neq \bar{\iota}_1$  (Lemma 2.15(d)). Hence  $b = |C_H(\iota_1)|/2 \geq 1$ , so  $F$  has a real place and  $\theta = (1 - b/d)/2 < 1/2$ . Moreover  $d = |H|$ , so  $b/d = |C_H(\iota_1)|/(2|H|)$  is half the reciprocal of the size of the conjugacy class of  $\iota_1$  in  $H$ . This class maps onto the class of  $\iota_1$  in  $\bar{G}_B$ , which has  $2^{15}$  elements by Lemma 2.15(c). Hence  $b/d \leq 2^{-16}$  and  $\theta \geq (1 - 2^{-16})/2 = 65535/131072$ .

*Local structure and splitting.* Let  $\mathfrak{P}$  be a prime of  $K$  above  $\mathfrak{p} \in S \cup \{\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2\}$ . Its decomposition group in  $H$  is conjugate to the image of the local map at  $\mathfrak{p}$ . Since  $U \subset D_3 G_B$ , Lemma 2.15(b) shows that this image is  $D$ ,  $C_2 \times C_2$  or  $C_4$ , the last one cyclic on the Frobenius, and that the prescribed local group injects into  $H$ . Therefore the completion  $K_{\mathfrak{P}}$  is  $L_2$  over  $B_{\mathfrak{p}_j} = \mathbb{Q}_2$ , the field  $\mathbb{Q}_p(\sqrt{u}, \sqrt{p})$  over  $B_{\mathfrak{p}} = \mathbb{Q}_p$  for  $p = 3, 5$ , and the unramified extension of degree four of  $B_{\mathfrak{t}_k}$ . Since  $B_{\mathfrak{t}_0}$  is unramified of degree two over  $\mathbb{Q}_7$  and  $B_{\mathfrak{t}_1} = B_{\mathfrak{t}_2} = \mathbb{Q}_{29}$ , the places of  $K$  have the types of Table 5. Every element of the decomposition group has elementary image in the span of the images of the local generators at  $\mathfrak{p}$ , whereas every conjugate of  $\iota_1$  has elementary image  $\bar{\iota}_1$ . By Lemma 2.15(d),  $\iota_1$  lies in no decomposition group of a prime above  $S \cup \{\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2\}$ , so the decomposition group of  $\mathfrak{P}$  in  $\text{Gal}(K/F) = \langle \iota_1 \rangle$  is trivial. Hence the places of  $F$  above 2, 3, 5, 7, 29 split in  $K$ , and they have the same local types.

*Ramification and root discriminant.* The extension  $K/B$ , and hence  $K/F$ , is unramified at every prime outside  $S$ , in particular at  $(\sqrt{241})$ . At primes above  $S$  the inertia group of  $K/F$  is trivial, because it lies in the decomposition group. So  $K/F$  is unramified at every finite place, its relative discriminant is trivial,  $|d_K| = |d_F|^2$ , and  $\text{rd}(K) = \text{rd}(F)$ . To compute  $\text{rd}(K)$  we use  $|d_K| = 241^{[K:B]} N \mathfrak{d}_{K/B}$ . Since  $K/B$  is Galois, for  $\mathfrak{p} \in S$  all primes  $\mathfrak{P}$  above  $\mathfrak{p}$  have the same ramification index  $e$ , residue degree and different exponent  $v_{\mathfrak{P}}(\mathfrak{d}_{K/B})$ , and  $v_{\mathfrak{p}}(\mathfrak{d}_{K/B}) = [K : B] v_{\mathfrak{P}}(\mathfrak{d}_{K/B})/e$ . For  $\mathfrak{p}$  above 3 or 5 we have  $e = 2$  and  $v_{\mathfrak{P}}(\mathfrak{d}_{K/B}) = 1$ , since over the unramified extension  $\mathbb{Q}_p(\sqrt{u})$  the different of  $\mathbb{Q}_p(\sqrt{u}, \sqrt{p})$  is generated by  $2\sqrt{p}$ . For  $\mathfrak{p}_1, \mathfrak{p}_2$ , Lemma 2.5(d) gives  $v_{\mathfrak{P}}(\mathfrak{d}_{K/B})/e = 9/4$ . Since  $[K : \mathbb{Q}] = 2[K : B]$ , the base contributes  $241^{1/2}$ , each of the two primes above 3 contributes  $3^{1/4}$ , each of the two above 5 contributes  $5^{1/4}$ , and each dyadic prime contributes  $2^{9/8}$ . Hence  $\text{rd}(K) = 2^{9/4} 3^{1/2} 5^{1/2} 241^{1/2} = \lambda$ .  $\square$

*Remark 2.25.* The Lean development [23] formalizes a version of this construction; Remark 4.28 states what it proves and the numerical hypothesis it assumes.

### 3. THE KUMMER FIELD AND ITS QUADRATIC $L$ -FUNCTIONS

The analytic estimate of Section 4 compares every field of Theorem 2.24 with one fixed subfield, the *Kummer field*

$$E = B(\sqrt{\alpha_0}, \dots, \sqrt{\alpha_7}) = B(\sqrt{V})$$

of Subsection 2.3, with  $\alpha_0, \dots, \alpha_7$  as in (2.1) and  $V$  as in Lemma 2.2(d). It is the maximal elementary abelian 2-extension of  $B$  unramified at the finite primes outside  $S$ , that is, the fixed field of the Frattini subgroup of  $G_B$  (Lemma 3.1); the Lean formalization calls it the genus field. This section shows that  $E$  lies in every field of Theorem 2.24. It then writes  $\zeta_E$  as the product of  $\zeta_B$  and 255 quadratic Hecke  $L$ -functions of  $B$ , and determines their conductors, gamma factors, root numbers and Euler factors.

For  $v, e \in \mathbb{F}_2^8$  put  $\langle v, e \rangle = \sum_{i=0}^7 v_i e_i \in \mathbb{F}_2$ ; as in Section 2, vectors are written as words  $v_0 v_1 \dots v_7$ . For  $e \in \mathbb{F}_2^8$  put

$$\alpha_e = \prod_{i=0}^7 \alpha_i^{e_i}, \quad B_e = B(\sqrt{\alpha_e}).$$

For a nonzero ideal  $\mathfrak{a}$  of a number field,  $N\mathfrak{a}$  denotes its absolute norm. The number of positive divisors of an integer  $n \geq 1$  is  $\tau(n)$ .

**3.1. The Characters of the Kummer Field.** By Subsection 2.3,  $[E : B] = 256$ , and the elementary image  $\bar{g} \in \mathbb{F}_2^8$ , defined by  $g(\sqrt{\alpha_i}) = (-1)^{\bar{g}_i} \sqrt{\alpha_i}$ , identifies  $\text{Gal}(E/B)$  with  $\mathbb{F}_2^8$ . For  $e \in \mathbb{F}_2^8$  let  $\chi_e$  be the character  $g \mapsto (-1)^{\langle \bar{g}, e \rangle}$  of  $\text{Gal}(E/B)$ . Since  $g(\sqrt{\alpha_e}) = \chi_e(g) \sqrt{\alpha_e}$ , the character  $\chi_e$  cuts out  $B_e$ , and the 256 characters  $\chi_e$  are all the characters of  $\text{Gal}(E/B)$ . For  $e \neq 0$  the extension  $B_e/B$  is quadratic, because the classes of  $\alpha_0, \dots, \alpha_7$  are independent in  $B^\times/B^{\times 2}$  (Lemma 2.2(d)).

For a pro-2 group  $H$ , the Frattini subgroup  $\Phi(H) = H^2[H, H]$  is  $D_2H$ .

**Lemma 3.1.** *The fixed field of  $D_2G_B = \Phi(G_B)$  is  $E$ . Consequently  $E$  is contained in every field  $K$  of Theorem 2.24.*

*Proof.* Let  $\mathcal{G}_B$  be the group of Subsection 2.3, so that  $\mathcal{G}_B/\Phi(\mathcal{G}_B) = \text{Gal}(E/B)$ , and let  $G_B = \mathcal{G}_B/N$  as in Definition 2.11. As observed after that definition,  $N \subseteq \Phi(\mathcal{G}_B)$ : the images of the Frattini subgroups of the local groups above 3 and 5 and the fourth powers of Frobenius elements lie in  $\Phi(\mathcal{G}_B)$ , and the kernel of the surjection  $\mathcal{G}_{\mathfrak{p}_j} \rightarrow D$  lies in  $\Phi(\mathcal{G}_{\mathfrak{p}_j})$  because both groups have generator rank three. Hence  $G_B/\Phi(G_B) = \mathcal{G}_B/\Phi(\mathcal{G}_B) = \text{Gal}(E/B)$ . A field  $K$  of Theorem 2.24 contains the fixed field of  $D_4G_B \subseteq D_2G_B$ , and hence contains  $E$ .  $\square$

**3.2. Local Data.** For a prime  $\mathfrak{p} \notin S$  of  $B$ , the Frobenius vector  $v_{\mathfrak{p}} \in \mathbb{F}_2^8$  is the elementary image of the Frobenius of  $\mathfrak{p}$  in  $\text{Gal}(E/B)$ . By Euler's criterion,  $(v_{\mathfrak{p}})_i = 1$  exactly when  $\alpha_i$  is not a square in  $\mathcal{O}_B/\mathfrak{p}$ . For a place  $\mathfrak{p}$  of  $B$  we write  $(\cdot, \cdot)_{\mathfrak{p}}$  for the quadratic Hilbert symbol of  $B_{\mathfrak{p}}$ .

**Proposition 3.2.** *Let  $e \in \mathbb{F}_2^8 \setminus \{0\}$ . For a prime  $\mathfrak{p}$  of  $B$  let  $a_{\mathfrak{p}}(e)$  be the exponent of  $\mathfrak{p}$  in the conductor of  $\chi_e$ , and if  $a_{\mathfrak{p}}(e) = 0$ , let  $\chi_e(\mathfrak{p}) \in \{\pm 1\}$  be the value of  $\chi_e$  at the Frobenius of  $\mathfrak{p}$ .*

- (1) *If  $\mathfrak{p} \notin S$ , then  $a_{\mathfrak{p}}(e) = 0$  and  $\chi_e(\mathfrak{p}) = (-1)^{\langle v_{\mathfrak{p}}, e \rangle}$ . Let  $p$  be the rational prime below  $\mathfrak{p}$ . If  $p$  splits in  $B$ , then  $\mathfrak{p} = (p, \sqrt{241} - r)$  with  $r^2 \equiv 241 \pmod{p}$ , and  $(v_{\mathfrak{p}})_i = 1$  exactly when the image of  $\alpha_i$  under  $\sqrt{241} \mapsto r$  is a quadratic nonresidue modulo  $p$ . If  $p$  is inert, then  $(v_{\mathfrak{p}})_i = 1$  exactly when  $N_{B/\mathbb{Q}}(\alpha_i)$  is a quadratic nonresidue modulo  $p$ . If  $p = 241$ , the same holds with  $\sqrt{241} \mapsto 0$ .*
- (2) *Let  $\mathfrak{p}$  lie above  $p \in \{3, 5\}$ , let  $u = -1$  if  $p = 3$  and  $u = 2$  if  $p = 5$ , and let  $\varpi = -\alpha_i$ , where  $\alpha_i$  generates  $\mathfrak{p}$ . If  $(u, \alpha_e)_{\mathfrak{p}} = -1$ , then  $a_{\mathfrak{p}}(e) = 1$ . Otherwise  $a_{\mathfrak{p}}(e) = 0$  and  $\chi_e(\mathfrak{p}) = (\varpi, \alpha_e)_{\mathfrak{p}}$ .*
- (3) *Let  $\mathfrak{p} = \mathfrak{p}_j$ . If  $(5, \alpha_e)_{\mathfrak{p}} = -1$ , then  $a_{\mathfrak{p}}(e) = 3$ . If  $(5, \alpha_e)_{\mathfrak{p}} = 1$  and  $(-1, \alpha_e)_{\mathfrak{p}} = -1$ , then  $a_{\mathfrak{p}}(e) = 2$ . Otherwise  $a_{\mathfrak{p}}(e) = 0$  and  $\chi_e(\mathfrak{p}) = (-2, \alpha_e)_{\mathfrak{p}}$ .*
- (4) *The real place  $v_k$  becomes complex in  $B_e$ , that is,  $\chi_e$  is nontrivial at  $v_k$ , exactly when  $\alpha_e < 0$  at  $v_k$ .*

*Proof.* For (1),  $\mathfrak{p}$  is odd and  $\alpha_e$  is a unit at  $\mathfrak{p}$ , so  $\mathfrak{p}$  is unramified in  $B_e$ . By Euler's criterion the Frobenius multiplies  $\sqrt{\alpha_e}$  by the quadratic residue symbol of  $\alpha_e$  modulo  $\mathfrak{p}$ , which is  $\prod_i (\alpha_i/\mathfrak{p})^{e_i}$ . For a split prime,  $\mathcal{O}_B/\mathfrak{p} \cong \mathbb{F}_p$  via  $\sqrt{241} \mapsto r$ . For an inert prime,  $\mathcal{O}_B/\mathfrak{p} \cong \mathbb{F}_{p^2}$  and the conjugation of  $B$  induces the Frobenius  $x \mapsto x^p$  of the residue field, so the residue of  $N_{B/\mathbb{Q}}(\alpha)$  is  $\bar{\alpha}^{p+1}$ . An element of  $\mathbb{F}_{p^2}^\times$  is a square exactly when  $\bar{\alpha}^{(p^2-1)/2} = (\bar{\alpha}^{p+1})^{(p-1)/2}$  equals 1, that is, when its norm is a square in  $\mathbb{F}_p$ . The prime above 241 has residue field  $\mathbb{F}_{241}$  and  $\sqrt{241} \mapsto 0$ .

For (2) and (3) we use local class field theory for  $B_{\mathfrak{p}} = \mathbb{Q}_p$ . The local Artin map sends  $b \in \mathbb{Q}_p^\times$  to the automorphism multiplying  $\sqrt{\alpha_e}$  by  $(b, \alpha_e)_{\mathfrak{p}}$ . Units map onto inertia, a uniformizer maps to a Frobenius, and the conductor exponent is the least  $n \geq 0$  such that  $b \mapsto (b, \alpha_e)_{\mathfrak{p}}$  is trivial on  $U^{(n)}$ , where  $U^{(0)} = \mathbb{Z}_p^\times$  and  $U^{(n)} = 1 + p^n \mathbb{Z}_p$  [19].

For  $p = 3, 5$ , the unit group modulo squares is  $\{1, u\}$ , and  $U^{(1)}$  consists of squares by Hensel's lemma. Hence  $a_{\mathfrak{p}}(e) \leq 1$ , with equality exactly when  $(u, \alpha_e)_{\mathfrak{p}} = -1$ . In the unramified case the Frobenius is the image of any uniformizer, such as  $\varpi$ .

For  $p = 2$ , the squares of  $\mathbb{Z}_2^\times$  form  $U^{(3)}$ . Moreover  $U^{(2)} = U^{(3)} \cup 5U^{(3)}$  and  $U^{(0)} = U^{(1)} = \{\pm 1, \pm 5\}U^{(3)}$ . The character is thus trivial on  $U^{(3)}$ . It is trivial on  $U^{(2)}$  exactly when  $(5, \alpha_e) = 1$ ,

and on  $U^{(0)}$  exactly when also  $(-1, \alpha_e) = 1$ . This gives the stated exponents; the exponent 1 cannot occur since  $U^{(0)} = U^{(1)}$ . In the unramified case the Frobenius is the image of the uniformizer  $-2$ .

For (4), the real place  $v_k$  splits in  $B_e$  exactly when  $\alpha_e > 0$  at  $v_k$ .  $\square$

Each condition in the proposition is linear in  $e$ . Indeed,  $(b, \alpha_e)_{\mathfrak{p}} = (-1)^{\langle w, e \rangle}$ , where  $w$  is the elementary image of the local generator that acts on square roots by  $(b, \cdot)_{\mathfrak{p}}$  in Table 4: at  $\mathfrak{p}_j$  the vectors  $\bar{x}_j, \bar{y}_j, \bar{z}_j$  belong to  $b = 5, -1, -2$ , and above 3 and 5 the images of the inertia and Frobenius generators belong to  $b = u$  and  $b = \varpi$ . Likewise  $\alpha_e < 0$  at  $v_k$  exactly when  $\langle \bar{\iota}_k, e \rangle = 1$ , where  $\bar{\iota}_k$  is the elementary image of the complex conjugation  $\iota_k$ . We put  $\nu_k(e) = \langle \bar{\iota}_k, e \rangle \in \{0, 1\}$ . The conductor of  $\chi_e$  is  $\mathfrak{f}_e = \prod_{\mathfrak{p} \in S} \mathfrak{p}^{a_{\mathfrak{p}}(e)}$ .

**Corollary 3.3.** *The ramification index and residue degree of a prime  $\mathfrak{p}$  of  $B$  in  $E/B$  are  $(4, 2)$  at  $\mathfrak{p}_1, \mathfrak{p}_2$ ;  $(2, 2)$  at  $\mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2$ ; and  $(1, 1)$  or  $(1, 2)$  at  $\mathfrak{p} \notin S$ , according as  $v_{\mathfrak{p}} = 0$  or not. In particular  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$  have type  $(1, 2)$ .*

*Proof.* For  $\mathfrak{p} \in S$ , local class field theory identifies the decomposition group of  $\mathfrak{p}$  in  $\text{Gal}(E/B) = \mathbb{F}_2^8$  with the image of  $\mathbb{Q}_p^\times$  under the local Artin map, and the inertia group with the image of the units. By the preceding paragraph, at  $\mathfrak{p}_j$  these are spanned by  $\bar{x}_j, \bar{y}_j, \bar{z}_j$  and by  $\bar{x}_j, \bar{y}_j$ ; the three vectors are independent (Table 4), so the decomposition group has order 8 and inertia has order 4. Above 3 and 5 the two elementary images are independent, which gives  $(2, 2)$ . For  $\mathfrak{p} \notin S$  the decomposition group is generated by  $v_{\mathfrak{p}}$ . The Frobenius vectors of  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$  are nonzero (Table 4).  $\square$

**3.3. The Zeta Function of the Kummer Field.** Let  $\chi_{241}$  be the Dirichlet character  $n \mapsto (n/241)$ .

**Proposition 3.4.** *For  $\text{Re } s > 1$ ,*

$$\zeta_E(s) = \zeta_B(s) \prod_{e \in \mathbb{F}_2^8 \setminus \{0\}} L(s, \chi_e), \quad \zeta_B(s) = \zeta(s) L(s, \chi_{241}),$$

where  $L(s, \chi_e) = \zeta_{B_e}(s)/\zeta_B(s)$  is the Hecke  $L$ -function of  $\chi_e$ .

*Proof.* Compare Euler factors at a prime  $\mathfrak{p}$  of  $B$ . Let  $T_{\mathfrak{p}} \subseteq Z_{\mathfrak{p}} \subseteq G = \text{Gal}(E/B)$  be its inertia and decomposition groups, and put  $f_{\mathfrak{p}} = |Z_{\mathfrak{p}}/T_{\mathfrak{p}}|$  and  $g_{\mathfrak{p}} = |G/Z_{\mathfrak{p}}|$ . The Euler factor of  $\zeta_E$  at  $\mathfrak{p}$  is  $(1 - X^{f_{\mathfrak{p}}})^{-g_{\mathfrak{p}}}$  with  $X = N\mathfrak{p}^{-s}$ . A character nontrivial on  $T_{\mathfrak{p}}$  contributes the factor 1. The characters trivial on  $T_{\mathfrak{p}}$  are those of  $G/T_{\mathfrak{p}}$ ; they restrict onto the  $f_{\mathfrak{p}}$  characters of the cyclic group  $Z_{\mathfrak{p}}/T_{\mathfrak{p}}$ , each with  $g_{\mathfrak{p}}$  extensions. Hence their factors multiply to  $\prod_{\zeta^{f_{\mathfrak{p}}}=1} (1 - \zeta X)^{-g_{\mathfrak{p}}} = (1 - X^{f_{\mathfrak{p}}})^{-g_{\mathfrak{p}}}$ . The same identity for  $B_e/B$  identifies  $L(s, \chi_e)$  with  $\zeta_{B_e}/\zeta_B$ . The second identity is the case of  $B/\mathbb{Q}$ . Since  $241 \equiv 1 \pmod{8}$ , the prime 2 splits in  $B$ , and quadratic reciprocity shows that the character of  $B/\mathbb{Q}$  is  $\chi_{241}$ .  $\square$

Put  $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$  and  $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$ , so that  $\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s) \Gamma_{\mathbb{R}}(s+1)$  by the duplication formula. For a number field  $M$  with discriminant  $\Delta_M$  and signature  $(r_1(M), r_2(M))$ , the completed Dedekind zeta function is

$$\Lambda_M(s) = |\Delta_M|^{s/2} \Gamma_{\mathbb{R}}(s)^{r_1(M)} \Gamma_{\mathbb{C}}(s)^{r_2(M)} \zeta_M(s).$$

**Proposition 3.5.** *Let  $e \neq 0$ , write  $L(s, \chi_e) = \sum_{n \geq 1} a_n(e) n^{-s}$ , and put*

$$Q_e = 241 N\mathfrak{f}_e, \quad \Lambda(s, \chi_e) = Q_e^{s/2} \Gamma_{\mathbb{R}}(s + \nu_1(e)) \Gamma_{\mathbb{R}}(s + \nu_2(e)) L(s, \chi_e).$$

- (1) *The coefficients  $a_n(e)$  are integers with  $|a_n(e)| \leq \tau(n)$ . The Euler factor at a rational prime  $p$  is  $\prod_{\mathfrak{p}|p} (1 - \chi_e(\mathfrak{p}) N\mathfrak{p}^{-s})^{-1}$ , with  $\chi_e(\mathfrak{p}) = 0$  when  $a_{\mathfrak{p}}(e) > 0$ .*
- (2) *The function  $\Lambda(s, \chi_e)$  equals  $\Lambda_{B_e}(s)/\Lambda_B(s)$ . It is entire of order at most one and satisfies  $\Lambda(s, \chi_e) = \Lambda(1-s, \chi_e)$ .*
- (3) *Exactly 63 of the characters have  $\nu_1(e) = \nu_2(e) = 0$ , exactly 64 have  $\nu_1(e) = \nu_2(e) = 1$ , and 128 have  $\nu_1(e) \neq \nu_2(e)$ .*

*Proof.* The Euler factor in (1) is Proposition 3.4 applied prime by prime. Above a rational prime there are at most two primes of  $B$ . So the local factor is either a product of at most two factors  $(1 - cp^{-s})^{-1}$ , or a single factor  $(1 - cp^{-2s})^{-1}$ , with  $c \in \{0, \pm 1\}$ . In both cases the coefficient of  $p^{-ks}$  has modulus at most  $k + 1 = \tau(p^k)$ . Multiplicativity gives (1).

For (2), the completed Dedekind zeta functions satisfy  $\Lambda_M(1-s) = \Lambda_M(s)$ , and  $L(s, \chi_e)$  extends to an entire function because  $\chi_e$  is nontrivial [12, 37]. The discriminants satisfy  $|\Delta_{B_e}| = |\Delta_B|^2 N\mathfrak{d}_{B_e/B}$ , where  $\mathfrak{d}_{B_e/B}$  is the relative discriminant, and the conductor–discriminant formula gives  $\mathfrak{d}_{B_e/B} = \mathfrak{f}_e$  for the quadratic extension  $B_e/B$ . Hence  $|\Delta_{B_e}|/|\Delta_B| = 241 N\mathfrak{f}_e = Q_e$ . If  $\nu_k(e) = 0$ , the place  $v_k$  splits into two real places of  $B_e$  and contributes  $\Gamma_{\mathbb{R}}(s)^2/\Gamma_{\mathbb{R}}(s) = \Gamma_{\mathbb{R}}(s)$ . If  $\nu_k(e) = 1$ , it becomes one complex place and contributes  $\Gamma_{\mathbb{C}}(s)/\Gamma_{\mathbb{R}}(s) = \Gamma_{\mathbb{R}}(s+1)$ . Thus  $\Lambda_{B_e}/\Lambda_B = \Lambda(s, \chi_e)$ . It is the completed Hecke  $L$ -function of  $\chi_e$ , which is entire of order at most one [12, 37].

For (3), the vectors  $\bar{t}_1, \bar{t}_2$  are linearly independent. So  $e \mapsto (\nu_1(e), \nu_2(e))$  maps  $\mathbb{F}_2^8$  onto  $\mathbb{F}_2^2$  with fibers of size 64. Removing  $e = 0$  from the fiber over  $(0, 0)$  gives the counts.  $\square$

The characters with  $\nu_1 = \nu_2$  have *pure* gamma factors  $\Gamma_{\mathbb{R}}(s + \nu)^2$ , and the others have *mixed* gamma factors  $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$ . Only the primes of  $S$  divide the conductors. Since  $N\mathfrak{f}_e = 2^{a_{\mathfrak{p}_1}(e)+a_{\mathfrak{p}_2}(e)} 3^{a_{\mathfrak{q}_1}(e)+a_{\mathfrak{q}_2}(e)} 5^{a_{\mathfrak{r}_1}(e)+a_{\mathfrak{r}_2}(e)}$ , the conductors are  $Q_e = 241 \cdot 2^i \cdot m$  with  $2^i \in \{1, 4, 8, 16, 32, 64\}$  and  $m \mid 225$ . The number of characters with given  $2^i$  and  $m$  is as follows.

(3.6)

| $2^i \backslash m$ | 1  | 3  | 5  | 9  | 15 | 25 | 45 | 75 | 225 | total |
|--------------------|----|----|----|----|----|----|----|----|-----|-------|
| 1                  | 0  | 2  | 2  | 1  | 4  | 1  | 2  | 2  | 1   | 15    |
| 4                  | 2  | 4  | 4  | 2  | 8  | 2  | 4  | 4  | 2   | 32    |
| 8                  | 4  | 8  | 8  | 4  | 16 | 4  | 8  | 8  | 4   | 64    |
| 16                 | 1  | 2  | 2  | 1  | 4  | 1  | 2  | 2  | 1   | 16    |
| 32                 | 4  | 8  | 8  | 4  | 16 | 4  | 8  | 8  | 4   | 64    |
| 64                 | 4  | 8  | 8  | 4  | 16 | 4  | 8  | 8  | 4   | 64    |
| total              | 15 | 32 | 32 | 16 | 64 | 16 | 32 | 32 | 16  | 255   |

No nontrivial character has  $N\mathfrak{f}_e = 1$ , since the narrow class number of  $B$  is one. The conductors range from  $Q_e = 723$  to  $Q_e = 241 \cdot 14400 = 3470400$ .

*Remark 3.7.* The supplementary program `lfun241.py` computes, for each  $e \neq 0$ , the conductor  $Q_e$ , the gamma type and the coefficients  $a_n(e)$  for  $n \leq N_e = 4\lfloor\sqrt{Q_e}\rfloor + 1$  from Propositions 3.2 and 3.5(1); these are the data used in Section 4. The supplementary program `check255.gp` compares all 255 of them with PARI/GP's own number fields and  $L$ -functions [41]. For each  $e$  it computes the maximal order of  $B_e$ , checks  $|\Delta_{B_e}| = 241 Q_e$  and the gamma type against the signature of  $B_e$ , and checks every coefficient  $a_n(e)$  with  $n \leq N_e$  against the Dirichlet series of  $\zeta_{B_e}/\zeta_B$ . All agree. It also prints the table (3.6), the counts of Proposition 3.5(3) and the range of  $Q_e$ . These comparisons check the implementation; the identification of the  $L$ -functions rests on Propositions 3.2 and 3.5.

#### 4. THE RELATIVE ZETA FACTOR AND ITS EXPLICIT CEILING

We bound the relative zeta value  $L_F(1)$  needed in the geometric construction. For a number field  $A$  we divide  $\log \zeta_A$  by  $[A : \mathbb{Q}]$ .

Let  $\mathcal{K}$  be the set of fields  $K$  described in Theorem 2.24. For  $K \in \mathcal{K}$  we use the notation of that theorem:  $\iota_1$  is a complex conjugation at a place of  $K$  above  $v_1$ ,  $F = K^{\langle \iota_1 \rangle}$ ,  $d = [F : \mathbb{Q}]$ , so that  $[K : \mathbb{Q}] = 2d$ ,  $(b, c)$  is the signature of  $F$ ,  $\theta = c/d$ , and  $\lambda = 2^{9/4}\sqrt{3615}$  is the root discriminant of both  $K$  and  $F$ . Put

$$\ell = \log \lambda = \frac{9}{4} \log 2 + \frac{1}{2} \log 3615 = 5.65600472355 \dots$$

Different choices of  $\iota_1$  are conjugate in  $\text{Gal}(K/B)$  and give conjugate fields  $F$ . The extension  $K/F$  is quadratic and unramified at every finite place,  $K$  is totally imaginary, and  $b \geq 1$ . Write  $L_F(s) = \zeta_K(s)/\zeta_F(s)$ , and let  $\gamma$  be Euler's constant. For a prime  $\mathfrak{p}$  of  $B$ , let  $e_K(\mathfrak{p})$  and  $f_K(\mathfrak{p})$  be its

ramification index and residue degree in the Galois extension  $K/B$ . A *ceiling* is an upper bound for  $d^{-1} \log L_F(1)$  that holds for all  $K \in \mathcal{K}$  of sufficiently large degree.

**Theorem 4.1.** *There is an  $m_0$  such that every  $K \in \mathcal{K}$  with  $[K : B] \geq m_0$  satisfies*

$$\frac{1}{d} \log L_F(1) < 0.04871285.$$

*The proof assumes neither the generalized Riemann hypothesis nor a Brauer–Siegel type asymptotic formula, and it does not give an effective value of  $m_0$ .*

The proof has three steps. Monotonicity of the completion of  $L_F$  reduces the value at  $s = 1$  to  $\zeta_K$  at a real point  $\sigma > 1$  (Subsection 4.1). The Tsfasman–Vlăduț inequality controls the cost of this shift (Subsection 4.2). Finally,  $\zeta_K(\sigma)$  is bounded prime by prime by the zeta function of the Kummer field  $E$  of Section 3, which is evaluated rigorously, with two refinements from the known local behavior of  $K$  (Subsections 4.3–4.7). Throughout,  $\sigma$  denotes a real number greater than 1 and  $\varepsilon = \sigma - 1$ ; in this section  $\varepsilon$  is not the fundamental unit of  $B$ .

**4.1. Reduction to  $s > 1$ .** For a number field  $A$ , write  $h_A$ ,  $\text{Reg}_A$ ,  $w_A$  and  $\Delta_A$  for its class number, regulator, number of roots of unity and discriminant. The residue of  $\zeta_A$  at  $s = 1$  is  $2^{r_1(A)}(2\pi)^{r_2(A)}h_A \text{Reg}_A / (w_A |\Delta_A|^{1/2})$ . Since  $K/F$  is unramified at the finite places,  $|\Delta_K| = |\Delta_F|^2$ . Hence

$$(4.2) \quad L_F(1) = \lim_{s \rightarrow 1} \frac{\zeta_K(s)}{\zeta_F(s)} = 2^c \pi^{d-c} \lambda^{-d/2} \frac{h_K \text{Reg}_K w_F}{h_F \text{Reg}_F w_K} > 0.$$

Let  $s > 1$  be real. At a prime of  $F$  of norm  $q$ , the Euler factor of  $L_F$  is  $(1 - q^{-s})^{-1}$  if the prime splits in  $K$  and  $(1 + q^{-s})^{-1}$  if it is inert. Its square is at most the Euler factor of  $\zeta_K$ , which is  $(1 - q^{-s})^{-2}$ , respectively  $(1 - q^{-2s})^{-1}$ . Consequently

$$(4.3) \quad \frac{\log L_F(s)}{d} \leq \frac{\log \zeta_K(s)}{2d} = \frac{\log \zeta_K(s)}{[K : \mathbb{Q}]}.$$

The function  $L_F$  is the Hecke  $L$ -function of the quadratic character of  $K/F$ . This character is unramified at every finite place and nontrivial at every real place of  $F$ , since  $K$  is totally imaginary. With  $\Gamma_{\mathbb{R}}$ ,  $\Gamma_{\mathbb{C}}$  and  $\Lambda_A$  as in Section 3, put

$$(4.4) \quad \mathcal{L}_F(s) = \frac{\Lambda_K(s)}{\Lambda_F(s)} = \lambda^{ds/2} \Gamma_{\mathbb{R}}(s+1)^b \Gamma_{\mathbb{C}}(s)^c L_F(s).$$

By Hecke’s theorem [12, 37],  $\mathcal{L}_F$  is entire of order at most one, and it satisfies  $\mathcal{L}_F(s) = \mathcal{L}_F(1-s)$  because both completed Dedekind zeta functions do.

**Lemma 4.5.** *The function  $\mathcal{L}_F(s)$  is positive and nondecreasing on  $[1, \infty)$ .*

*Proof.* The Euler product shows that  $\mathcal{L}_F$  has no zeros with  $\text{Re } s > 1$ , and the functional equation excludes  $\text{Re } s < 0$ . So every zero  $\rho$  satisfies  $0 \leq \text{Re } \rho \leq 1$ . There is no zero on  $[1, \infty)$ , by the Euler product and by (4.2). As  $\mathcal{L}_F(s) > 0$  for large real  $s$ , it is positive on  $[1, \infty)$ .

Put  $\Xi(z) = \mathcal{L}_F(\frac{1}{2} + z)$ . It is entire of order at most one, even, and real on the real axis. Its zeros  $z = \rho - \frac{1}{2}$  satisfy  $|\text{Re } z| \leq \frac{1}{2}$ , and  $\sum_{z \neq 0} |z|^{-2} < \infty$ . Group the nonzero zeros into pairs  $\{z_0, -z_0\}$ , and let  $k \geq 0$  be the order of  $\Xi$  at 0. By Hadamard’s theorem for functions of order at most one,

$$\frac{\Xi'}{\Xi}(z) - \frac{k}{z} - \sum_{\{z_0, -z_0\}} \frac{2z}{z^2 - z_0^2}$$

is constant, where the series converges absolutely and locally uniformly away from the zeros. Since  $\Xi$  is even,  $\Xi'/\Xi$  is odd, so the constant is zero. For real  $x \geq \frac{1}{2}$  the logarithmic derivative is real, so

it equals the sum of the real parts. If  $z_0 = a + iy$ , then

$$\operatorname{Re} \frac{2x}{x^2 - z_0^2} = \frac{2x(x^2 - a^2 + y^2)}{(x^2 - a^2 + y^2)^2 + 4a^2y^2} \geq 0,$$

because  $|a| \leq \frac{1}{2} \leq x$ . Since also  $k/x \geq 0$ , the logarithmic derivative is nonnegative, and  $\log \Xi$  is nondecreasing on  $[\frac{1}{2}, \infty)$ .  $\square$

Let  $0 < \varepsilon \leq 1$ . By Lemma 4.5,  $\mathcal{L}_F(1) \leq \mathcal{L}_F(1 + \varepsilon)$ . In (4.4) the gamma factors change by

$$\frac{\Gamma_{\mathbb{R}}(2 + \varepsilon)}{\Gamma_{\mathbb{R}}(2)} = \pi^{-\varepsilon/2} \Gamma\left(1 + \frac{\varepsilon}{2}\right), \quad \frac{\Gamma_{\mathbb{C}}(1 + \varepsilon)}{\Gamma_{\mathbb{C}}(1)} = (2\pi)^{-\varepsilon} \Gamma(1 + \varepsilon).$$

Taking logarithms, dividing by  $d$ , and using  $b/d = 1 - 2\theta$ ,  $c/d = \theta$  and (4.3), we obtain

$$(4.6) \quad \frac{\log L_F(1)}{d} \leq \frac{\log \zeta_K(1 + \varepsilon)}{[K : \mathbb{Q}]} + g_{\infty}(\varepsilon, \theta),$$

$$(4.7) \quad g_{\infty}(\varepsilon, \theta) = \frac{\varepsilon}{2}(\ell - \log \pi) - \varepsilon\theta \log 2 + (1 - 2\theta) \log \Gamma\left(1 + \frac{\varepsilon}{2}\right) + \theta \log \Gamma(1 + \varepsilon).$$

Since  $g_{\infty}$  is affine in  $\theta$ ,  $g_{\infty}(\varepsilon, \theta) \rightarrow 0$  as  $\varepsilon \downarrow 0$ , uniformly for  $0 \leq \theta \leq \frac{1}{2}$ .

The term  $g_{\infty}$  does not appear in the final estimate. Applying (4.6) at  $\varepsilon = 1/300$  would cost  $g_{\infty}(1/300, \theta) \approx 0.0054$  for  $\theta$  near  $\frac{1}{2}$ . Instead, Proposition 4.11 uses (4.6) only in the limit  $\varepsilon \downarrow 0$  along a sequence of fields, where  $g_{\infty}$  vanishes, and the Tsfasman–Vlăduț inequality pays for the passage from  $1 + \varepsilon$  to 1; the final cost is about 0.00215. This limit is also the reason why  $m_0$  is not effective.

**4.2. The Tsfasman–Vlăduț Inequality over  $B$ .** For  $q > 1$  and  $s \geq 1$  put  $g_s(q) = -\log(1 - q^{-s})$ .

**Lemma 4.8.** *Let  $M$  be a finite Galois extension of  $B$  and  $s > 1$ . For a prime  $\mathfrak{p}$  of  $B$  let  $e_M(\mathfrak{p})$  and  $f_M(\mathfrak{p})$  be its ramification index and residue degree in  $M/B$ . Then*

$$(4.9) \quad \frac{\log \zeta_M(s)}{[M : \mathbb{Q}]} = \sum_{\mathfrak{p}} \frac{g_s(N\mathfrak{p}^{f_M(\mathfrak{p})})}{2e_M(\mathfrak{p})f_M(\mathfrak{p})},$$

$$(4.10) \quad -\frac{1}{[M : \mathbb{Q}]} \frac{\zeta'_M}{\zeta_M}(2) = \frac{1}{[M : \mathbb{Q}]} \sum_{\mathfrak{p}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}^2 - 1} = \sum_{\mathfrak{p}} \frac{\log N\mathfrak{p}}{2e_M(\mathfrak{p})(N\mathfrak{p}^{2f_M(\mathfrak{p})} - 1)},$$

where  $\mathfrak{p}$  runs over the primes of  $M$ . Each summand on the right does not increase when  $e_M(\mathfrak{p})$  or  $f_M(\mathfrak{p})$  is replaced by a larger real number. If  $M \subseteq M'$  are both Galois over  $B$ , then  $e_M(\mathfrak{p}) \mid e_{M'}(\mathfrak{p})$  and  $f_M(\mathfrak{p}) \mid f_{M'}(\mathfrak{p})$ .

*Proof.* Above  $\mathfrak{p}$  there are  $[M : B]/(ef)$  primes of  $M$ , each of norm  $N\mathfrak{p}^f$ , and  $[M : \mathbb{Q}] = 2[M : B]$ . This gives both formulas. The first equality in (4.10) follows from  $-\zeta'_M/\zeta_M(s) = \sum_{\mathfrak{p}} \log N\mathfrak{p}/(N\mathfrak{p}^s - 1)$ . For monotonicity in  $f$ , write  $g_s(x^f)/f = \sum_{j \geq 1} x^{-fjs}/(fj)$  for  $x > 1$ ; each term decreases in  $f$ . The divisibilities hold because ramification indices and residue degrees are multiplicative in towers.  $\square$

Put

$$\kappa_{\infty} = \frac{\ell - \gamma - \log(4\pi)}{4}, \quad w(q) = 2 \log q \sum_{m \geq 1} \frac{1}{q^m + 1}.$$

**Proposition 4.11.** *Let  $\sigma = 1 + \varepsilon$  with  $0 < \varepsilon \leq 1$ , and let  $Y_*$  be a real number with  $[K : \mathbb{Q}]^{-1} \log \zeta_K(\sigma) \leq Y_*$  for every  $K \in \mathcal{K}$ . For each prime  $\mathfrak{p}$  of  $B$ , let  $e_{\mathfrak{p}}^0, f_{\mathfrak{p}}^0 \geq 1$  be integers, called floors, with  $e_K(\mathfrak{p}) \geq e_{\mathfrak{p}}^0$  and  $f_K(\mathfrak{p}) \geq f_{\mathfrak{p}}^0$  for every  $K \in \mathcal{K}$ , and let*

$$\mathcal{D} \geq \sum_{\mathfrak{p}} \frac{\log N\mathfrak{p}}{2e_{\mathfrak{p}}^0(N\mathfrak{p}^{2f_{\mathfrak{p}}^0} - 1)}.$$

Then every  $C' > Y_* + \varepsilon(\kappa_\infty + \mathcal{D})$  has the following property: there is an  $m_0$  such that  $d^{-1} \log L_F(1) < C'$  for every  $K \in \mathcal{K}$  with  $[K : B] \geq m_0$ .

*Proof.* Suppose not. Then there are fields  $K_j \in \mathcal{K}$  of strictly increasing degree, hence pairwise non-isomorphic, with  $d_j^{-1} \log L_{F_j}(1) \geq C'$ , where  $[K_j : \mathbb{Q}] = 2d_j$ . For a prime power  $q$  let  $N_q(K)$  be the number of primes of  $K$  of norm  $q$ . If  $q = p^f$ , then  $N_q(K) \leq [K : \mathbb{Q}]/f$ , since the local degrees above  $p$  add up to  $[K : \mathbb{Q}]$ . By a diagonal argument we may assume that  $\beta_q = \lim_j N_q(K_j)/(2d_j)$  exists for every  $q$ .

Tsfasman and Vlăduţ call a sequence  $(A_i)$  of pairwise non-isomorphic number fields asymptotically exact if its genus  $g_{A_i} = \log |\Delta_{A_i}|^{1/2}$  tends to infinity and the limits  $\phi_q = \lim N_q(A_i)/g_{A_i}$ ,  $\phi_{\mathbb{R}} = \lim r_1(A_i)/g_{A_i}$  and  $\phi_{\mathbb{C}} = \lim r_2(A_i)/g_{A_i}$  exist. Their unconditional Basic Inequality' [42, Section 3.2, Proposition 3.1] states that every such sequence satisfies

$$2 \sum_q \phi_q \log q \sum_{m \geq 1} \frac{1}{q^m + 1} + \phi_{\mathbb{R}} \left( \frac{\gamma}{2} + \frac{1}{2} + \log 2\sqrt{\pi} \right) + \phi_{\mathbb{C}} (\gamma + \log 4\pi) \leq 1.$$

For our sequence  $g_{K_j} = d_j \ell$  and  $K_j$  is totally imaginary, so  $\phi_q = 2\beta_q/\ell$ ,  $\phi_{\mathbb{R}} = 0$  and  $\phi_{\mathbb{C}} = 1/\ell$ , and the inequality becomes

$$(4.12) \quad \sum_q \beta_q w(q) \leq \frac{\ell - \gamma - \log(4\pi)}{2} = 2\kappa_\infty.$$

Put  $Z(s) = \sum_q \beta_q g_s(q)$  for  $s \geq 1$ . Since  $g_1(q) \leq 1/(q-1)$  and  $w(q) \geq 2 \log q/(q+1)$ , we have  $g_1(q) \leq 3w(q)/(2 \log 2)$ , so  $Z(1)$  is finite. For fixed real  $s > 1$ ,  $(2d_j)^{-1} \log \zeta_{K_j}(s) = \sum_q (N_q(K_j)/2d_j) g_s(q)$  tends to  $Z(s)$  by dominated convergence: for  $q = p^f$  the terms are at most  $g_s(q)/f \leq 2q^{-s}$ , and  $\sum_q q^{-s} < \infty$ . By (4.6) with  $s-1$  in place of  $\varepsilon$ , for  $1 < s \leq 2$ ,

$$C' \leq \limsup_j \frac{\log L_{F_j}(1)}{d_j} \leq Z(s) + \sup_{0 \leq \theta \leq 1/2} g_\infty(s-1, \theta).$$

As  $s \downarrow 1$ ,  $Z(s)$  increases to  $Z(1)$  and the supremum tends to zero. Hence  $C' \leq Z(1)$ .

Next, for every prime power  $q$ ,

$$\begin{aligned} g_1(q) - g_{1+\varepsilon}(q) &= \int_1^{1+\varepsilon} \frac{\log q}{q^u - 1} du \leq \frac{\varepsilon \log q}{q - 1}, \\ \frac{\log q}{q - 1} - \frac{w(q)}{2} &= \log q \sum_{m \geq 1} \frac{1}{q^m (q^m + 1)} \leq \frac{\log q}{q^2 - 1}. \end{aligned}$$

With (4.12) these give

$$Z(1) \leq Z(1 + \varepsilon) + \varepsilon \left( \kappa_\infty + \sum_q \beta_q \frac{\log q}{q^2 - 1} \right).$$

By Fatou's lemma, (4.10) for  $M = K_j$ , and the monotonicity in Lemma 4.8,

$$\sum_q \beta_q \frac{\log q}{q^2 - 1} \leq \liminf_j \sum_{\mathfrak{p}} \frac{\log N_{\mathfrak{p}}}{2e_{K_j}(\mathfrak{p})(N_{\mathfrak{p}}^{2f_{K_j}(\mathfrak{p})} - 1)} \leq \mathcal{D}.$$

Finally  $Z(1 + \varepsilon) = \lim_j (2d_j)^{-1} \log \zeta_{K_j}(1 + \varepsilon) \leq Y_*$ . Thus  $C' \leq Z(1) \leq Y_* + \varepsilon(\kappa_\infty + \mathcal{D})$ , a contradiction.  $\square$

**4.3. Comparison with the Kummer Field.** We use the degree-two relation space  $R_2$  of Lemma 2.14, the restricted square  $v^{[2]}$  of a vector  $v \in \mathbb{F}_2^8$  as in Section 2, and the Frobenius vectors  $v_{\mathfrak{p}}$  of Section 3.

**Lemma 4.13.** *Let  $K \in \mathcal{K}$ , and let  $\mathfrak{p} \notin S$  be a prime of  $B$  with  $v_{\mathfrak{p}}^{[2]} \notin R_2$ . Then  $e_K(\mathfrak{p}) = 1$  and  $f_K(\mathfrak{p}) \geq 4$ .*

*Proof.* The field  $K$  is unramified over  $B$  outside  $S$ , so  $e_K(\mathfrak{p}) = 1$ , and a prime  $\mathfrak{P}$  of  $K$  above  $\mathfrak{p}$  has a Frobenius  $\text{Frob}_{\mathfrak{P}} \in \text{Gal}(K/B)$  of order  $f_K(\mathfrak{p})$ . Choose  $g \in G_B$  mapping to  $\text{Frob}_{\mathfrak{P}}$ . Its image in  $G_B/D_2G_B = \text{Gal}(E/B)$  is the Frobenius of  $\mathfrak{p}$  in  $E$  (Lemma 3.1), whose elementary image is  $v_{\mathfrak{p}}$ ; it is nonzero, since  $0^{[2]} = 0 \in R_2$ . By Lemma 2.14, whose relation space  $R_2$  is complete by Lemma 2.12,  $g$  has order at least 4 in  $G_B/D_3G_B$ . The projection  $G_B \rightarrow G_B/D_3G_B$  factors through  $\text{Gal}(K/B)$ , because  $K$  contains the fixed field of  $D_4G_B \subseteq D_3G_B$ . So  $\text{Frob}_{\mathfrak{P}}$  has order at least 4.  $\square$

Put

$$(4.14) \quad \Delta_{\text{sel}}(\sigma) = 2\left(\frac{g_{\sigma}(2^2)}{16} - \frac{g_{\sigma}(2^4)}{64}\right) + 2\left(\frac{g_{\sigma}(29^2)}{4} - \frac{g_{\sigma}(29^4)}{8}\right) + \frac{g_{\sigma}(7^4)}{4} - \frac{g_{\sigma}(7^8)}{8},$$

$$(4.15) \quad \Delta_{\text{cen}}(\sigma) = \sum_{\mathfrak{p} \in \mathcal{P}_4} \left( \frac{g_{\sigma}(\text{N}\mathfrak{p}^2)}{4} - \frac{g_{\sigma}(\text{N}\mathfrak{p}^4)}{8} \right),$$

where  $\mathcal{P}_4$  is the set of primes  $\mathfrak{p} \notin S$  of  $B$ , other than  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$ , with  $\text{N}\mathfrak{p} \leq 10^6$  and  $v_{\mathfrak{p}}^{[2]} \notin R_2$ . Every term of both sums is nonnegative, since  $g_{\sigma}(x^2) \leq g_{\sigma}(x)$ .

**Proposition 4.16.** *Every  $K \in \mathcal{K}$  satisfies*

$$\frac{\log \zeta_K(\sigma)}{[K : \mathbb{Q}]} \leq \frac{\log \zeta_E(\sigma)}{512} - \Delta_{\text{sel}}(\sigma) - \Delta_{\text{cen}}(\sigma).$$

*Proof.* By Lemma 3.1,  $E \subseteq K$ , and both are Galois over  $B$ . Apply (4.9) to  $M = E$  and  $M = K$ . By Lemma 4.8, each term for  $K$  is at most the corresponding term for  $E$ . For the primes listed below we replace the term for  $K$  by an upper bound obtained from its known type, and keep the difference.

The types in  $E/B$  are given by Corollary 3.3:  $(4, 2)$  at  $\mathfrak{p}_1, \mathfrak{p}_2$ , and  $(1, 2)$  at  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$  and at the primes of  $\mathcal{P}_4$ . By Theorem 2.24, the primes of  $K$  above 2, 29 and 7 have absolute types  $(8, 4)$ ,  $(1, 4)$  and  $(1, 8)$ . The primes  $\mathfrak{p}_j, \mathfrak{t}_1$  and  $\mathfrak{t}_2$  have residue degree one over  $\mathbb{Q}$ , and  $\mathfrak{t}_0$  has residue degree two. So the types in  $K/B$  are  $(8, 4)$  at  $\mathfrak{p}_j$  and  $(1, 4)$  at  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$ . By Lemma 4.13, the primes of  $\mathcal{P}_4$  have  $e_K = 1$  and  $f_K \geq 4$ . The differences of the terms of  $E$  and the upper bounds for the terms of  $K$  are exactly the summands of (4.14) and (4.15). For example, at  $\mathfrak{p}_j$  the term of  $E$  is  $g_{\sigma}(2^2)/(2 \cdot 4 \cdot 2)$  and that of  $K$  is  $g_{\sigma}(2^4)/(2 \cdot 8 \cdot 4)$ .  $\square$

**4.4. A Degree-Two Approximate Functional Equation.** We evaluate  $\zeta_E(\sigma)$  through Proposition 3.4. The factors  $L(\sigma, \chi_e)$  are computed from an approximate functional equation. For real  $\mu$  and  $x, y > 0$  define the kernels

$$H_{\mu}(x) = \int_1^{\infty} u^{\mu-1} e^{-xu} du, \quad I_{\mu}(y) = \int_1^{\infty} u^{\mu} K_0(yu) du,$$

where  $K_0$  is the modified Bessel function of the second kind, and the envelopes

$$\mathcal{E}_1(x) = \frac{e^{-x}}{x} \left(1 + \frac{1}{x}\right), \quad \mathcal{E}_2(y) = \sqrt{\frac{\pi}{2}} \frac{e^{-y}}{y^{3/2}} \left(1 + \frac{1}{y}\right).$$

Numerical evaluation of  $L$ -functions from their functional equations is developed in [3]. Part (1) of the next lemma treats mixed gamma factors, and part (2), for pure gamma factors, uses the kernel  $K_0$ . We prove both.

**Lemma 4.17.** *Let  $a_n$  be real numbers with  $|a_n| \leq \tau(n)$ , let  $Q > 0$  and  $\nu_1, \nu_2 \in \{0, 1\}$ , and let  $L(s) = \sum_{n \geq 1} a_n n^{-s}$ . Suppose that  $\Lambda(s) = Q^{s/2} \Gamma_{\mathbb{R}}(s + \nu_1) \Gamma_{\mathbb{R}}(s + \nu_2) L(s)$  extends to an entire function of order at most one with  $\Lambda(1-s) = \Lambda(s)$ . Put  $t = 2\pi/\sqrt{Q}$  and let  $\sigma > 1$  be real.*

(1) *If  $\nu_1 \neq \nu_2$ , then*

$$L(\sigma) = \frac{t^\sigma}{\Gamma(\sigma)} \sum_{n \geq 1} a_n (H_\sigma(tn) + H_{1-\sigma}(tn)).$$

(2) *If  $\nu_1 = \nu_2 = \nu$ , then*

$$L(\sigma) = \frac{4(t/2)^{\sigma+\nu}}{\Gamma((\sigma+\nu)/2)^2} \sum_{n \geq 1} a_n n^\nu (I_{\sigma+\nu-1}(tn) + I_{\nu-\sigma}(tn)).$$

*The series converge absolutely.*

*Proof.* In case (1) put  $k(x) = e^{-x}$  and  $\nu = 0$ , and in case (2) put  $k(x) = K_0(x)$ . Their Mellin transforms are  $\Gamma(w)$  and  $2^{w-2}\Gamma(w/2)^2$  for  $\operatorname{Re} w > 0$  [25, 10.43.19]. Let  $\Theta(u) = \sum_n a_n n^\nu k(tnu)$  for  $u > 0$ . For  $\operatorname{Re} w > 1 + \nu$ , termwise integration gives

$$\mathcal{M}(w) = \int_0^\infty \Theta(u) u^{w-1} du = \begin{cases} t^{-w} \Gamma(w) L(w) = \frac{1}{2} \Lambda(w) & \text{in case (1),} \\ 2^{w-2} t^{-w} \Gamma(w/2)^2 L(w-\nu) = \frac{1}{4} Q^{\nu/2} \Lambda(w-\nu) & \text{in case (2).} \end{cases}$$

In case (1) we used  $\Gamma_{\mathbb{R}}(w) \Gamma_{\mathbb{R}}(w+1) = \Gamma_{\mathbb{C}}(w)$  and  $Q^{w/2} (2\pi)^{-w} = t^{-w}$ . In both cases  $\mathcal{M}$  is entire and  $\mathcal{M}(w) = \mathcal{M}(1+2\nu-w)$ .

The function  $L = \Lambda / (Q^{s/2} \Gamma_{\mathbb{R}}(s + \nu_1) \Gamma_{\mathbb{R}}(s + \nu_2))$  of the complex variable  $s$  is entire of finite order. Fix  $\eta > 0$ . The function  $L$  is bounded on  $\operatorname{Re} s = 1 + \eta$ , and, by the functional equation and Stirling's formula, of polynomial growth on  $\operatorname{Re} s = -\eta$ . By the Phragmén–Lindelöf principle it has polynomial growth in  $|\operatorname{Im} s|$  on every vertical strip. With Stirling's formula this shows that  $\mathcal{M}(w)$  decays exponentially in  $|\operatorname{Im} w|$ , uniformly on vertical strips. Mellin inversion on a line  $\operatorname{Re} w = c > 1 + \nu$  and a shift to  $\operatorname{Re} w = 1 + 2\nu - c$  are therefore justified, and no residues occur. With the functional equation of  $\mathcal{M}$  they give

$$\Theta(u) = u^{-1-2\nu} \Theta(1/u) \quad (u > 0).$$

In particular  $\Theta(u)$  decays rapidly as  $u \downarrow 0$ . Splitting the Mellin integral at  $u = 1$  and substituting  $u \mapsto 1/u$  below 1 gives

$$\mathcal{M}(\sigma + \nu) = \int_1^\infty \Theta(u) (u^{\sigma+\nu-1} + u^{\nu-\sigma}) du.$$

Termwise integration, justified by the exponential decay of  $k$ , turns the right side into  $\sum_n a_n n^\nu$  times  $H_\sigma(tn) + H_{1-\sigma}(tn)$  in case (1), and times  $I_{\sigma+\nu-1}(tn) + I_{\nu-\sigma}(tn)$  in case (2). On the left,  $\mathcal{M}(\sigma) = t^{-\sigma} \Gamma(\sigma) L(\sigma)$  in case (1), and  $\mathcal{M}(\sigma + \nu) = 2^{\sigma+\nu-2} t^{-\sigma-\nu} \Gamma((\sigma+\nu)/2)^2 L(\sigma)$  in case (2).  $\square$

**Lemma 4.18.** *Let  $\mu \in \mathbb{R}$ .*

- (1) *The functions  $H_\mu$  and  $I_\mu$  are Laplace transforms of positive measures on  $[1, \infty)$ . They are completely monotone on  $(0, \infty)$ , extend holomorphically to  $\operatorname{Re} z > 0$ , and satisfy  $|H(z)| \leq H(\operatorname{Re} z)$  there, for  $H = H_\mu$  or  $H = I_\mu$ .*
- (2) *If  $\mu \leq 2$ , then  $H_\mu \leq \mathcal{E}_1$ . If  $\mu \leq 3/2$ , then  $I_\mu \leq \mathcal{E}_2$ .*
- (3) *They satisfy*

$$xH'_\mu(x) + \mu H_\mu(x) = -e^{-x}, \quad yI'_\mu(y) + (\mu+1)I_\mu(y) = -K_0(y),$$

*and  $K_0$  satisfies  $y^2 K_0'' + yK_0' - y^2 K_0 = 0$  and  $K_0' = -K_1$ .*

*Proof.* The definition exhibits  $H_\mu$  as the Laplace transform of  $u^{\mu-1} du$  on  $[1, \infty)$ . Taking the order 0 in [25, 10.32.9] and substituting  $v = \cosh r$  in its integral gives  $K_0(x) = \int_1^\infty e^{-xv} (v^2 - 1)^{-1/2} dv$ . Substituting this into  $I_\mu$ , and then  $v = r/u$  in the inner integral, shows

$$I_\mu(y) = \int_1^\infty e^{-yr} \left( \int_1^r u^\mu (r^2 - u^2)^{-1/2} du \right) dr,$$

with a nonnegative inner integral. Differentiation under the integral gives complete monotonicity. Since the measures are positive and  $|e^{-zr}| = e^{-r \operatorname{Re} z}$ , we get  $|H(z)| \leq H(\operatorname{Re} z)$  for  $\operatorname{Re} z > 0$ , which proves (1).

For (2),  $u^{\mu-1} \leq u$  for  $u \geq 1$  and  $\mu \leq 2$ , and  $\int_1^\infty u e^{-xu} du = \mathcal{E}_1(x)$ . For  $I_\mu$  we use  $K_0(x) \leq K_{1/2}(x) = \sqrt{\pi/(2x)} e^{-x}$  [25, 10.39.2]. It holds because the modified Bessel function of order  $\eta \geq 0$  is  $\int_0^\infty e^{-x \cosh r} \cosh(\eta r) dr$  [25, 10.32.9], which increases with  $\eta$ . Since  $u^{\mu-1/2} \leq u$  for  $\mu \leq 3/2$ ,

$$I_\mu(y) \leq \sqrt{\frac{\pi}{2y}} \int_1^\infty u^{\mu-1/2} e^{-yu} du \leq \mathcal{E}_2(y).$$

For (3), differentiate under the integral and integrate by parts:  $xH'_\mu(x) = -\int_1^\infty u^\mu x e^{-xu} du = -e^{-x} - \mu H_\mu(x)$ , and  $yI'_\mu(y) = \int_1^\infty u^{\mu+1} \frac{d}{du} K_0(yu) du = -K_0(y) - (\mu+1)I_\mu(y)$ . The last two identities are Bessel's equation and a standard derivative [25, 10.25.1, 10.29.3].  $\square$

In the application  $1 < \sigma \leq 3/2$ . The four exponents  $\sigma$ ,  $1 - \sigma$ ,  $\sigma + \nu - 1$  and  $\nu - \sigma$  in Lemma 4.17 are then covered by the envelopes:  $\mathcal{E}_1$  bounds  $H_\sigma$  and  $H_{1-\sigma}$ , and  $\mathcal{E}_2$  bounds  $I_{\sigma+\nu-1}$  and  $I_{\nu-\sigma}$ .

**Lemma 4.19.** *Let  $1 < \sigma \leq 3/2$ , let  $|a_n| \leq \tau(n)$ , let  $t > 0$ ,  $N \geq 1$  and  $\omega > 1$ .*

(1) *If  $t(N+1) \geq \omega - 1$ , then for  $\mu \in \{\sigma, 1 - \sigma\}$*

$$\sum_{n>N} |a_n| H_\mu(tn) \leq \zeta(\omega)^2 (N+1)^\omega \mathcal{E}_1(t(N+1)).$$

(2) *If  $\nu \in \{0, 1\}$  and  $t(N+1) \geq \omega + \nu - \frac{3}{2}$ , then for  $\mu \in \{\sigma + \nu - 1, \nu - \sigma\}$*

$$\sum_{n>N} |a_n| n^\nu I_\mu(tn) \leq \zeta(\omega)^2 (N+1)^{\omega+\nu} \mathcal{E}_2(t(N+1)).$$

*Proof.* The logarithmic derivative of  $x^\omega \mathcal{E}_1(x) = x^{\omega-1} e^{-x} (1 + 1/x)$  is  $(\omega-1)/x - 1 - 1/(x(x+1)) < 0$  for  $x \geq \omega - 1$ . So  $n^\omega \mathcal{E}_1(tn) \leq (N+1)^\omega \mathcal{E}_1(t(N+1))$  for  $n > N$ . By Lemma 4.18(2),

$$\sum_{n>N} |a_n| H_\mu(tn) \leq \sum_{n>N} \frac{\tau(n)}{n^\omega} n^\omega \mathcal{E}_1(tn) \leq (N+1)^\omega \mathcal{E}_1(t(N+1)) \sum_{n \geq 1} \frac{\tau(n)}{n^\omega},$$

and the last sum is  $\zeta(\omega)^2$ . Part (2) is the same, since  $y^{\omega+\nu} \mathcal{E}_2(y)$  is a multiple of  $y^{\omega+\nu-3/2} e^{-y} (1 + 1/y)$ , which decreases for  $y \geq \omega + \nu - \frac{3}{2}$ .  $\square$

**4.5. Kernel Enclosures and Finite Sums.** We enclose the kernels on a geometric grid. Let  $H$  be  $H_\mu$  or  $I_\mu$  with  $\mu$  as in Lemma 4.19, and let  $\mathcal{E}$  be its envelope,  $\mathcal{E}_1$  or  $\mathcal{E}_2$ . For  $J \geq 0$  and  $\xi > 0$  put

$$\rho_J(\xi) = \mathcal{E}(\xi/2) \frac{16^{-J-1}}{1 - 1/16}.$$

**Lemma 4.20.** *Let  $\xi > 0$  and let  $h_j = H^{(j)}(\xi)/j!$  be the Taylor coefficients of  $H$  at  $\xi$ .*

(1)  *$(-1)^j h_j \geq 0$  and  $|h_j| \leq \mathcal{E}(\xi/2) (2/\xi)^j$  for all  $j \geq 0$ .*

(2) *For  $|x - \xi| \leq \xi/32$ ,  $|H(x) - \sum_{j=0}^J h_j (x - \xi)^j| \leq \rho_J(\xi)$ . For  $x = 31\xi/32$  the difference lies in  $[0, \rho_J(\xi)]$ .*

- (3) If  $H = H_\mu$ , then  $(j+1)\xi h_{j+1} = -k_j - (j+\mu)h_j$  with  $k_j = (-1)^j e^{-\xi}/j!$ . If  $H = I_\mu$ , then  $(j+1)\xi h_{j+1} = -k_j - (j+\mu+1)h_j$ , where  $k_j$  are the Taylor coefficients of  $K_0$  at  $\xi$ :  $k_0 = K_0(\xi)$ ,  $k_1 = -K_1(\xi)$ , and

$$(j+1)(j+2)\xi^2 k_{j+2} = -(j+1)(2j+1)\xi k_{j+1} - (j^2 - \xi^2)k_j + 2\xi k_{j-1} + k_{j-2},$$

with  $k_{-1} = k_{-2} = 0$ .

*Proof.* Complete monotonicity gives the signs in (1). The disc  $|z - \xi| \leq \xi/2$  lies in  $\operatorname{Re} z \geq \xi/2$ , where  $|H(z)| \leq H(\xi/2) \leq \mathcal{E}(\xi/2)$  by Lemma 4.18. Cauchy's estimate gives the bound on  $h_j$ . For  $|x - \xi| \leq \xi/32$ , the omitted terms have modulus at most  $\mathcal{E}(\xi/2)16^{-j}$ , and their sum over  $j > J$  is at most  $\rho_J(\xi)$ . For  $x < \xi$  every term  $h_j(x - \xi)^j$  is nonnegative by (1), so the remainder is nonnegative. Part (3) follows by comparing coefficients of  $(x - \xi)^j$  in the differential equations of Lemma 4.18(3), written at  $x = \xi + (x - \xi)$ .  $\square$

The grid points are  $\xi_i = 64(31/32)^i$  for  $0 \leq i \leq i_1$ , where  $i_1 = 1878$  is the least index with  $\xi_{i_1} \leq 2^{-80}$ . Taylor polynomials are formed at  $\xi_0, \dots, \xi_{i_1-1}$ . The initial values  $e^{-\xi_i}$ ,  $K_0(\xi_i)$  and  $K_1(\xi_i)$  of part (3) of Lemma 4.20 are enclosed with the rigorous exponential and Bessel functions of Arb [15], which is now part of FLINT [38] and is called through its Python interface. We start from  $H(\xi_0) \in [0, \mathcal{E}(64)]$ , which holds by Lemma 4.18. Given an enclosure of  $H(\xi_i)$ , part (3) of Lemma 4.20 gives enclosures of  $h_0, \dots, h_J$  at  $\xi_i$ . Part (2) then gives

$$H(\xi_{i+1}) \in \sum_{j=0}^J h_j(-\xi_i/32)^j + [0, \rho_J(\xi_i)].$$

This backward recursion is valid because the true Taylor coefficients are obtained from the true value  $H(\xi_i)$  by the same recurrence.

Now let  $t > 0$ ,  $\nu \in \{0, 1\}$ , integers  $a_1, \dots, a_N$  and  $J \geq 0$  be given, with  $tN < \xi_0$  and  $t > \xi_{i_1}$ . For  $0 \leq i < i_1$  let  $\mathcal{B}_i = \{n : 1 \leq n \leq N, \xi_{i+1} < tn \leq \xi_i\}$ . These sets partition  $\{1, \dots, N\}$ . For nonempty  $\mathcal{B}_i$ , let  $\bar{n}_i$  be the integer part of the average of its least and largest elements, and put

$$M_{i,j} = \sum_{n \in \mathcal{B}_i} a_n n^\nu (n - \bar{n}_i)^j, \quad A_i = \sum_{n \in \mathcal{B}_i} |a_n| n^\nu, \quad P_{i,j} = t^j \sum_{k=j}^J \binom{k}{j} h_{i,k}(t\bar{n}_i - \xi_i)^{k-j},$$

where  $h_{i,k}$  are the Taylor coefficients of  $H$  at  $\xi_i$ . Since  $tn - \xi_i = t(n - \bar{n}_i) + (t\bar{n}_i - \xi_i)$  and  $|tn - \xi_i| < \xi_i/32$  for  $n \in \mathcal{B}_i$ , Lemma 4.20(2) gives

$$(4.21) \quad \left| \sum_{n=1}^N a_n n^\nu H(tn) - \sum_i \sum_{j=0}^J P_{i,j} M_{i,j} \right| \leq \sum_i \rho_J(\xi_i) A_i.$$

The moments  $M_{i,j}$  and the masses  $A_i$  are exact integers. In the application below  $t \geq 2\pi/\sqrt{3470400} > 0.0033$ , so every argument  $tn$  exceeds 0.0033, and the part of the grid below that point is not used.

**4.6. The Certified Value at  $\sigma = 301/300$ .** From now on  $\sigma = 301/300$ . Let  $e \neq 0$ . By Proposition 3.5,  $L(s, \chi_e)$  satisfies the hypotheses of Lemma 4.17 with  $Q = Q_e$  and  $\nu_k = \nu_k(e)$ . Let  $\Pi_e$  be the prefactor of that lemma in the relevant case. With  $t = 2\pi/\sqrt{Q_e}$  and

$$N_e = 4\lfloor \sqrt{Q_e} \rfloor + 1,$$

split each of the two series of the lemma at  $N_e$ . The finite parts are enclosed by (4.21) with  $J = 20$ . The tails are bounded by Lemma 4.19: here  $N_e$  lies between 105 and 7449, and  $t(N_e + 1)$  between 24.7 and 25.5, for every  $e$  (the supplementary program `check255.gp` prints both ranges), and  $\omega$  is

chosen among  $\frac{101}{100}, \frac{21}{20}, \frac{11}{10}, \frac{9}{8}, \frac{6}{5}, \frac{5}{4}, \frac{4}{3}, \frac{3}{2}, 2$  subject to the hypothesis of that lemma, to minimize the bound. Writing  $\mathcal{S}_1, \mathcal{S}_2$  for the two finite approximations and

$$\mathcal{R}_e = \sum_i \rho_{20}(\xi_i) A_i + \zeta(\omega)^2 (N_e + 1)^{\omega+\nu} \mathcal{E}(t(N_e + 1)),$$

with  $\nu = 0$  in the mixed case, we obtain

$$(4.22) \quad |L(\sigma, \chi_e) - \Pi_e(\mathcal{S}_1 + \mathcal{S}_2)| \leq 2\Pi_e \mathcal{R}_e.$$

The same bound  $\mathcal{R}_e$  serves both series, since their kernels have the same envelope.

The coefficients  $a_n(e)$  for  $n \leq N_e$  are computed exactly from the Euler factors of Proposition 3.2 by the supplementary program `lfun241.py`. The supplementary program `afe241.py` evaluates (4.22) in Arb midpoint-radius interval arithmetic [15, 38] with 256 bits of precision, rounding every quantity outward. Its kernel and moment routines are those of the module `nonpositive-afe.py`, imported unchanged from an earlier supplementary archive of the author (Section 1.6); they implement Lemmas 4.17–4.20 and (4.21) as stated here. Every enclosure of  $L(\sigma, \chi_e)$  is positive. The factor  $\zeta_B(\sigma) = \zeta(\sigma)L(\sigma, \chi_{241})$  is evaluated with

$$L(\sigma, \chi_{241}) = 241^{-\sigma} \sum_{a=1}^{240} \left( \frac{a}{241} \right) \zeta\left(\sigma, \frac{a}{241}\right),$$

using Arb's rigorous Hurwitz zeta function [14]. This gives  $\zeta_B(\sigma) = 725.51409164864\dots$  and  $L(\sigma, \chi_{241}) = 2.41373420241\dots$ . Summing the logarithms, by Proposition 3.4, proves the following.

**Proposition 4.23.** *For  $\sigma = 301/300$ ,*

$$0.08264460177456 < \frac{\log \zeta_E(\sigma)}{512} < 0.08264460807138.$$

The enclosure computed by `afe241.py` is

$$[0.08264460177456168\dots, 0.08264460807137072\dots],$$

and the proposition states it rounded outward. This is a computer-assisted enclosure: the inequalities follow from Lemmas 4.17–4.20, (4.21), the exact coefficients, and outward-rounded interval arithmetic. As a check independent of the approximate functional equation, the supplementary programs `yecheck.gp` and `lvalues255.gp` evaluate the same values numerically with PARI/GP [41]: each of the 255 values  $L(\sigma, \chi_e)$  lies in its enclosure, and PARI's value  $0.08264460492293\dots$  of  $\log \zeta_E(\sigma)/512$  lies in the enclosure above.

**4.7. The Frobenius Census and the Prime Floors.** The set  $\Sigma_2 = \{v \in \mathbb{F}_2^8 : v^{[2]} \in R_2\}$  is computed by linear algebra over  $\mathbb{F}_2$  from the relations of Lemma 2.14, by the supplementary program `census241.py`; the data file `sqinR.txt`, which `ceiling241.py` reads, records the same set. It has 21 elements: zero and

$$\begin{array}{cccccc} 00100000 & 11110010 & 11010010 & 00010000 & 10000001 \\ 10010001 & 00001000 & 10100111 & 10101111 & 00000100 \\ 11101011 & 11101111 & 00000010 & 01100101 & 01100111 \\ 00000001 & 01011010 & 01011011 & 10111010 & 11000101. \end{array}$$

In the notation of Table 4, these are the elementary images of the local involutions  $x_j, y_j$  and  $x_j y_j$  for  $j = 1, 2$ , of the nonidentity elements of the four local groups  $C_2 \times C_2$  above 3 and 5, and of  $\iota_1$  and  $\iota_2$ . Each of these images lies in  $\Sigma_2$  by Lemma 2.14, since an involution has trivial square; the computation shows that  $\Sigma_2$  contains no other vectors.

The *Frobenius census* classifies the 78616 primes  $\mathfrak{p} \notin S$  of  $B$ , other than  $\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2$ , with  $N\mathfrak{p} \leq 10^6$ , by their Frobenius vectors, computed from Proposition 3.2(1) with exact modular arithmetic. The supplementary program `census241.py` performs this classification and finds that 246 of these primes have  $v_{\mathfrak{p}} = 0$ , that 5994 have  $v_{\mathfrak{p}} \in \Sigma_2 \setminus \{0\}$ , and that the remaining 72376 form  $\mathcal{P}_4$ . The

supplementary program `ceiling241.py` uses the same classification and evaluates (4.14) and (4.15) in Arb interval arithmetic:

$$(4.24) \quad \Delta_{\text{sel}}(\sigma) = 0.03445342991613\dots, \quad \Delta_{\text{cen}}(\sigma) = 0.00162984111205\dots$$

For Proposition 4.11 we use the following floors, valid for every  $K \in \mathcal{K}$  by Theorem 2.24, Corollary 3.3, Lemma 4.13 and  $E \subseteq K$ :

$$(4.25) \quad (e_{\mathfrak{p}}^0, f_{\mathfrak{p}}^0) = \begin{cases} (8, 4) & \mathfrak{p} = \mathfrak{p}_1, \mathfrak{p}_2, \\ (2, 2) & \mathfrak{p} = \mathfrak{q}_1, \mathfrak{q}_2, \mathfrak{r}_1, \mathfrak{r}_2, \\ (1, 4) & \mathfrak{p} \in \{\mathfrak{t}_0, \mathfrak{t}_1, \mathfrak{t}_2\} \cup \mathcal{P}_4, \\ (1, 2) & N\mathfrak{p} \leq 10^6, v_{\mathfrak{p}} \in \Sigma_2 \setminus \{0\}, \\ (1, 1) & \text{otherwise.} \end{cases}$$

The primes of norm greater than  $10^6$  receive  $(1, 1)$ . At most two primes of  $B$  have a given norm, and  $x \mapsto \log x/(x^2 - 1)$  decreases on  $[2, \infty)$ . So their contribution to the sum defining  $\mathcal{D}$  is at most

$$\sum_{m > 10^6} \frac{\log m}{m^2 - 1} \leq \int_{10^6}^{\infty} \frac{\log x}{x^2 - 1} dx \leq \frac{\log 10^6 + 1}{10^6} (1 + 10^{-6}).$$

The finite part of the sum is evaluated in interval arithmetic by `ceiling241.py`. Hence the following value is admissible:

$$(4.26) \quad \mathcal{D} = 0.0085105512261\dots < 0.0085105513.$$

Finally,  $\kappa_{\infty} = (\ell - \gamma - \log(4\pi))/4 = 0.63694120292\dots$

#### 4.8. Proof of the Ceiling.

*Proof of Theorem 4.1.* Let  $\sigma = 301/300$  and  $\varepsilon = 1/300$ . By Proposition 4.16, Proposition 4.23 and (4.24), every  $K \in \mathcal{K}$  satisfies  $[K : \mathbb{Q}]^{-1} \log \zeta_K(\sigma) \leq Y_*$  with

$$Y_* = 0.0826446081 - 0.0344534299 - 0.0016298411 = 0.0465613371.$$

Here the enclosure of Proposition 4.23 is rounded up and the two nonnegative corrections are rounded down. The floors (4.25) satisfy the hypotheses of Proposition 4.11, with  $\mathcal{D}$  as in (4.26). Using  $\kappa_{\infty} < 0.6369412030$ , we obtain

$$Y_* + \varepsilon(\kappa_{\infty} + \mathcal{D}) < 0.0465613371 + \frac{0.6369412030 + 0.0085105513}{300} < 0.0487128430 < 0.04871285.$$

The interval evaluation of the same expression from the unrounded enclosures, by `ceiling241.py`, gives the upper bound 0.0487128429. Proposition 4.11 with  $C' = 0.04871285$  proves the theorem.  $\square$

*Remark 4.27.* The selected primes alone limit what this method can give. For a sequence of fields as in the proof of Proposition 4.11, the types of the primes above 2, 3, 5, 7, 29 in  $K$  are known exactly, and these primes contribute to  $Z(1)$  exactly

$$\frac{2g_1(2^4)}{64} + \frac{2g_1(3^2)}{8} + \frac{2g_1(5^2)}{8} + \frac{2g_1(29^4)}{8} + \frac{g_1(7^8)}{8} = 0.0416684615\dots,$$

a value printed by `ceiling241.py`. The proof of Proposition 4.11 shows that  $Z(1) \leq Y_* + \varepsilon(\kappa_{\infty} + \mathcal{D})$  for all admissible choices. So no choice of  $\sigma$ ,  $Y_*$  or floors in that proposition gives a ceiling below this number. The ceiling 0.04871285 exceeds it by 0.0070444.

*Remark 4.28* (The hypothesis of the Lean formalization). The Lean formalization [23] proves, under one explicit hypothesis that it does not prove, that there are finite planar sets  $U_j$  with  $|U_j| \rightarrow \infty$  and  $u(U_j)/|U_j|^{1.0427} \rightarrow \infty$ . The hypothesis is

$$(4.29) \quad \frac{\log \zeta_E(1 + \frac{1}{300})}{512} + \frac{1}{300} \left( \frac{\ell - \gamma - \log(4\pi)}{4} - \frac{1}{512} \frac{\zeta'_E}{\zeta_E}(2) \right) < 0.0852.$$

In the formalization,  $E$  is the subfield of an algebraic closure of  $\mathbb{Q}$  generated by  $\sqrt{241}$  and square roots of  $\alpha_0, \dots, \alpha_7$ ; it is isomorphic to the Kummer field  $E$ , and the formalization calls it the genus field.

The left side of (4.29) is a coarser, fixed-base form of the bound of this section. By (4.10) for  $M = E$ ,  $-(\zeta'_E/\zeta_E)(2)/512$  is the sum defining  $\mathcal{D}$  in Proposition 4.11 when the floors are the types of  $E$ . These floors are valid since  $E \subseteq K$ . The choice  $Y_* = \log \zeta_E(\sigma)/512$  is valid by Lemma 4.8. So the left side of (4.29) is the ceiling given by Proposition 4.11 without the refinements of Proposition 4.16 and with the local types of  $E$  in place of those of  $K$ . The supplementary program `h241_receipt.py` bounds it by  $0.0848335193 < 0.0852$ . It uses the upper endpoint in Proposition 4.23, the value of  $\kappa_\infty$ , and  $-(\zeta'_E/\zeta_E)(2)/512 < 0.0197321539$ , which it obtains by summing (4.10) in interval arithmetic over the primes of norm at most  $10^6$  and bounding the rest as above.

From (4.29), the formalization subtracts corrections at the primes above 2, 7 and 29 and at the primes above 41, 47, 53, 59, 61, 67, 79, 83 and 97, which lie in  $\mathcal{P}_4$ . It obtains the weaker ceiling 0.0495 for the tower fields. This suffices for the exponent 1.0427 but not for 1.04273. The ceiling 0.04871285 of Theorem 4.1 uses the complete census and the floors of  $K$ ; this refinement is not formalized.

## 5. NORM-ONE MASS AND THE GEOMETRIC TRANSFER

This section and Sections 6 and 7 turn arithmetic data of a quadratic extension into planar point sets. Let  $K/F$  be a quadratic extension of number fields, let  $\iota$  be the nontrivial automorphism of  $K$  over  $F$ , and let  $(b, c)$  be the signature of  $F$ . Throughout these three sections we assume:

- (G1)  $K$  is totally imaginary, and  $b \geq 1$ ;
- (G2)  $K/F$  is unramified at every finite place;
- (G3)  $\text{rd}(K) = \text{rd}(F) = \lambda$ .

Put  $d = [F : \mathbb{Q}] = b + 2c$ ,  $\theta = c/d$  and  $\ell = \log \lambda$ . Then  $[K : \mathbb{Q}] = 2d$ ,  $|\Delta_K| = \lambda^{2d}$ ,  $|\Delta_F| = \lambda^d$  and  $0 \leq \theta < 1/2$ , where  $\Delta_K$  and  $\Delta_F$  are the absolute discriminants. Theorem 2.24 provides such extensions, with  $\lambda = 2^{9/4}\sqrt{3615}$  and  $\theta \geq \theta_*$ ; there  $\iota$  is the complex conjugation  $\iota_1$ . The arguments of these three sections use no Galois structure over  $B$  or over  $\mathbb{Q}$ . For a nonzero fractional ideal  $\mathfrak{a}$  of  $K$  or of  $F$ ,  $N\mathfrak{a}$  denotes its absolute norm, so that  $N(x\mathcal{O}_K) = |N_{K/\mathbb{Q}}x|$  [18, Propositions 4.1 and 4.2]. We take the unit theorem and the covolumes of ideal lattices from [18], and the Herbrand quotient, Hilbert's Theorem 90 and the analytic class-number formula from [19].

The argument has five steps. Proposition 5.2 expresses the mass of the units of relative norm one through the class-number formula. Lemma 5.5 uses the split primes and one ideal class to produce many elements of relative norm one in a single fractional ideal. Lemmas 5.7 and 5.9 average over the norm-one units and over translates of the ideal lattice, which counts the resulting unit displacements exactly. Lemmas 5.11 and 5.13 bound the number of lattice points in a window uniformly, by Poisson summation. Finally, Proposition 5.18 takes the window to be a sublevel set of an energy and combines these steps.

**5.1. Coordinates.** A real place of  $F$  extends to  $K$  either as two real places or as one complex place, and by (G1) the second case occurs. A complex place of  $F$  has two extensions to  $K$ , since  $\mathbb{C}$  has no quadratic extension. Thus  $K$  has exactly  $d$  complex places. For each real place  $v$  of  $F$ , let  $\sigma_v : K \rightarrow \mathbb{C}$  be one of the two embeddings inducing the place of  $K$  above  $v$ . Number the complex places of  $F$  from 1 to  $c$ . For the  $j$ -th one fix an embedding  $\sigma_j^+ : K \rightarrow \mathbb{C}$  whose restriction  $\tau_j$  to  $F$  induces it, and put  $\sigma_j^- = \sigma_j^+ \circ \iota$ . The embeddings  $\sigma_j^\pm$  induce the two places of  $K$  above this place: they are distinct, and they are not complex conjugate, because  $\tau_j$  is not real. We identify  $K \otimes_{\mathbb{Q}} \mathbb{R}$  with  $\mathbb{C}^d$  through the map  $x \mapsto x_\infty$ , where

$$x_\infty = ((\sigma_v(x))_v, (\sigma_j^+(x), \sigma_j^-(x))_j),$$

write  $\mathfrak{a}_\infty = \{x_\infty : x \in \mathfrak{a}\}$  for  $\mathfrak{a} \subset K$ , and identify  $\mathbb{C}^d$  with  $\mathbb{R}^{2d}$  through real and imaginary parts. Volumes are Lebesgue measure in these coordinates, and  $\langle x, \xi \rangle = \operatorname{Re} \sum_w x_w \overline{\xi_w}$  is the real inner product, where  $w$  runs over the  $d$  coordinates.

At a real place  $v$  of  $F$ , the automorphism  $\iota$  induces complex conjugation on the completion  $\mathbb{C}$  of  $K$ , so  $\sigma_v(\iota x) = \overline{\sigma_v(x)}$ . Writing  $v(y) \in \mathbb{R}$  for the image of  $y \in F$  under the real embedding  $v$ , we obtain for  $x \in K$

$$(5.1) \quad |\sigma_v(x)|^2 = v(N_{K/F}x), \quad \sigma_j^+(x) \sigma_j^-(x) = \tau_j(N_{K/F}x).$$

For  $x \in K^\times$  put

$$l_j(x) = \frac{1}{2}(\log |\sigma_j^+(x)| - \log |\sigma_j^-(x)|), \quad l(x) = (l_1(x), \dots, l_c(x)) \in \mathbb{R}^c.$$

If  $N_{K/F}\beta = 1$ , then by (5.1)  $|\sigma_v(\beta)| = 1$  for every real place  $v$ , and  $|\sigma_j^\pm(\beta)| = e^{\pm l_j(\beta)}$  for every  $j$ . Accordingly the  $b$  coordinates  $\sigma_v$  are called the *compact coordinates*: there the elements of relative norm one lie on the unit circle. The  $c$  pairs  $(\sigma_j^+, \sigma_j^-)$  are the *pair coordinates*: there they have reciprocal moduli. A compact coordinate, or a pair of coordinates, is called a *block*. The planar sets will be images under one compact coordinate, which is injective on  $K$  and sends every element of relative norm one to a unit vector. We count ordered pairs of points, called ordered edges, and divide by two at the end.

**5.2. The Unit Mass.** Let  $\mathcal{O}_K^\times$  and  $\mathcal{O}_F^\times$  be the unit groups, and put

$$\mathcal{O}_K^1 = \ker(N_{K/F} : \mathcal{O}_K^\times \rightarrow \mathcal{O}_F^\times), \quad I_N = [\mathcal{O}_F^\times : N_{K/F}(\mathcal{O}_K^\times)].$$

Let  $w_K$  be the number of roots of unity in  $K$ , let  $h_{\text{rel}} = h_K/h_F$ , let  $\kappa$  be the order of the kernel of the map  $\operatorname{Cl}(F) \rightarrow \operatorname{Cl}(K)$  induced by extension of ideals, and let  $L_F(s) = \zeta_K(s)/\zeta_F(s)$ . At  $s = 1$ ,  $L_F(1)$  denotes the quotient of the residues of  $\zeta_K$  and  $\zeta_F$ . The regulators  $\operatorname{Reg}_K$  and  $\operatorname{Reg}_F$  are the ordinary ones, which use twice the logarithm of the absolute value at a complex place [18, Chapter 5].

**Proposition 5.2.** *The roots of unity of  $K$  lie in  $\mathcal{O}_K^1$ , and the restriction of  $l$  to  $\mathcal{O}_K^1$  has kernel of order  $w_K$  and image a full lattice in  $\mathbb{R}^c$ . Let  $\operatorname{Reg}^1$  be the covolume of this lattice, with  $\operatorname{Reg}^1 = 1$  if  $c = 0$ . Then*

$$\kappa = I_N/2^{b-1}, \quad \operatorname{Reg}_K / \operatorname{Reg}_F = 2^{c-1} I_N \operatorname{Reg}^1,$$

and consequently

$$(5.3) \quad \frac{w_K}{h_{\text{rel}} \kappa \operatorname{Reg}^1} = \frac{2^{d-1} \pi^{b+c}}{\lambda^{d/2} L_F(1)}.$$

*Proof. Signs.* Fix a real place  $v_0$  of  $F$ . By (5.1),  $v_0(N_{K/F}x) = |\sigma_{v_0}(x)|^2 > 0$  for  $x \in K^\times$ , so  $-1$  is not a norm from  $K^\times$ . If  $\zeta \in K$  is a root of unity, then  $N_{K/F}\zeta$  is a root of unity of  $F$ , hence  $\pm 1$  because  $F$  has a real place, and it is positive at  $v_0$ . Thus  $N_{K/F}\zeta = 1$ .

*Logarithmic embeddings.* For  $x \in \mathcal{O}_K^\times$  and  $y \in \mathcal{O}_F^\times$  let

$$\operatorname{Log}_K(x) = ((2 \log |\sigma_v x|)_v, (2 \log |\sigma_j^+ x|, 2 \log |\sigma_j^- x|)_j), \quad \operatorname{Log}_F(y) = ((\log |v(y)|)_v, (2 \log |\tau_j(y)|)_j).$$

Their images are lattices in the hyperplanes of coordinate sum zero, the kernel of  $\operatorname{Log}_F$  is  $\{\pm 1\}$  [18, Proposition 5.8], and  $\operatorname{Reg}_K$  and  $\operatorname{Reg}_F$  are the covolumes of the images after one coordinate is deleted; the choice of the deleted coordinate does not matter.

*The lattice  $l(\mathcal{O}_K^1)$ .* Let  $\beta \in \mathcal{O}_K^1$ . Then  $|\sigma_v(\beta)| = 1$  and  $|\sigma_j^\pm(\beta)| = e^{\pm l_j(\beta)}$ , so

$$\operatorname{Log}_K(\beta) = (0, (2l_j(\beta), -2l_j(\beta))_j).$$

Hence  $l(\beta) = 0$  exactly when every conjugate of  $\beta$  has absolute value one, that is, when  $\beta$  is a root of unity [18, Corollary 5.6]. By the first step the kernel of  $l$  on  $\mathcal{O}_K^1$  has order  $w_K$ . The norm of  $y \in \mathcal{O}_F^\times$  is  $y^2$ , so  $N_{K/F}(\mathcal{O}_K^\times)$  has finite index in  $\mathcal{O}_F^\times$  and rank  $b + c - 1$ . By the unit theorem [18,

Theorem 5.1],  $\mathcal{O}_K^\times$  has rank  $d - 1$ , so  $\mathcal{O}_K^1$  has rank  $c$ . As  $\text{Log}_K(\mathcal{O}_K^1)$  is discrete, so is  $l(\mathcal{O}_K^1)$ , and it is a full lattice in  $\mathbb{R}^c$ .

*The capitulation kernel.* The Herbrand quotient  $h(M) = |\widehat{H}^0(\langle \iota \rangle, M)|/|H^1(\langle \iota \rangle, M)|$  [19, Chapter II, §3] of the units satisfies  $2h(\mathcal{O}_K^\times) = \prod_v [K_w : F_v]$ , where  $v$  runs over the archimedean places of  $F$  and  $w$  lies above  $v$  [19, Chapter VII, Proposition 3.1]. Each real place contributes 2 and each complex place 1, so  $h(\mathcal{O}_K^\times) = 2^{b-1}$ . Since  $\widehat{H}^0(\langle \iota \rangle, \mathcal{O}_K^\times) = \mathcal{O}_F^\times / N_{K/F}(\mathcal{O}_K^\times)$  has order  $I_N$ , we get  $|H^1(\langle \iota \rangle, \mathcal{O}_K^\times)| = I_N / 2^{b-1}$ . This group is  $\mathcal{O}_K^1$  modulo the units  $\gamma / \iota \gamma$  with  $\gamma \in \mathcal{O}_K^\times$ . If  $\mathfrak{a}$  is a fractional ideal of  $F$  with  $\mathfrak{a}\mathcal{O}_K = x\mathcal{O}_K$ , the class of the unit  $x/\iota x$  depends neither on the choice of  $x$  nor on the choice of  $\mathfrak{a}$  in its ideal class. This defines a homomorphism from  $\ker(\text{Cl}(F) \rightarrow \text{Cl}(K))$  to  $H^1(\langle \iota \rangle, \mathcal{O}_K^\times)$ . It is injective: if  $x/\iota x = \gamma/\iota \gamma$ , then  $y = x/\gamma$  lies in  $F$  and  $\mathfrak{a}\mathcal{O}_K = y\mathcal{O}_K$ , so  $\mathfrak{a} = y\mathcal{O}_F$  by unique factorization. It is surjective: by Hilbert's Theorem 90 [19, Chapter II, Corollary 1.23], an element of  $\mathcal{O}_K^1$  has the form  $x/\iota x$  with  $x \in K^\times$ , and then the fractional ideal  $x\mathcal{O}_K = (\iota x)\mathcal{O}_K$  is invariant under  $\iota$ . By (G2), every  $\iota$ -invariant fractional ideal of  $K$  is extended from  $F$ : at a split prime  $\mathfrak{P} \iota \mathfrak{P}$  invariance forces equal exponents, an inert prime of  $F$  stays prime, and no prime ramifies. Hence  $\kappa = I_N / 2^{b-1}$ .

*Regulators.* Delete the coordinate of  $v_0$  from both logarithmic embeddings, and let  $\Lambda_K \subset \mathbb{R}^{d-1}$  and  $\Lambda_F \subset \mathbb{R}^{b+c-1}$  be the resulting lattices, of covolumes  $\text{Reg}_K$  and  $\text{Reg}_F$ . By (5.1),  $\text{Log}_F(N_{K/F}x)$  is obtained from  $\text{Log}_K(x)$  by keeping the coordinates of the real places and adding the two coordinates of each pair. This induces a linear map  $\phi: \mathbb{R}^{d-1} \rightarrow \mathbb{R}^{b+c-1}$  such that  $\phi(\Lambda_K)$  is the image of  $N_{K/F}(\mathcal{O}_K^\times)$  in  $\Lambda_F$ . The substitution  $(x_j, y_j) \mapsto (x_j, x_j + y_j)$  on each pair has determinant one and turns  $\phi$  into a coordinate projection whose kernel consists of the  $c$  coordinates  $x_j$ . The covolume of a lattice in a product of two coordinate spaces is the covolume of its intersection with the first factor times the covolume of its projection to the second, whenever both are lattices.

A unit  $x \in \mathcal{O}_K^\times$  lies in the kernel of  $\phi$  exactly when  $\text{Log}_F(N_{K/F}x) = 0$ , because the deleted coordinate is minus the sum of the others. Then  $N_{K/F}x = \pm 1$ , and  $N_{K/F}x = 1$  by the first step. So the intersection of  $\Lambda_K$  with the kernel is the image of  $\mathcal{O}_K^1$ , which after the substitution is  $2l(\mathcal{O}_K^1)$ , of covolume  $2^c \text{Reg}^1$ . Since  $-1 \notin N_{K/F}(\mathcal{O}_K^\times)$ , the projection  $\phi(\Lambda_K)$  has index  $[\mathcal{O}_F^\times : \{\pm 1\}N_{K/F}(\mathcal{O}_K^\times)] = I_N/2$  in  $\Lambda_F$ . Hence  $\text{Reg}_K = 2^c \text{Reg}^1 \cdot \text{Reg}_F I_N/2$ .

*The mass formula.* The analytic class-number formula [19, Chapter V, Theorem 2.4], applied to  $K$  (with  $r_1 = 0$  and  $r_2 = d$ ) and to  $F$  (with  $r_1 = b$ ,  $r_2 = c$  and  $w_F = 2$ ), gives

$$L_F(1) = \frac{(2\pi)^d h_K \text{Reg}_K}{w_K \lambda^d} \cdot \frac{2\lambda^{d/2}}{2^b (2\pi)^c h_F \text{Reg}_F} = \frac{2^{c+1} \pi^{b+c} h_{\text{rel}}}{w_K \lambda^{d/2}} \frac{\text{Reg}_K}{\text{Reg}_F}.$$

Substituting  $\text{Reg}_K / \text{Reg}_F = 2^{c-1} I_N \text{Reg}^1 = 2^{b+c-2} \kappa \text{Reg}^1$  gives (5.3).  $\square$

**5.3. Norm-One Elements from Split Primes.** Let  $\mathcal{R}$  be a finite set of rational primes. We say that  $K/F$  has *uniform local types*  $(e_r, f_r)_{r \in \mathcal{R}}$  if, for every  $r \in \mathcal{R}$ , every prime of  $F$  above  $r$  splits in  $K$  and has absolute ramification index  $e_r$  and absolute residue degree  $f_r$ . Then  $F$  has exactly  $d/(e_r f_r)$  primes above  $r$ , each of norm  $r^{f_r}$ . Given integers  $k_r \geq 0$ , define

$$(5.4) \quad H = \sum_{r \in \mathcal{R}} \frac{k_r \log r}{e_r}, \quad J = \sum_{r \in \mathcal{R}} \frac{\log(k_r + 1)}{e_r f_r}.$$

**Lemma 5.5.** *Assume uniform local types and fix integers  $k_r \geq 0$ . There are a fractional ideal  $I$  of  $K$  with  $N(I) = e^{-dH}$ , an integer  $t \geq e^{dJ}/(\kappa h_{\text{rel}})$ , and elements  $\beta_1, \dots, \beta_t \in I$  of relative norm one whose cosets  $\beta_i \mathcal{O}_K^1$  are pairwise disjoint.*

*Proof.* For each prime  $\mathfrak{p}$  of  $F$  above a prime  $r \in \mathcal{R}$ , write  $\mathfrak{p}\mathcal{O}_K = \mathfrak{P}_{\mathfrak{p}} \iota \mathfrak{P}_{\mathfrak{p}}$  and  $k_{\mathfrak{p}} = k_r$ . For integer vectors  $\mathbf{j} = (j_{\mathfrak{p}})$  with  $0 \leq j_{\mathfrak{p}} \leq k_{\mathfrak{p}}$  put

$$\mathfrak{a}_{\mathbf{j}} = \prod_{\mathfrak{p}} \mathfrak{P}_{\mathfrak{p}}^{j_{\mathfrak{p}}}, \quad \mathfrak{z}_{\mathbf{j}} = \prod_{\mathfrak{p}} \mathfrak{P}_{\mathfrak{p}}^{j_{\mathfrak{p}}} (\iota \mathfrak{P}_{\mathfrak{p}})^{k_{\mathfrak{p}} - j_{\mathfrak{p}}}.$$

There are  $\prod_{\mathfrak{p}}(k_{\mathfrak{p}} + 1) = e^{dJ}$  such vectors. The quotient of  $\text{Cl}(K)$  by the image of  $\text{Cl}(F)$  has order  $h_K \kappa / h_F = \kappa h_{\text{rel}}$ , so one class of this quotient contains the classes of  $\mathfrak{A}_{\mathbf{j}}$  for a set of  $t \geq e^{dJ} / (\kappa h_{\text{rel}})$  vectors  $\mathbf{j}$ . Fix one of them,  $\mathbf{j}_0$ , and put  $I = \mathfrak{J}_{\mathbf{j}_0}^{-1}$ . For each  $\mathbf{j}$  in the set, write  $\mathfrak{A}_{\mathbf{j}} \mathfrak{A}_{\mathbf{j}_0}^{-1} = x \mathfrak{a} \mathcal{O}_K$  with  $x \in K^\times$  and  $\mathfrak{a}$  a fractional ideal of  $F$ , and put  $\beta_{\mathbf{j}} = x / \iota x$ . Then  $N_{K/F} \beta_{\mathbf{j}} = 1$ , and since  $\iota$  fixes  $\mathfrak{a} \mathcal{O}_K$ ,

$$\beta_{\mathbf{j}} \mathcal{O}_K = \mathfrak{A}_{\mathbf{j}} \mathfrak{A}_{\mathbf{j}_0}^{-1} (\iota \mathfrak{A}_{\mathbf{j}})^{-1} \iota \mathfrak{A}_{\mathbf{j}_0} = \mathfrak{J}_{\mathbf{j}} I.$$

As  $\mathfrak{J}_{\mathbf{j}}$  is integral,  $\beta_{\mathbf{j}} \in I$ . The ideals  $\mathfrak{J}_{\mathbf{j}} I$  are distinct for distinct  $\mathbf{j}$ , and every element of  $\beta_{\mathbf{j}} \mathcal{O}_K^1$  generates  $\mathfrak{J}_{\mathbf{j}} I$ , so the cosets are disjoint. Finally, since each  $\mathfrak{p}$  splits in  $K$ ,  $N(I)^{-1} = \prod_{\mathfrak{p}} (N\mathfrak{p})^{k_{\mathfrak{p}}} = \prod_r r^{f_r k_r d / (e_r f_r)} = e^{dH}$ .  $\square$

**5.4. Unit Averaging and Translation.** In this subsection and the next,  $I$  is a fractional ideal of  $K$  and  $\beta_1, \dots, \beta_t \in I$  are elements of relative norm one whose cosets  $\beta_i \mathcal{O}_K^1$  are pairwise disjoint; Lemma 5.5 provides such data. For  $h \in \mathbb{R}^c$  let  $W_h: \mathbb{C}^d \rightarrow \mathbb{C}^d$  multiply the coordinate  $\sigma_j^+$  by  $e^{-h_j}$  and  $\sigma_j^-$  by  $e^{h_j}$ , and fix the compact coordinates. It is a real diagonal map of determinant one. Put  $\Lambda_h = W_h I_\infty$ . By [18, Proposition 4.26], extended to fractional ideals by scaling and [18, Proposition 4.2],

$$\text{covol}(\mathfrak{a}_\infty) = 2^{-d} |\Delta_K|^{1/2} N \mathfrak{a}$$

for every fractional ideal  $\mathfrak{a}$  of  $K$ . Hence  $\text{covol}(\Lambda_h) = 2^{-d} \lambda^d N(I)$ , independently of  $h$ ; for the ideal of Lemma 5.5,

$$(5.6) \quad \text{covol}(\Lambda_h) = 2^{-d} \lambda^d e^{-dH}.$$

If  $N_{K/F} \beta = 1$ , then  $W_h \beta_\infty$  has compact coordinates of modulus one and pair coordinates of moduli  $e^{\pm(l_j(\beta) - h_j)}$ . Let  $\Pi^1$  be a fundamental parallelepiped of the lattice  $l(\mathcal{O}_K^1)$ ; its volume is  $\text{Reg}^1$ . When  $c = 0$ , integrals over  $\mathbb{R}^0$  and over  $\Pi^1$  are evaluations at the single point.

**Lemma 5.7.** *For every Borel function  $\Phi: \mathbb{R}^c \rightarrow [0, \infty]$ ,*

$$(5.8) \quad \frac{1}{\text{Reg}^1} \int_{\Pi^1} \sum_{i=1}^t \sum_{\beta \in \beta_i \mathcal{O}_K^1} \Phi(l(\beta) - h) dh = \frac{t w_K}{\text{Reg}^1} \int_{\mathbb{R}^c} \Phi(u) du.$$

*Proof.* The map  $l$  is a homomorphism on  $K^\times$ , so  $l(\beta_i \mathcal{O}_K^1) = l(\beta_i) + l(\mathcal{O}_K^1)$ , and by Proposition 5.2 each point of this translate of  $l(\mathcal{O}_K^1)$  is the image of exactly  $w_K$  elements of  $\beta_i \mathcal{O}_K^1$ . The translates  $\Pi^1 - \lambda'$ , with  $\lambda' \in l(\mathcal{O}_K^1)$ , tile  $\mathbb{R}^c$ , so by Tonelli's theorem  $\int_{\Pi^1} \sum_{\lambda' \in l(\mathcal{O}_K^1)} \Phi(l(\beta_i) + \lambda' - h) dh = \int_{\mathbb{R}^c} \Phi(u) du$  for each  $i$ .  $\square$

A *window* is a bounded Borel set  $\Omega \subset \mathbb{C}^d$  of positive volume that is invariant under rotation of each coordinate separately. For  $u \in \mathbb{R}^c$  let  $\nu(u) \in \mathbb{C}^d$  have compact coordinates 1 and pair coordinates  $(e^{u_j}, e^{-u_j})$ , and put

$$\Phi_\Omega(u) = |\Omega \cap (\Omega - \nu(u))|.$$

If  $x \in \mathbb{C}^d$  has compact coordinates of modulus one and pair coordinates of moduli  $e^{\pm u_j}$ , then a rotation of the coordinates preserves  $\Omega$  and maps  $\nu(u)$  to  $x$ , so  $|\Omega \cap (\Omega - x)| = \Phi_\Omega(u)$ . In particular  $|\Omega \cap (\Omega - W_h \beta_\infty)| = \Phi_\Omega(l(\beta) - h)$  when  $N_{K/F} \beta = 1$ . The function  $\Phi_\Omega$  is continuous, bounded by  $|\Omega|$ , and vanishes when some  $e^{|u_j|}$  exceeds the diameter of  $\Omega$ .

Fix a  $\mathbb{Z}$ -basis of  $I_\infty$ , and for  $y \in [0, 1)^{2d}$  let  $y_I \in \mathbb{C}^d$  be the point with coordinate vector  $y$  in this basis. For  $h \in \mathbb{R}^c$  let  $\mathcal{X}(h, y) = (\Lambda_h + W_h y_I) \cap \Omega$  and  $N(h, y) = |\mathcal{X}(h, y)|$ , and let  $\mathcal{E}(h, y)$  be the number of pairs  $(x, \beta)$  with  $x \in \mathcal{X}(h, y)$ ,  $\beta \in \bigcup_i \beta_i \mathcal{O}_K^1$  and  $x + W_h \beta_\infty \in \Omega$ . Since  $\beta \in I$ , the point  $x + W_h \beta_\infty$  then lies in  $\mathcal{X}(h, y)$ .

**Lemma 5.9.** *Let  $\Omega$  be a window and suppose  $N(h, y) \leq N_{\max} < \infty$  for all  $(h, y)$ . Then there is a pair  $(h, y)$  with  $N(h, y) > 0$  and*

$$(5.10) \quad \frac{\mathcal{E}(h, y)}{N(h, y)} \geq \frac{tw_K}{\text{Reg}^1} \frac{\int_{\mathbb{R}^c} \Phi_\Omega(u) du}{|\Omega|}.$$

*Proof.* For fixed  $x$ , distinct  $\beta$  give distinct points  $x + W_h \beta_\infty \neq x$ , so  $\mathcal{E} \leq N^2 \leq N_{\max}^2$ . Both  $N$  and  $\mathcal{E}$  are countable sums of indicator functions of Borel sets in  $(h, y)$ . For fixed  $h$ , the map  $y \mapsto W_h y_I$  has Jacobian  $\text{covol}(\Lambda_h)$  and maps  $[0, 1)^{2d}$  onto a fundamental domain of  $\Lambda_h$ , so tiling gives

$$\int_{[0, 1)^{2d}} N(h, y) dy = \frac{|\Omega|}{\text{covol}(\Lambda_h)}, \quad \int_{[0, 1)^{2d}} \mathcal{E}(h, y) dy = \frac{1}{\text{covol}(\Lambda_h)} \sum_i \sum_{\beta \in \beta_i \mathcal{O}_K^1} \Phi_\Omega(l(\beta) - h).$$

Average over  $h \in \Pi^1$  with respect to  $dh/\text{Reg}^1$  and apply (5.8). For the probability measure  $dh dy/\text{Reg}^1$  on  $\Pi^1 \times [0, 1)^{2d}$  this gives  $\mathbb{E} \mathcal{E} = \rho \mathbb{E} N$  and  $\mathbb{E} N = |\Omega|/\text{covol}(\Lambda_h) > 0$ , where  $\rho$  is the right side of (5.10). If  $\mathcal{E} < \rho N$  held wherever  $N > 0$ , then, since  $\mathcal{E} = 0$  where  $N = 0$  and  $\{N > 0\}$  has positive measure, we would get  $\mathbb{E} \mathcal{E} < \rho \mathbb{E} N$ .  $\square$

**5.5. Uniform Lattice Counting.** For a lattice  $\Lambda \subset \mathbb{C}^d$  let  $\Lambda^* = \{\xi \in \mathbb{C}^d : \langle \xi, x \rangle \in \mathbb{Z} \text{ for all } x \in \Lambda\}$  be its dual lattice, and let  $\|\xi\|_* = \sum_w |\xi_w|$ , a norm on  $\mathbb{C}^d$ . We use the Fourier transform  $\widehat{\Psi}(\xi) = \int_{\mathbb{C}^d} \Psi(x) e^{-2\pi i \langle x, \xi \rangle} dx$ .

**Lemma 5.11.** *Let  $\mathfrak{a}$  be a fractional ideal of  $K$  and  $h \in \mathbb{R}^c$ . Every nonzero  $\xi$  in the dual lattice of  $W_h \mathfrak{a}_\infty$  satisfies*

$$\prod_w |\xi_w| \geq \frac{2^d}{\lambda^d (N\mathfrak{a})^{1/2}}, \quad \|\xi\|_* \geq \frac{2d}{\lambda (N\mathfrak{a})^{1/(2d)}}.$$

*In particular, if  $N(I) = e^{-dH}$ , as for the ideal of Lemma 5.5, every nonzero dual vector of  $\Lambda_h$  satisfies*

$$(5.12) \quad \prod_w |\xi_w| \geq \mu^d, \quad \mu = 2e^{H/2-\ell}, \quad \|\xi\|_* \geq d\mu.$$

*Proof.* First let  $h = 0$ . For  $x, y \in K$  we have  $\langle x_\infty, \overline{y_\infty} \rangle = \text{Re} \sum_w \sigma_w(xy) = \frac{1}{2} \text{Tr}_{K/\mathbb{Q}}(xy)$ , where the bar is coordinatewise complex conjugation and  $\sigma_w$  is the embedding of the coordinate  $w$ , because these embeddings and their complex conjugates are all the embeddings of  $K$ . Let  $\mathfrak{a}^\vee = \{y \in K : \text{Tr}_{K/\mathbb{Q}}(y\mathfrak{a}) \subset \mathbb{Z}\}$ . It is an  $\mathcal{O}_K$ -module, and since the trace form is nondegenerate, the dual basis of a  $\mathbb{Z}$ -basis of  $\mathfrak{a}$  is a  $\mathbb{Z}$ -basis of  $\mathfrak{a}^\vee$ . So  $\mathfrak{a}^\vee$  is a fractional ideal, and the dual lattice of  $\mathfrak{a}_\infty$  is  $(2\mathfrak{a}^\vee)_\infty$ . Dual lattices have reciprocal covolumes, and complex conjugation preserves volume. With the covolume formula above and  $N(2\mathfrak{a}^\vee) = 2^{2d} N(\mathfrak{a}^\vee)$ , this gives  $N(\mathfrak{a}^\vee) = \lambda^{-2d} (N\mathfrak{a})^{-1}$ . If  $0 \neq y \in \mathfrak{a}^\vee$ , then  $y\mathcal{O}_K \subset \mathfrak{a}^\vee$  and  $|N_{K/\mathbb{Q}} y| = N(y\mathcal{O}_K) \geq N(\mathfrak{a}^\vee)$  [18, Propositions 4.1 and 4.2]. For  $\xi = \overline{(2y)_\infty}$ ,

$$\prod_w |\xi_w| = |N_{K/\mathbb{Q}}(2y)|^{1/2} \geq (2^{2d} \lambda^{-2d} (N\mathfrak{a})^{-1})^{1/2}.$$

For general  $h$ , the dual lattice of  $W_h \mathfrak{a}_\infty$  is the image of the dual lattice of  $\mathfrak{a}_\infty$  under  $W_{-h}$ , because  $W_h$  is real diagonal, and  $W_{-h}$  preserves  $\prod_w |\xi_w|$ . The bound for  $\|\xi\|_*$  follows from the arithmetic-geometric mean inequality. For (5.12) take  $\mathfrak{a} = I$ .  $\square$

**Lemma 5.13.** *Let  $\Lambda \subset \mathbb{C}^d$  be a lattice such that  $\prod_w |\xi_w| \geq \mu^d$  for every nonzero  $\xi \in \Lambda^*$ , where  $\mu > 0$ . Let  $\Psi: \mathbb{C}^d \rightarrow [0, \infty)$  be smooth, bounded and integrable, with  $\int \Psi > 0$  and all partial derivatives bounded, and suppose that*

$$|\widehat{\Psi}(\xi)| \leq M^d e^{-\sigma \|\xi\|_*} \int \Psi \quad (\xi \in \mathbb{C}^d)$$

for some  $M \geq 1$  and  $\sigma > 0$ . If

$$(5.14) \quad \sigma\mu \geq \log M + 2\log 5 + 2,$$

then for every  $y \in \mathbb{C}^d$ ,

$$(5.15) \quad \sum_{x \in \Lambda + y} \Psi(x) \leq \frac{2}{\text{covol}(\Lambda)} \int \Psi.$$

*Proof.* A nonzero  $\xi \in \Lambda^*$  has no zero coordinate, and the arithmetic-geometric mean inequality gives  $\|\xi\|_* \geq d\mu$ . Hence distinct points of  $\Lambda^*$  are at  $\|\cdot\|_*$ -distance at least  $d\mu$ , and the open  $\|\cdot\|_*$ -balls of radius  $d\mu/2$  about them are disjoint. For  $j \geq 1$ , the balls about the points with  $j d\mu \leq \|\xi\|_* < (j+1)d\mu$  lie in the ball of radius  $(j+3/2)d\mu$  about 0. Comparing volumes in  $\mathbb{R}^{2d}$  bounds the number of these points by  $(2j+3)^{2d} \leq 5^{2dj}$ . By (5.14) and  $M \geq 1$ ,

$$M^d \sum_{\xi \in \Lambda^* \setminus \{0\}} e^{-\sigma\|\xi\|_*} \leq \sum_{j \geq 1} e^{d(\log M + 2j \log 5 - j\sigma\mu)} \leq \sum_{j \geq 1} e^{-2dj} \leq 1.$$

For  $\tau > 0$  put  $\Psi_\tau(x) = \Psi(x)e^{-\tau|x|^2}$ , a Schwartz function. Its Fourier transform is  $\widehat{\Psi} * G_\tau$ , where  $G_\tau(\eta) = (\pi/\tau)^d e^{-\pi^2|\eta|^2/\tau}$  is a probability density on  $\mathbb{R}^{2d}$ . Hence  $|\widehat{\Psi}_\tau(\xi)| \leq m_\tau M^d e^{-\sigma\|\xi\|_*} \int \Psi$ , with  $m_\tau = \int e^{\sigma\|\eta\|_*} G_\tau(\eta) d\eta$ . The function  $y \mapsto \sum_{x \in \Lambda} \Psi_\tau(x+y)$  is smooth and  $\Lambda$ -periodic, its Fourier coefficient at  $\xi \in \Lambda^*$  is  $\widehat{\Psi}_\tau(\xi)/\text{covol}(\Lambda)$ , and these coefficients are absolutely summable. This gives the Poisson summation formula

$$\sum_{x \in \Lambda} \Psi_\tau(x+y) = \frac{1}{\text{covol}(\Lambda)} \sum_{\xi \in \Lambda^*} \widehat{\Psi}_\tau(\xi) e^{2\pi i \langle \xi, y \rangle}.$$

Using  $\widehat{\Psi}_\tau(0) = \int \Psi_\tau$  and the bound above, the sum is at most  $(\int \Psi_\tau + m_\tau \int \Psi)/\text{covol}(\Lambda)$ . As  $\tau \rightarrow 0$ , we have  $m_\tau \rightarrow 1$  by dominated convergence, since  $G_\tau$  is the image of  $G_1$  under  $\eta \mapsto \sqrt{\tau} \eta$ ; moreover  $\Psi_\tau$  increases to  $\Psi$ . Monotone convergence gives (5.15).  $\square$

**Lemma 5.16.** *If  $0 < a_{\mathbb{R}} \leq 1$  and  $\Psi_{\mathbb{R}}(z) = e^{-a_{\mathbb{R}}|z|^2}$  on  $\mathbb{C}$ , then  $\widehat{\Psi}_{\mathbb{R}}(\xi) = (\pi/a_{\mathbb{R}})e^{-\pi^2|\xi|^2/a_{\mathbb{R}}}$  and  $|\widehat{\Psi}_{\mathbb{R}}(\xi)| \leq 2e^{-|\xi|} \int \Psi_{\mathbb{R}}$ .*

*Proof.* The formula is the Gaussian integral, and  $\int \Psi_{\mathbb{R}} = \pi/a_{\mathbb{R}}$ . Since  $a_{\mathbb{R}} \leq 1$ , it suffices that  $\pi^2 x^2 - x + \log 2 \geq 0$  for  $x \geq 0$ , and the minimum of the left side is  $\log 2 - 1/(4\pi^2) > 0$ .  $\square$

**5.6. Profiles at Real and Complex Places.** Fix  $0 < \delta < 1$  and put  $p = 2/(1+\delta)$ , the Lebesgue exponent, so that  $1 < p < 2$  and  $p(1+\delta) = 2$ . A *real-place profile* is a positive radial function  $f_{\mathbb{R}}$  on  $\mathbb{C}$ , used at the compact coordinates, and a *complex-place profile* is a positive function  $g_{\mathbb{C}}$  on  $\mathbb{C}^2$  that depends only on  $|z|$  and  $|w|$ , used at the pair coordinates. Define

$$A_{\mathbb{R}} = \int_{\mathbb{C}} f_{\mathbb{R}}^p, \quad I_{\mathbb{R}} = \int_{\mathbb{C}} f_{\mathbb{R}}(z) f_{\mathbb{R}}(z+1) dz, \quad A_{\mathbb{C}} = \int_{\mathbb{C}^2} g_{\mathbb{C}}^p,$$

$$I_{\mathbb{C}} = \int_{\mathbb{R}} \int_{\mathbb{C}^2} g_{\mathbb{C}}(z, w) g_{\mathbb{C}}(z + e^u, w + e^{-u}) dz dw du,$$

$$J_{\mathbb{R}} = \log I_{\mathbb{R}} - (1+\delta) \log A_{\mathbb{R}}, \quad J_{\mathbb{C}} = \log I_{\mathbb{C}} - (1+\delta) \log A_{\mathbb{C}}.$$

When these integrals are finite and positive, the *endpoint laws* are the probability densities  $f_{\mathbb{R}}(z)f_{\mathbb{R}}(z+1)/I_{\mathbb{R}}$  on  $\mathbb{C}$  and  $g_{\mathbb{C}}(z, w)g_{\mathbb{C}}(z + e^u, w + e^{-u})/I_{\mathbb{C}}$  on  $\mathbb{C}^2 \times \mathbb{R}$ . Put  $V_{\mathbb{R}} = -\log f_{\mathbb{R}}$  and  $V_{\mathbb{C}} = -\log g_{\mathbb{C}}$ ; these are the *energies* of the profiles.

**Definition 5.17.** Let  $M \geq 1$  and  $\sigma > 0$ . The pair  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  is *admissible with Fourier constants*  $(M, \sigma)$  if:

- (P1)  $f_{\mathbb{R}}^p$  and  $g_{\mathbb{C}}^p$  are smooth and bounded, and all their partial derivatives are bounded;
- (P2)  $A_{\mathbb{R}}, I_{\mathbb{R}}, A_{\mathbb{C}}$  and  $I_{\mathbb{C}}$  are finite and positive;

- (P3)  $V_{\mathbb{R}}$  and  $V_{\mathbb{C}}$  are bounded below and coercive, that is, their sublevel sets are bounded;  
(P4)  $V_{\mathbb{R}}(z)$  and  $V_{\mathbb{C}}(z, w)$  have finite second moments under the endpoint laws;  
(P5)  $|\widehat{f_{\mathbb{R}}^p}(\xi)| \leq M e^{-\sigma\|\xi\|} A_{\mathbb{R}}$  and  $|\widehat{g_{\mathbb{C}}^p}(\xi_1, \xi_2)| \leq M^2 e^{-\sigma(\|\xi_1\| + \|\xi_2\|)} A_{\mathbb{C}}$  for all  $\xi, \xi_1, \xi_2 \in \mathbb{C}$ .

For an admissible pair, the tensor weight  $\Psi = \prod_v f_{\mathbb{R}}^p \prod_j g_{\mathbb{C}}^p$  on  $\mathbb{C}^d$ , with one factor for each block, satisfies the hypotheses of Lemma 5.13 other than (5.14). It is smooth and bounded, with bounded derivatives,  $\int \Psi = A_{\mathbb{R}}^b A_{\mathbb{C}}^c$ , and its Fourier transform is the product of the transforms of the factors, so that  $|\widehat{\Psi}(\xi)| \leq M^{b+2c} e^{-\sigma\|\xi\|} \int \Psi = M^d e^{-\sigma\|\xi\|} \int \Psi$ . Section 7 proves admissibility for the profiles we use.

**Proposition 5.18.** *Let  $0 < \delta < 1$  and  $C \in \mathbb{R}$ . Let  $(K_j/F_j)_{j \geq 1}$  be quadratic extensions satisfying (G1)–(G3) with a common  $\lambda$  and with  $d_j = [F_j : \mathbb{Q}] \rightarrow \infty$ , such that  $\log L_{F_j}(1) \leq d_j C$  and such that every  $K_j/F_j$  has the same uniform local types  $(e_r, f_r)_{r \in \mathbb{R}}$ . Fix integers  $k_r \geq 0$ , let  $H$  and  $J$  be as in (5.4), and let  $\mu = 2e^{H/2-\ell}$ . Let  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  be admissible with Fourier constants  $(M, \sigma)$  satisfying (5.14). Put*

$$(5.19) \quad \mathcal{M}(\theta) = J - \delta H - (1/2 - \delta)\ell - C + (1 - \delta) \log 2 + (1 - \theta) \log \pi + (1 - 2\theta)J_{\mathbb{R}} + \theta J_{\mathbb{C}}.$$

*If  $\inf_j \mathcal{M}(\theta_j) > 0$ , where  $\theta_j$  is the value of  $\theta$  for  $F_j$ , then there are finite sets  $U_j \subset \mathbb{R}^2$  with  $|U_j| \rightarrow \infty$  and  $u(U_j)/|U_j|^{1+\delta} \rightarrow \infty$ .*

*Proof.* Let  $m_{\mathbb{R}}$  and  $m_{\mathbb{C}}$  be the means of  $V_{\mathbb{R}}(z)$  and  $V_{\mathbb{C}}(z, w)$  under the endpoint laws, and let  $\text{Var}_{\mathbb{R}}$  and  $\text{Var}_{\mathbb{C}}$  be their variances, which are finite by (P4). Fix  $\eta > 0$  with  $4\eta < \inf_j \mathcal{M}(\theta_j)$ , and consider one extension  $K/F$  of the sequence, with  $d$  large. Take  $I, t$  and  $\beta_1, \dots, \beta_t$  from Lemma 5.5, so that  $\Lambda_h$  and the counts  $N(h, y)$  and  $\mathcal{E}(h, y)$  are defined; the window  $\Omega$  is chosen next. For  $x \in \mathbb{C}^d$  with compact coordinates  $x_v$  and pair coordinates  $(x_j^+, x_j^-)$ , let  $V(x) = \sum_v V_{\mathbb{R}}(x_v) + \sum_j V_{\mathbb{C}}(x_j^+, x_j^-)$  be the total energy, put  $\bar{m} = (bm_{\mathbb{R}} + cm_{\mathbb{C}})/d$ , and let

$$\Omega = \{x \in \mathbb{C}^d : V(x) \leq d(\bar{m} + \eta)\}.$$

By (P3) this is a bounded Borel set, and it is invariant under coordinate rotations because the profiles are radial in each coordinate. With  $\Psi = \prod_v f_{\mathbb{R}}^p \prod_j g_{\mathbb{C}}^p = e^{-pV}$  as above, we have  $\mathbf{1}_{\Omega} \leq e^{pd(\bar{m}+\eta)} \Psi$ , so

$$|\Omega| \leq e^{pd(\bar{m}+\eta)} A_{\mathbb{R}}^b A_{\mathbb{C}}^c,$$

and Lemmas 5.11 and 5.13 give, for all  $(h, y)$ ,

$$N(h, y) \leq \frac{2e^{pd(\bar{m}+\eta)} A_{\mathbb{R}}^b A_{\mathbb{C}}^c}{\text{covol}(\Lambda_h)} = 2e^{dY_{\eta}}, \quad Y_{\eta} = H + \log 2 - \ell + (1 - 2\theta) \log A_{\mathbb{R}} + \theta \log A_{\mathbb{C}} + p(\bar{m} + \eta),$$

by (5.6),  $b/d = 1 - 2\theta$  and  $c/d = \theta$ . In particular  $|\Omega| > 0$  for large  $d$ , by the next paragraph, so  $\Omega$  is a window.

We bound the overlap from below. Let  $f = e^{-V} = \prod_v f_{\mathbb{R}} \prod_j g_{\mathbb{C}}$ . The function  $(x, u) \mapsto f(x)f(x + \nu(u))/(I_{\mathbb{R}}^b I_{\mathbb{C}}^c)$  is the product of the endpoint densities of the blocks, hence a probability density on  $\mathbb{C}^d \times \mathbb{R}^c$ . Under it, the energies  $V(x)$  and  $V(x + \nu(u))$  of the two endpoints are sums of independent terms, one for each block, and  $V(x)$  has mean  $d\bar{m}$ . On a compact block the map  $z \mapsto -z - 1$ , and on a pair block the map  $(z, w, u_j) \mapsto (-z - e^{u_j}, -w - e^{-u_j}, u_j)$ , preserves the endpoint density and carries the first endpoint to minus the second. Since the profiles are even,  $V(x + \nu(u))$  has the same law as  $V(x)$ . By Chebyshev's inequality, each of the two energies differs from  $d\bar{m}$  by more than  $\eta d$  with probability at most  $\max(\text{Var}_{\mathbb{R}}, \text{Var}_{\mathbb{C}})/(\eta^2 d)$ . So the indicator  $\chi$  of the event that both energies lie in  $[d(\bar{m} - \eta), d(\bar{m} + \eta)]$  has expectation at least  $1/2$  for all large  $d$ , uniformly in the signature. Where  $\chi = 1$ , both endpoints lie in  $\Omega$  and  $f(x)f(x + \nu(u)) \leq e^{-2d(\bar{m}-\eta)}$ . Therefore

$$\int_{\mathbb{R}^c} \Phi_{\Omega}(u) du \geq \int \chi dx du \geq e^{2d(\bar{m}-\eta)} I_{\mathbb{R}}^b I_{\mathbb{C}}^c \mathbb{E}\chi \geq \frac{1}{2} I_{\mathbb{R}}^b I_{\mathbb{C}}^c e^{2d(\bar{m}-\eta)}.$$

Now take a pair  $(h, y)$  from Lemma 5.9. By  $t \geq e^{dJ}/(\kappa h_{\text{rel}})$ , Proposition 5.2,  $\log L_F(1) \leq dC$  and  $b + c = (1 - \theta)d$ ,

$$\frac{tw_K}{\text{Reg}^1} \geq \frac{e^{dJ} 2^{d-1} \pi^{b+c}}{\lambda^{d/2} L_F(1)} \geq \frac{1}{2} \exp(d[J + \log 2 + (1 - \theta) \log \pi - \ell/2 - C]).$$

Combined with the bounds for  $|\Omega|$  and for the overlap, (5.10) shows that the set  $\mathcal{X} = \mathcal{X}(h, y)$  has  $N = |\mathcal{X}| > 0$  points and at least  $\frac{1}{4} e^{dX_\eta} N$  ordered edges, where

$$\begin{aligned} X_\eta &= J + \log 2 + (1 - \theta) \log \pi - \ell/2 - C + (1 - 2\theta)(\log I_{\mathbb{R}} - \log A_{\mathbb{R}}) \\ &\quad + \theta(\log I_{\mathbb{C}} - \log A_{\mathbb{C}}) + (2 - p)\bar{m} - (2 + p)\eta. \end{aligned}$$

Since  $N \leq 2e^{dY_\eta}$  and  $p(1 + \delta) = 2$ ,

$$\frac{\mathcal{E}(h, y)}{N^{1+\delta}} \geq \frac{e^{dX_\eta}}{4 \cdot 2^\delta e^{\delta dY_\eta}} = 2^{-2-\delta} e^{d(\mathcal{M}(\theta) - 4\eta)}.$$

Finally let  $v_0$  be a real place of  $F$ , and let  $U = \{x_{v_0} : x \in \mathcal{X}\} \subset \mathbb{C} = \mathbb{R}^2$ . Two distinct points of  $\mathcal{X}$  differ by  $W_h \gamma_\infty$  with  $0 \neq \gamma \in I$ , whose  $v_0$ -coordinate  $\sigma_{v_0}(\gamma)$  is nonzero, so  $|U| = N$ . An ordered edge  $(x, \beta)$  gives two points of  $U$  at distance  $|\sigma_{v_0}(\beta)| = 1$ , and distinct ordered edges give distinct ordered pairs of points. Hence  $u(U) \geq \mathcal{E}(h, y)/2$  and

$$\frac{u(U)}{|U|^{1+\delta}} \geq 2^{-3-\delta} e^{d(\mathcal{M}(\theta) - 4\eta)},$$

which tends to infinity along the sequence. Since  $u(U) \leq |U|^2/2$  and  $\delta < 1$ , also  $|U| \rightarrow \infty$ . These sets are the  $U_j$ . The conclusion concerns a sequence of cardinalities, not every large cardinality.  $\square$

## 6. WEIGHTED WINDOWS AT SPLIT FINITE PLACES

We now place locally constant weights at the selected finite places. This refines the choice of ideals in Lemma 5.5 in the same way that the archimedean profiles refine a ball. We keep the hypotheses (G1)–(G3) and the notation of Section 5, and put

$$\text{covol}_0 = \text{covol}((\mathcal{O}_K)_\infty) = (\lambda/2)^d.$$

**6.1. The Local Functional.** Let  $L$  be a nonarchimedean local field with valuation ring  $\mathcal{O}$ , uniformizer  $\varpi$  and residue field of cardinality  $Q$ . We use the additive Haar measure on  $L$  with  $|\mathcal{O}| = 1$ , the product measure on  $L^2$ , and the multiplicative Haar measure  $d\omega$  on  $\mathcal{O}^\times$  of total mass one. For  $n \in \mathbb{Z}$  and  $\omega \in \mathcal{O}^\times$  put

$$s(n, \omega) = (\varpi^n \omega, \varpi^{-n} \omega^{-1}) \in L^2, \quad Tg(z) = \sum_{n \in \mathbb{Z}} \int_{\mathcal{O}^\times} g(z + s(n, \omega)) d\omega.$$

The two coordinates of  $s(n, \omega)$  have product one; they model the finite coordinates of an element of relative norm one at a split place. For a nonnegative, locally constant, compactly supported function  $g$  on  $L^2$  with  $g \neq 0$ , and  $0 < \delta < 1$ ,  $p = 2/(1 + \delta)$ , put

$$(6.1) \quad A(g) = \int_{L^2} g^p, \quad Z(g) = \int_{L^2} g Tg, \quad \mathcal{F}_\delta(g) = \frac{Z(g)}{A(g)^{1+\delta}}.$$

Only finitely many  $n$  contribute to  $Z(g)$ : if  $g(z)g(z + s(n, \omega)) \neq 0$  then both coordinates of  $s(n, \omega)$  lie in the compact set  $\text{supp } g - \text{supp } g$ , which bounds  $n$  from above and from below.

For  $i \in \mathbb{Z}$  let  $\mathcal{B}_i = \varpi^{-i} \mathcal{O}$ , a ball of measure  $Q^i$ . Write  $[x]_+ = \max(x, 0)$ .

**Lemma 6.2.** *For all integers  $i_1, j_1, i_2, j_2$ ,*

$$(6.3) \quad \int_{L^2} \mathbf{1}_{\mathcal{B}_{i_1} \times \mathcal{B}_{j_1}} T \mathbf{1}_{\mathcal{B}_{i_2} \times \mathcal{B}_{j_2}} = Q^{\min(i_1, i_2) + \min(j_1, j_2)} [\max(i_1, i_2) + \max(j_1, j_2) + 1]_+.$$

*Proof.* Two balls of an ultrametric space are disjoint or nested. Hence  $\mathcal{B}_{i_1} \cap (\mathcal{B}_{i_2} - x)$  has measure  $Q^{\min(i_1, i_2)}$  if  $x \in \mathcal{B}_{\max(i_1, i_2)}$ , and is empty otherwise. The left side of (6.3) is therefore  $Q^{\min(i_1, i_2) + \min(j_1, j_2)}$  times the measure of the set of  $(n, \omega)$  with  $\varpi^n \omega \in \mathcal{B}_{\max(i_1, i_2)}$  and  $\varpi^{-n} \omega^{-1} \in \mathcal{B}_{\max(j_1, j_2)}$ . These conditions say  $-\max(i_1, i_2) \leq n \leq \max(j_1, j_2)$  and do not involve  $\omega$ , and the unit integral has mass one.  $\square$

Put  $\mathcal{S}_0 = \mathcal{O}$  and  $\mathcal{S}_i = \mathcal{B}_i \setminus \mathcal{B}_{i-1}$  for  $i \geq 1$ , the *shells*, of measures  $m_0 = 1$  and  $m_i = Q^i - Q^{i-1}$ .

**Definition 6.4.** Let  $k \geq 0$  be an integer, and let  $w = (w_{ij})_{i,j \geq 0}$  be nonnegative weights, finitely many of them nonzero, with  $w_{00} > 0$ . The *shell profile* of type  $(Q, k, w)$  on  $L^2$  is

$$g = \sum_{i,j \geq 0} w_{ij} \mathbf{1}_{\mathcal{S}_i \times \varpi^{-k} \mathcal{S}_j}.$$

A shell profile is nonnegative, locally constant and compactly supported. It is invariant under  $(x, y) \mapsto (\omega x, \omega' y)$  for all  $\omega, \omega' \in \mathcal{O}^\times$ , since multiplication by a unit preserves valuations and hence each shell; in particular it is even. It is also invariant under translation by the group

$$(6.5) \quad \mathcal{C} = \mathcal{O} \times \varpi^{-k} \mathcal{O}, \quad |\mathcal{C}| = Q^k,$$

whatever the number of shells: adding an element of  $\mathcal{O}$  does not change the shell of a point. We call  $\mathcal{C}$  the *period* of  $g$ . Define the symmetric matrix  $\Delta = (\Delta_{ii'})_{i,i' \geq 0}$  by

$$\Delta_{ii'} = m_{\min(i, i')} \quad (i \neq i'), \quad \Delta_{00} = 0, \quad \Delta_{ii} = im_i - Q^{i-1} \quad (i \geq 1),$$

and the kernel

$$\mathcal{K}_{(i,j),(i',j')} = (k+1)m_i m_j \mathbf{1}_{i=i', j=j'} + m_j \Delta_{ii'} \mathbf{1}_{j=j'} + m_i \Delta_{jj'} \mathbf{1}_{i=i'}.$$

**Lemma 6.6.** *The shell profile  $g$  of type  $(Q, k, w)$  satisfies*

$$(6.7) \quad A(g) = Q^k \sum_{i,j} m_i m_j w_{ij}^p, \quad Z(g) = Q^k \sum_{(i,j),(i',j')} w_{ij} \mathcal{K}_{(i,j),(i',j')} w_{i'j'}.$$

*In particular  $\mathcal{F}_\delta(g)$  depends only on the type, and the hard window  $w_{00} = 1$ ,  $w_{ij} = 0$  otherwise, has  $\mathcal{F}_\delta(g) = (k+1)Q^{-\delta k}$ .*

*Proof.* The formula for  $A(g)$  holds because the sets  $\mathcal{S}_i \times \varpi^{-k} \mathcal{S}_j$  are disjoint of measure  $Q^k m_i m_j$ . For  $Z(g)$ , write  $\mathbf{1}_{\mathcal{S}_0} = \mathbf{1}_{\mathcal{B}_0}$  and  $\mathbf{1}_{\mathcal{S}_i} = \mathbf{1}_{\mathcal{B}_i} - \mathbf{1}_{\mathcal{B}_{i-1}}$  for  $i \geq 1$ , and note  $\varpi^{-k} \mathcal{B}_j = \mathcal{B}_{j+k}$ . Only balls with nonnegative indices occur, so by (6.3) and  $k \geq 0$  the positive part can be dropped, and the Gram value of  $\mathcal{B}_{i_1} \times \mathcal{B}_{j_1+k}$  and  $\mathcal{B}_{i_2} \times \mathcal{B}_{j_2+k}$ , for  $i_1, j_1, i_2, j_2 \geq 0$ , is

$$Q^k [(k+1)Q^{\min(i_1, i_2)} Q^{\min(j_1, j_2)} + \max(i_1, i_2) Q^{\min(i_1, i_2)} Q^{\min(j_1, j_2)} + Q^{\min(i_1, i_2)} \max(j_1, j_2) Q^{\min(j_1, j_2)}].$$

For a function  $\varphi$  of two nonnegative integers put  $(\partial\varphi)(i, i') = \varphi(i, i') - \varphi(i-1, i') - \varphi(i, i'-1) + \varphi(i-1, i'-1)$ , where terms with an index  $-1$  are omitted; this is the expansion of the shells into balls. Apply  $\partial$  in  $(i_1, i_2)$  and in  $(j_1, j_2)$ . A direct computation gives, for  $i, i' \geq 0$ , that  $\partial Q^{\min(i_1, i_2)}$  at  $(i, i')$  is  $m_i \mathbf{1}_{i=i'}$ , and that  $\partial(\max(i_1, i_2) Q^{\min(i_1, i_2)})$  is  $\Delta_{ii'}$ . For example, when  $1 \leq i < i'$  the latter is  $i'Q^i - i'Q^{i-1} - (i'-1)Q^i + (i'-1)Q^{i-1} = m_i$ , and when  $i = i' \geq 1$  it is  $iQ^i - 2iQ^{i-1} + (i-1)Q^{i-1} = im_i - Q^{i-1}$ . This gives the formula for  $Z(g)$ . For the hard window only  $\mathcal{K}_{(0,0),(0,0)} = k+1$  contributes.  $\square$

All entries of  $\mathcal{K}$  are nonnegative, since  $\Delta_{ii} = Q^{i-1}(iQ - i - 1) \geq 0$  for  $i \geq 1$ . Hence  $Z(g) \geq (k+1)Q^k w_{00}^2 > 0$  for every shell profile.

For product weights  $w_{ij} = x_i x'_j$ , with vectors  $x, x'$  of nonnegative numbers, put

$$P_x = \sum_i m_i x_i^p, \quad S_x = \sum_i m_i x_i^2, \quad R_x = \sum_{i,i'} x_i \Delta_{ii'} x_{i'},$$

and define  $P_{x'}, S_{x'}, R_{x'}$  in the same way. Then (6.7) becomes

$$(6.8) \quad A(g) = Q^k P_x P_{x'}, \quad Z(g) = Q^k [(k+1)S_x S_{x'} + R_x S_{x'} + S_x R_{x'}].$$

For example, the one-shell choice  $x = x' = (1, t)$  has  $S_x = 1 + (Q-1)t^2$  and  $R_x = 2t + (Q-2)t^2$ , and its gain over the hard window is

$$(6.9) \quad \log \frac{(k+1)S_x^2 + 2S_x R_x}{k+1} - 2(1+\delta) \log(1 + (Q-1)t^p).$$

Since  $p > 1$ , its right derivative at  $t = 0$  is  $4/(k+1) > 0$ , so a sufficiently small first shell always improves on the hard window. The certificate uses the exact finite functional (6.8).

**6.2. Finite Periods and Lattice Counting.** Let  $\mathcal{P}$  be a finite set of finite places of  $F$  that split in  $K$ . For  $v \in \mathcal{P}$  write  $v\mathcal{O}_K = \mathfrak{P}_v \iota \mathfrak{P}_v$ , where we also write  $v$  for the prime ideal of  $F$ . The completion of  $K$  at  $\mathfrak{P}_v$  is  $F_v$ ; let  $j_v: K \rightarrow F_v$  be the completion map. The map  $x \mapsto (j_v(x), j_v(\iota x))$  records the completions at  $\mathfrak{P}_v$  and at  $\iota \mathfrak{P}_v$ , and  $\iota$  exchanges the two coordinates. If  $N_{K/F}\beta = 1$ , then  $j_v(\beta)j_v(\iota\beta) = 1$ . Let  $\mathcal{O}_v$ ,  $\varpi_v$  and  $Q_v$  be the valuation ring, a uniformizer and the residue cardinality of  $F_v$ . Put

$$\mathbb{A}_{\mathcal{P}} = \prod_{v \in \mathcal{P}} (F_v \times F_v),$$

with the Haar measure that gives  $\prod_v (\mathcal{O}_v \times \mathcal{O}_v)$  measure one, and let  $\Gamma = \mathcal{O}_{K, \mathcal{P}}$  be the ring of elements of  $K$  that are integral at every prime other than the  $\mathfrak{P}_v$  and  $\iota \mathfrak{P}_v$ . We embed  $\Gamma$  in  $\mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$  by  $\gamma \mapsto (\gamma_{\infty}, \gamma_f)$ , where  $\gamma_f = (j_v(\gamma), j_v(\iota\gamma))_v$ . A subgroup of  $\mathbb{A}_{\mathcal{P}}$  of the form

$$\mathcal{C}_f = \prod_{v \in \mathcal{P}} (\varpi_v^{a_v} \mathcal{O}_v \times \varpi_v^{b_v} \mathcal{O}_v), \quad a_v, b_v \in \mathbb{Z},$$

is called *rectangular*; its measure is  $|\mathcal{C}_f| = \prod_v Q_v^{-a_v - b_v}$ .

**Lemma 6.10.** *Let  $\mathcal{C}_f$  be rectangular and put  $\mathfrak{L} = \prod_v \mathfrak{P}_v^{a_v} (\iota \mathfrak{P}_v)^{b_v}$ .*

- (a) *The elements  $\gamma \in \Gamma$  with  $\gamma_f \in \mathcal{C}_f$  form  $\mathfrak{L}$ , and  $N\mathfrak{L} = |\mathcal{C}_f|^{-1}$ , so  $\text{covol}(\mathfrak{L}_{\infty}) = \text{covol}_0 / |\mathcal{C}_f|$ .*
- (b) *Every coset of  $\mathcal{C}_f$  in  $\mathbb{A}_{\mathcal{P}}$  contains  $\gamma_f$  for some  $\gamma \in \Gamma$ .*
- (c)  *$\Gamma$  is a lattice in  $\mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$ . If  $P_{\mathfrak{L}}$  is a fundamental parallelepiped of  $\mathfrak{L}_{\infty}$ , then  $P_{\mathfrak{L}} \times \mathcal{C}_f$  is a fundamental domain for  $\Gamma$ , of measure  $\text{covol}_0$ .*

*Proof.* (a) The valuation of  $j_v(\gamma)$  is the exponent of  $\mathfrak{P}_v$  in  $\gamma$ , and that of  $j_v(\iota\gamma)$  is the exponent of  $\iota \mathfrak{P}_v$ . Together with integrality at the other primes,  $\gamma_f \in \mathcal{C}_f$  says exactly that  $\gamma \in \mathfrak{L}$ . As  $N\mathfrak{P}_v = N(\iota \mathfrak{P}_v) = Q_v$ , we get  $N\mathfrak{L} = \prod_v Q_v^{a_v + b_v} = |\mathcal{C}_f|^{-1}$ , and the covolume follows from the formula of Section 5.

(b) Choose  $a'_v \leq a_v$  and  $b'_v \leq b_v$  such that the given coset lies in  $\mathcal{C}'_f = \prod_v (\varpi_v^{a'_v} \mathcal{O}_v \times \varpi_v^{b'_v} \mathcal{O}_v)$ , and let  $\mathfrak{L}' \supset \mathfrak{L}$  be the corresponding ideal. The map  $\gamma \mapsto \gamma_f$  induces a homomorphism  $\mathfrak{L}'/\mathfrak{L} \rightarrow \mathcal{C}'_f/\mathcal{C}_f$ . It is injective by (a), applied to  $\mathcal{C}_f$ . Both groups are finite of the same order:  $|\mathcal{C}'_f/\mathcal{C}_f| = |\mathcal{C}'_f|/|\mathcal{C}_f|$ , and  $|\mathfrak{L}'/\mathfrak{L}| = N\mathfrak{L}/N\mathfrak{L}'$  [18, Proposition 4.2], which is the same number by (a). So the map is bijective, and every coset of  $\mathcal{C}_f$  in  $\mathcal{C}'_f$  contains  $\gamma_f$  for some  $\gamma \in \mathfrak{L}' \subset \Gamma$ . No principality is needed.

(c) Given  $(z, c) \in \mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$ , by (b) there is  $\gamma \in \Gamma$  with  $c - \gamma_f \in \mathcal{C}_f$ , and then a unique  $\gamma' \in \mathfrak{L}$  with  $z - (\gamma + \gamma')_{\infty} \in P_{\mathfrak{L}}$ . If  $(z, c)$  had two such representations, the difference of the two elements of  $\Gamma$  would have finite part in  $\mathcal{C}_f$ , hence lie in  $\mathfrak{L}$  by (a), and then it would vanish because  $P_{\mathfrak{L}}$  is a fundamental domain. Thus  $P_{\mathfrak{L}} \times \mathcal{C}_f$  is a fundamental domain, of measure  $(\text{covol}_0 / |\mathcal{C}_f|) \cdot |\mathcal{C}_f| = \text{covol}_0$ . Finally  $\Gamma$  is discrete: for a bounded open set  $Y \ni 0$  in  $\mathbb{C}^d$ , its intersection with the neighborhood  $Y \times \mathcal{C}_f$  of 0 consists of the finitely many  $\gamma \in \mathfrak{L}$  with  $\gamma_{\infty} \in Y$ .  $\square$

For  $h \in \mathbb{R}^c$  let  $\Gamma_h = \{(W_h \gamma_\infty, \gamma_f) : \gamma \in \Gamma\}$ . It is again a lattice, with fundamental domains  $W_h P_{\mathfrak{L}} \times \mathcal{C}_f$  of measure  $\text{covol}_0$ . For a rectangular  $\mathcal{C}_f$  put

$$H_f = d^{-1} \log |\mathcal{C}_f|.$$

By Lemma 6.10(a),  $N\mathfrak{L} = e^{-dH_f}$ , and Lemma 5.11 gives the dual product bound  $\mu_f^d$  for every  $W_h \mathfrak{L}_\infty$ , with  $\mu_f = 2e^{H_f/2-\ell}$ . Accordingly, the Fourier condition used in this section is

$$(6.11) \quad \sigma \cdot 2e^{H_f/2-\ell} \geq \log M + 2 \log 5 + 2.$$

**Lemma 6.12.** *Let  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  be admissible with Fourier constants  $(M, \sigma)$ , and let  $\Psi_\infty = \prod_v f_{\mathbb{R}}^p \prod_j g_{\mathbb{C}}^p$  on  $\mathbb{C}^d$ . Let  $\Psi_f \geq 0$  be a function on  $\mathbb{A}_{\mathcal{P}}$  that is invariant under translation by a rectangular group  $\mathcal{C}_f$  and vanishes outside a finite union of its cosets. If (6.11) holds, with  $H_f$  defined by this  $\mathcal{C}_f$ , then for every  $h \in \mathbb{R}^c$  and every  $z_0 \in \mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$ ,*

$$(6.13) \quad \sum_{\gamma \in \Gamma_h + z_0} \Psi_\infty(\gamma_\infty) \Psi_f(\gamma_f) \leq \frac{2}{\text{covol}_0} \int \Psi_\infty \int \Psi_f,$$

where  $\gamma_\infty$  and  $\gamma_f$  are the two components of  $\gamma$ .

*Proof.* Group the points by the coset of  $\mathcal{C}_f$  containing their finite component. If two points of  $\Gamma_h + z_0$  have finite components in the same coset, their difference comes from an element of  $\mathfrak{L}$ , by Lemma 6.10(a). So the archimedean components of the points in one coset form a translate of  $W_h \mathfrak{L}_\infty$ , or the empty set. By Lemma 5.13, with the dual bound  $\mu_f$  and the Fourier bound for  $\Psi_\infty$  from Section 5, the sum of  $\Psi_\infty$  over such a translate is at most  $2|\mathcal{C}_f| \int \Psi_\infty / \text{covol}_0$ . The values of  $\Psi_f$  on its finitely many nonzero cosets sum to  $\int \Psi_f / |\mathcal{C}_f|$ .  $\square$

If  $\alpha \in \mathbb{A}_{\mathcal{P}}$  has coordinates  $(\varpi_v^{n_v}, \varpi_v^{-n_v})$ , multiplication by  $\alpha$  preserves the Haar measure and maps rectangular groups to rectangular groups of the same measure. So (6.11) is unchanged, and Lemma 6.12 applies after such a rescaling of  $\Psi_f$ , even though the rescaling does not preserve  $\Gamma$ .

**6.3. Class Selection and Unit Averaging.** For  $u \in \mathbb{R}^c$  and  $n = (n_v) \in \mathbb{Z}^{\mathcal{P}}$  let  $\nu(u, n) \in \mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$  have archimedean part  $\nu(u)$  and finite part  $(\varpi_v^{n_v}, \varpi_v^{-n_v})_v$ .

**Lemma 6.14.** *Let  $\mathcal{N} \subset \mathbb{Z}^{\mathcal{P}}$  be finite, and let  $\mathcal{W} : \mathcal{N} \rightarrow [0, \infty)$  have positive sum. There are  $n_0 \in \mathcal{N}$  with  $\mathcal{W}(n_0) > 0$ , a subset  $\mathcal{N}' \subset \mathcal{N}$  with*

$$\sum_{n \in \mathcal{N}'} \mathcal{W}(n) \geq \frac{1}{\kappa h_{\text{rel}}} \sum_{n \in \mathcal{N}} \mathcal{W}(n),$$

and elements  $\beta_n \in \Gamma$ , for  $n \in \mathcal{N}'$ , with  $N_{K/F} \beta_n = 1$  and

$$\beta_n \mathcal{O}_K = \prod_v \mathfrak{P}_v^{n_v - n_{0,v}} (\iota \mathfrak{P}_v)^{-(n_v - n_{0,v})}.$$

The cosets  $\beta_n \mathcal{O}_K^1$  are pairwise disjoint, and  $\beta_{n_0} \mathcal{O}_K^1$  is the set of all elements of relative norm one that generate this ideal.

*Proof.* Map  $n$  to the class of  $\mathfrak{A}(n) = \prod_v \mathfrak{P}_v^{n_v}$  in  $\text{Cl}(K)$  modulo the image of  $\text{Cl}(F)$ , a group of order  $\kappa h_{\text{rel}}$ . Let  $\mathcal{N}'$  be a fiber of largest total weight, and choose  $n_0 \in \mathcal{N}'$  with  $\mathcal{W}(n_0) > 0$ . For  $n \in \mathcal{N}'$  write  $\mathfrak{A}(n) \mathfrak{A}(n_0)^{-1} = x_n \mathfrak{a}_n \mathcal{O}_K$  with  $x_n \in K^\times$  and  $\mathfrak{a}_n$  a fractional ideal of  $F$ , and put  $\beta_n = x_n / \iota x_n$ . As in the proof of Lemma 5.5,  $N_{K/F} \beta_n = 1$  and  $\beta_n$  generates the displayed ideal, which is supported on the primes  $\mathfrak{P}_v, \iota \mathfrak{P}_v$ ; hence  $\beta_n \in \Gamma$ . Two elements of relative norm one generating the same ideal differ by an element of  $\mathcal{O}_K^1$ , and distinct labels give distinct ideals.  $\square$

If  $\beta \in \beta_n \mathcal{O}_K^1$ , then  $j_v(\beta)$  has valuation  $n_v - n_{0,v}$  and  $j_v(\iota \beta) = j_v(\beta)^{-1}$ . Let  $\varpi_0 \in \mathbb{A}_{\mathcal{P}}$  have coordinates  $(\varpi_v^{n_{0,v}}, \varpi_v^{-n_{0,v}})$ . Then  $\varpi_0 \beta_f$  has coordinates  $(\varpi_v^{n_v} \omega_v, \varpi_v^{-n_v} \omega_v^{-1})$  with units  $\omega_v \in \mathcal{O}_v^\times$ .

#### 6.4. The Finite-Place Transfer.

**Proposition 6.15.** *Let  $0 < \delta < 1$  and  $C \in \mathbb{R}$ . Let  $(K_j/F_j)_{j \geq 1}$  be quadratic extensions satisfying (G1)–(G3) with a common  $\lambda$ , with  $d_j = [F_j : \mathbb{Q}] \rightarrow \infty$  and  $\log L_{F_j}(1) \leq d_j C$ . For each  $j$  let  $\mathcal{P}_j$  be a finite set of finite places of  $F_j$  that split in  $K_j$ , and for  $v \in \mathcal{P}_j$  let  $g_v$  be a shell profile on  $F_v^2$  of type  $(Q_v, k_v, w^{(v)})$ , where the types range over a fixed finite set independent of  $j$ . Let  $\mathcal{F}_{\delta,v} = \mathcal{F}_{\delta}(g_v)$ , let  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  be admissible with Fourier constants  $(M, \sigma)$ , and suppose that (6.11) holds for every  $j$ , with  $\mathcal{C}_f = \prod_{v \in \mathcal{P}_j} (\mathcal{O}_v \times \varpi_v^{-k_v} \mathcal{O}_v)$ , that is, with  $H_f = d_j^{-1} \sum_{v \in \mathcal{P}_j} k_v \log Q_v$ . Put*

$$(6.16) \quad \begin{aligned} \mathcal{M}_{\text{fin}}(\theta) = & \frac{1}{d} \sum_{v \in \mathcal{P}} \log \mathcal{F}_{\delta,v} - (1/2 - \delta)\ell - C + (1 - \delta) \log 2 + (1 - \theta) \log \pi \\ & + (1 - 2\theta)J_{\mathbb{R}} + \theta J_{\mathbb{C}}, \end{aligned}$$

*evaluated for  $K_j/F_j$ . If these margins have a positive lower bound over  $j$ , then there are finite sets  $U_j \subset \mathbb{R}^2$  with  $|U_j| \rightarrow \infty$  and  $u(U_j)/|U_j|^{1+\delta} \rightarrow \infty$ .*

*Proof.* Fix  $\eta > 0$  with  $4\eta$  less than the lower bound of the margins, and consider one extension  $K/F$  of the sequence, with  $d$  large.

*The profile and its window.* On  $\mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$  let  $f = \prod_v f_{\mathbb{R}} \prod_j g_{\mathbb{C}} \prod_{v \in \mathcal{P}} g_v$ , with one factor for each archimedean block and each finite place, let  $V = -\log f$  on the set where  $f > 0$ , and put

$$A = \int f^p = A_{\mathbb{R}}^b A_{\mathbb{C}}^c \prod_{v \in \mathcal{P}} A(g_v), \quad Z = I_{\mathbb{R}}^b I_{\mathbb{C}}^c \prod_{v \in \mathcal{P}} Z(g_v).$$

At a finite place, the *endpoint law* is the probability density  $g_v(z)g_v(z+s(n, \omega))/Z(g_v)$  on  $F_v^2 \times \mathbb{Z} \times \mathcal{O}_v^{\times}$ ; the energy  $-\log g_v$  of its first endpoint takes finitely many values. The map  $(z, n, \omega) \mapsto (-z - s(n, \omega), n, \omega)$  preserves this law and carries the first endpoint to minus the second, and  $g_v$  is even. Together with the archimedean endpoint laws of Section 5, the product of all these laws is a probability law under which the energies  $V$  of the two endpoints have the same distribution and are sums of independent terms. Let  $d\bar{m}$  be their mean. The residue characteristics of the places in  $\mathcal{P}$  lie in a fixed finite set, so  $|\mathcal{P}| = O(d)$ , and since the types range over a finite set, the variance of either energy is  $O(d)$ . As in the proof of Proposition 5.18, with probability at least  $1/2$  for large  $d$ , both energies lie in  $[d(\bar{m} - \eta), d(\bar{m} + \eta)]$ .

Let  $\Omega$  be the set of points of  $\mathbb{C}^d \times \mathbb{A}_{\mathcal{P}}$  where  $f > 0$  and  $V \leq d(\bar{m} + \eta)$ . It is a bounded Borel set: the archimedean energies are coercive and bounded below, and the finite part lies in the compact support of  $\prod_v g_v$ , on which the finite energies are bounded. It is invariant under rotations of the archimedean coordinates, under  $(x_v, y_v) \mapsto (\omega x_v, \omega' y_v)$  for units at each  $v \in \mathcal{P}$ , and under translation by  $\mathcal{C}_f$ . For  $u \in \mathbb{R}^c$  and  $n \in \mathbb{Z}^{\mathcal{P}}$  let  $\Phi(u, n) = |\Omega \cap (\Omega - \nu(u, n))|$ , and let  $\mathcal{W}(n) = \int_{\mathbb{R}^c} \Phi(u, n) du$ . By the unit invariance, the overlap of  $\Omega$  with its translate by a vector whose archimedean part has the moduli of  $\nu(u)$  and whose finite part is  $(\varpi_v^{n_v} \omega_v, \varpi_v^{-n_v} \omega_v^{-1})_v$  equals  $\Phi(u, n)$ . Only finitely many  $n$  have  $\mathcal{W}(n) \neq 0$ . Undoing the endpoint density on the event of the previous paragraph, as in the proof of Proposition 5.18, gives

$$\sum_n \mathcal{W}(n) \geq \frac{1}{2} Z e^{2d(\bar{m}-\eta)}, \quad |\Omega| \leq A e^{pd(\bar{m}+\eta)}.$$

*Class selection and rescaling.* Apply Lemma 6.14 to the finitely many  $n$  with  $\mathcal{W}(n) > 0$ , obtaining  $n_0, \mathcal{N}'$  and  $\beta_n$ , and let  $\varpi_0$  be as above. Put  $\Omega' = \varpi_0^{-1} \Omega$ , where  $\varpi_0$  acts on the finite part. Then  $|\Omega'| = |\Omega|$ , and  $\Omega'$  is invariant under the rectangular group  $\mathcal{C}'_f = \varpi_0^{-1} \mathcal{C}_f$ , of measure  $|\mathcal{C}_f|$ . For  $n \in \mathcal{N}'$ ,  $\beta \in \beta_n \mathcal{O}_K^1$  and  $h \in \mathbb{R}^c$ , the element  $\beta_h = (W_h \beta_{\infty}, \beta_f) \in \Gamma_h$  satisfies

$$|\Omega' \cap (\Omega' - \beta_h)| = |\Omega \cap (\Omega - (W_h \beta_{\infty}, \varpi_0 \beta_f))| = \Phi(l(\beta) - h, n),$$

by the last paragraph of the previous subsection and the unit invariance.

*Points and edges.* For  $h \in \mathbb{R}^c$  and  $z_0$  in the fundamental domain  $W_h P_{\mathcal{L}'} \times \mathcal{C}'_f$  of  $\Gamma_h$  given by Lemma 6.10, where  $\mathcal{L}'$  corresponds to  $\mathcal{C}'_f$ , let  $\mathcal{X}(h, z_0) = (\Gamma_h + z_0) \cap \Omega'$  and  $N(h, z_0) = |\mathcal{X}(h, z_0)|$ , and let  $\mathcal{E}(h, z_0)$  count the pairs  $(x, \beta)$  with  $x \in \mathcal{X}(h, z_0)$ ,  $\beta \in \bigcup_{n \in \mathcal{N}'} \beta_n \mathcal{O}_K^1$  and  $x + \beta_h \in \Omega'$ . Parametrize  $z_0$  by  $y \in [0, 1)^{2d}$  through a basis of  $\mathcal{L}'_\infty$ , as in Section 5, and by the Haar probability on  $\mathcal{C}'_f$ . Since  $\mathbf{1}_{\Omega'} \leq e^{pd(\bar{m}+\eta)}(f \circ \varpi_0)^p$  and  $(f \circ \varpi_0)^p$  is the product of an archimedean factor and a function invariant under  $\mathcal{C}'_f$ , Lemma 6.12 gives

$$N(h, z_0) \leq N_{\max} = \frac{2Ae^{pd(\bar{m}+\eta)}}{\text{covol}_0}$$

for all  $(h, z_0)$ . By tiling, the average of  $N(h, z_0)$  over  $z_0$  is  $|\Omega'|/\text{covol}_0$ , and the average of  $\mathcal{E}(h, z_0)$  is  $\text{covol}_0^{-1} \sum_{n \in \mathcal{N}'} \sum_{\beta \in \beta_n \mathcal{O}_K^1} \Phi(l(\beta) - h, n)$ . Averaging over  $h \in \Pi^1$  as in Lemma 5.7, the average of  $\mathcal{E}$  becomes

$$\frac{w_K}{\text{Reg}^1 \text{covol}_0} \sum_{n \in \mathcal{N}'} \mathcal{W}(n) \geq \frac{w_K}{\kappa h_{\text{rel}} \text{Reg}^1 \text{covol}_0} \sum_n \mathcal{W}(n).$$

Measurability and integrability hold as in Lemma 5.9, since  $\mathcal{E} \leq N^2 \leq N_{\max}^2$ . The argument of that lemma yields a pair  $(h, z_0)$  with  $N = N(h, z_0) > 0$  and

$$\frac{\mathcal{E}(h, z_0)}{N} \geq \frac{w_K}{\kappa h_{\text{rel}} \text{Reg}^1} \frac{\sum_n \mathcal{W}(n)}{|\Omega|} \geq \frac{w_K}{\kappa h_{\text{rel}} \text{Reg}^1} \frac{Z e^{2d(\bar{m}-\eta)}}{2Ae^{pd(\bar{m}+\eta)}}.$$

With  $N \leq N_{\max}$  and  $p(1+\delta) = 2$ ,

$$\frac{\mathcal{E}(h, z_0)}{N^{1+\delta}} \geq \frac{w_K \text{covol}_0^\delta}{2^{1+\delta} \kappa h_{\text{rel}} \text{Reg}^1} \frac{Z}{A^{1+\delta}} e^{-4\eta d}.$$

By (5.3) and  $\log L_F(1) \leq dC$ , the first factor is at least  $2^{-2-\delta} 2^{d(1-\delta)} \pi^{(1-\theta)d} \lambda^{-(1/2-\delta)d} e^{-dC}$ , and  $Z/A^{1+\delta} = \exp(bJ_{\mathbb{R}} + cJ_{\mathbb{C}} + \sum_v \log \mathcal{F}_{\delta,v})$ . Hence  $\mathcal{E}/N^{1+\delta} \geq 2^{-2-\delta} e^{d(\mathcal{M}_{\text{fin}}(\theta) - 4\eta)}$ .

*The planar set.* Let  $v_0$  be a real place of  $F$  and let  $U$  be the set of  $v_0$ -coordinates of the points of  $\mathcal{X}(h, z_0)$ . Distinct points differ by  $(W_h \gamma_\infty, \gamma_f)$  with  $0 \neq \gamma \in \Gamma$ , whose  $v_0$ -coordinate is nonzero, so  $|U| = N$ . Each counted pair  $(x, \beta)$  gives two points at distance  $|\sigma_{v_0}(\beta)| = 1$ , and distinct pairs give distinct ordered pairs of points. As in the proof of Proposition 5.18,  $u(U) \geq \mathcal{E}/2$ , the ratio  $u(U)/|U|^{1+\delta}$  tends to infinity, and the bound  $u(U) \leq |U|^2/2$  forces  $|U| \rightarrow \infty$ .  $\square$

**6.5. Uniform Local Types.** Suppose that  $K/F$  has uniform local types  $(e_r, f_r)_{r \in \mathcal{R}}$  in the sense of Section 5, that  $\mathcal{P}$  is the set of all places of  $F$  above  $\mathcal{R}$ , and that at every place above  $r$  we use a shell profile of the same type  $(r^{f_r}, k_r, w^{(r)})$ . Write  $\mathcal{F}_{\delta,r}$  for its functional. Since there are  $d/(e_r f_r)$  places above  $r$ ,

$$(6.17) \quad \frac{1}{d} \sum_{v \in \mathcal{P}} \log \mathcal{F}_{\delta,v} = \sum_{r \in \mathcal{R}} \frac{\log \mathcal{F}_{\delta,r}}{e_r f_r}, \quad H_f = \sum_{r \in \mathcal{R}} \frac{k_r \log r}{e_r} = H.$$

In particular (6.11) coincides with (5.14), whatever the shell weights. For hard windows, Lemma 6.6 gives  $\log \mathcal{F}_{\delta,r} = \log(k_r + 1) - \delta k_r f_r \log r$ , and the sum in (6.17) is  $J - \delta H$ . Thus Proposition 6.15 contains Proposition 5.18, and the shell weights replace the term  $J - \delta H$  of (5.19) by the finite functional without changing the Fourier condition.

## 7. ARCHIMEDEAN PROFILES

We use a Gaussian at the compact coordinates, and a positive polynomial modification of a Student profile at the pair coordinates. For  $(z, w) \in \mathbb{C}^2$  put

$$t = (1 + a|z|^2)^{-1}, \quad t' = (1 + a|w|^2)^{-1}, \quad g_{\mathbb{C}}(z, w) = t^s t'^s P(t, t'),$$

where

$$s = 1.14600585, \quad a = 0.0000128633241758,$$

and

$$P(t, t') = \sum_{i,j=0}^3 \beta_{ij} b_i^3(t) b_j^3(t'), \quad b_i^m(t) = \binom{m}{i} t^i (1-t)^{m-i},$$

with the symmetric matrix of Bernstein coefficients

$$(7.1) \quad (\beta_{ij}) = \begin{pmatrix} 1 & 2.4129438977 & 2.7721799210 & 6.6857798061 \\ 2.4129438977 & 3.8532983888 & 4.6371750119 & 8.1307168183 \\ 2.7721799210 & 4.6371750119 & 5.4628959084 & 9.5512647588 \\ 6.6857798061 & 8.1307168183 & 9.5512647588 & 13.8659430610 \end{pmatrix}.$$

All finite decimals here are exact rational numbers. The Bernstein polynomials  $b_i^m$  are nonnegative on  $[0, 1]$  and sum to one, so  $\min \beta \leq P \leq \max \beta$  on  $[0, 1]^2$ , where  $\min \beta$  and  $\max \beta$  are the smallest and largest coefficients.

The shape of  $g_{\mathbb{C}}$  is adapted to the displacements at a pair of coordinates. By (5.1) they have moduli  $(e^u, e^{-u})$ , so as  $u$  varies they run along a hyperbola. A profile of width  $a^{-1/2}$  in each coordinate keeps a large overlap with its translate for all  $u$  with  $e^{|u|} \lesssim a^{-1/2}$ , a range of length about  $\log(1/a)$ ; this produces the factor  $\log(1/a)$  in Proposition 7.13, against a mass penalty of order  $\delta \log(1/a)$ . For Student profiles the overlaps along this hyperbola reduce to beta integrals and digamma values (Lemmas 7.4–7.8), which makes certified bounds possible, and the polynomial  $P$  couples the two radii.

In this section we prove that these profiles are admissible in the sense of Definition 5.17, and we derive bounds for  $J_{\mathbb{R}}$  and  $J_{\mathbb{C}}$  that reduce to finite computations with rational data. Section 8 carries out these computations at  $\delta = 0.04273$ . Throughout,  $0 < \delta < 1$  and  $p = 2/(1 + \delta)$ . We write  $\psi = \Gamma'/\Gamma$ ,  $\gamma$  for Euler's constant,  $(x)_n$  for the rising factorial,  $H_n = \sum_{k=1}^n 1/k$  with  $H_0 = 0$ , and  $\text{Beta}(\alpha, \alpha')$  for the beta distribution on  $(0, 1)$  with density proportional to  $t^{\alpha-1}(1-t)^{\alpha'-1}$ .

### 7.1. The Gaussian at the Compact Coordinates.

**Lemma 7.2.** *Let  $a_{\mathbb{R}} = 2p\delta$  and  $f_{\mathbb{R}}(z) = e^{-a_{\mathbb{R}}|z|^2/p}$ , so that  $f_{\mathbb{R}}^p = e^{-a_{\mathbb{R}}|z|^2}$ . Then*

$$(7.3) \quad J_{\mathbb{R}} = \log(p/2) + \delta \log(a_{\mathbb{R}}/\pi) - a_{\mathbb{R}}/(2p) = \log(p/2) + \delta(\log(a_{\mathbb{R}}/\pi) - 1),$$

and  $a_{\mathbb{R}}$  maximizes  $J_{\mathbb{R}}$  over the Gaussians  $e^{-a'|z|^2/p}$ ,  $a' > 0$ .

*Proof.* For  $f = e^{-a'|z|^2/p}$  we have  $A_{\mathbb{R}} = \pi/a'$  and, since  $|z|^2 + |z+1|^2 = 2|z+1/2|^2 + 1/2$ ,  $I_{\mathbb{R}} = (\pi p/(2a'))e^{-a'/(2p)}$ . Hence  $J_{\mathbb{R}} = \log(p/2) + \delta \log(a'/\pi) - a'/(2p)$ , a concave function of  $\log a'$  with its maximum at  $a' = 2p\delta$ .  $\square$

By (5.19), the margin is affine in  $\theta$  with slope  $-\log \pi - 2J_{\mathbb{R}} + J_{\mathbb{C}}$ . This slope depends on the profiles; for ours it is positive (Section 8).

### 7.2. The Pair Convolution.

**Lemma 7.4.** *Let  $\alpha, \alpha' > 0$  with  $A = \alpha + \alpha' - 1 > 0$ , and let  $h \in \mathbb{C}$ . Then*

$$(7.5) \quad \int_{\mathbb{C}} (1 + |z|^2)^{-\alpha} (1 + |z+h|^2)^{-\alpha'} dz = \frac{\pi}{A} \mathbb{E}(1 + T(1-T)|h|^2)^{-A}, \quad T \sim \text{Beta}(\alpha, \alpha').$$

*Proof.* Insert  $(1 + |z|^2)^{-\alpha} = \Gamma(\alpha)^{-1} \int_0^\infty r^{\alpha-1} e^{-r(1+|z|^2)} dr$  and the analogous formula with a variable  $r'$ . Since  $r|z|^2 + r'|z+h|^2 = (r+r')|z+r'h/(r+r')|^2 + rr'|h|^2/(r+r')$ , the integral of  $e^{-r|z|^2-r'|z+h|^2}$  over  $z \in \mathbb{C}$  is  $\pi(r+r')^{-1} e^{-rr'|h|^2/(r+r')}$ ; the factors  $e^{-r}$  and  $e^{-r'}$  remain in the Laplace integrals. Substitute  $r = RT$ ,  $r' = R(1-T)$ , with Jacobian  $R$ , and integrate over  $R$ . This gives

$$\frac{\pi \Gamma(A)}{\Gamma(\alpha)\Gamma(\alpha')} \int_0^1 T^{\alpha-1} (1-T)^{\alpha'-1} (1 + T(1-T)|h|^2)^{-A} dT,$$

which equals the right side of (7.5) because  $\Gamma(\alpha + \alpha') = A\Gamma(A)$ . All integrands are nonnegative, so Tonelli's theorem justifies the interchanges.  $\square$

For  $A, B > 0$  and  $y > 0$  define

$$Q_{AB}(y) = \int_{\mathbb{R}} (1 + ye^{2v})^{-A} (1 + ye^{-2v})^{-B} dv.$$

**Lemma 7.6.** *Let  $A, B > 0$ .*

(a) *For  $y > 0$ ,*

$$(7.7) \quad \left| Q_{AB}(y) - \log(1/y) + \frac{\psi(A) + \psi(B) + 2\gamma}{2} \right| \leq \frac{(A+B)y}{2}.$$

(b) *For  $0 < y < 1$ , with absolutely convergent series,*

$$Q_{AB}(y) = \sum_{n \geq 0} \frac{(A)_n (B)_n}{(n!)^2} y^{2n} \left[ \log(1/y) + \psi(n+1) - \frac{\psi(A+n) + \psi(B+n)}{2} \right].$$

(c) *Suppose  $A, B \geq 1$ ,  $0 < y < 1$ ,  $AB y^2 < 1$  and  $2 \log(1/y) \geq H_{\lceil A \rceil - 1} + H_{\lceil B \rceil - 1}$ . Then every term of the series in (b) is nonnegative, and for every integer  $m \geq 0$  its remainder  $\mathcal{R}_m(y)$  after the terms  $n \leq m$  satisfies*

$$0 \leq \mathcal{R}_m(y) \leq \log(1/y) \frac{(AB y^2)^{m+1}}{1 - AB y^2}.$$

*Proof.* (a) Split the integral at  $v = 0$ . On  $v \geq 0$ , first replace  $(1 + ye^{-2v})^{-B}$  by 1. Since  $0 \leq 1 - (1 + ye^{-2v})^{-B} \leq B y e^{-2v}$ , this changes the half-integral by an amount in  $[0, By/2]$ . The substitution  $x = ye^{2v}$  gives  $\int_0^\infty (1 + ye^{2v})^{-A} dv = \frac{1}{2} \int_y^\infty (1+x)^{-A} dx/x$ , and

$$\int_y^\infty \frac{dx}{x(1+x)^A} = \log \frac{1}{y} + c_A + \int_0^y (1 - (1+x)^{-A}) \frac{dx}{x}, \quad c_A = \int_0^\infty ((1+x)^{-A} - \mathbf{1}_{x < 1}) \frac{dx}{x}.$$

The last integral lies in  $[0, Ay]$ , because  $0 \leq 1 - (1+x)^{-A} \leq Ax$ ; in particular it tends to 0 with  $y$ . On the other hand, the substitution  $t = 1/(1+x)$  gives

$$\int_y^\infty \frac{dx}{x(1+x)^A} = \int_0^{1/(1+y)} \frac{t^{A-1} - 1}{1-t} dt + \log \frac{1+y}{y}.$$

Letting  $y \rightarrow 0$  in the two expressions yields  $c_A = \int_0^1 (t^{A-1} - 1)(1-t)^{-1} dt = -\psi(A) - \gamma$  [25, Equation (5.9.16)]. Thus the half  $v \geq 0$  equals  $\frac{1}{2}(\log(1/y) - \psi(A) - \gamma)$  plus a gain in  $[0, Ay/2]$  minus a loss in  $[0, By/2]$ . The half  $v \leq 0$  is the same with  $A$  and  $B$  exchanged. Adding the halves proves (a).

(b) The substitutions  $x = ye^{2v}$  and then  $t = x/(x+y^2)$  give

$$2Q_{AB}(y) = \int_0^1 t^{B-1} (1-t)^{A-1} (1 - (1-y^2)t)^{-A} dt.$$

By Euler's integral [25, Equation (15.6.1)], the right side is  $\Gamma(A)\Gamma(B)\mathbf{F}(A, B; A+B; 1-y^2)$ , where  $\mathbf{F}$  is Olver's regularized hypergeometric function. The expansion [25, Equation (15.8.10)], in the case  $c = a + b$ , gives

$$\mathbf{F}(A, B; A+B; 1-y^2) = \frac{1}{\Gamma(A)\Gamma(B)} \sum_{n \geq 0} \frac{(A)_n (B)_n}{(n!)^2} y^{2n} [2\psi(n+1) - \psi(A+n) - \psi(B+n) - \log y^2]$$

for  $0 < y < 1$ . Dividing by two proves (b). The coefficients grow at most polynomially in  $n$ , so the series converges absolutely.

(c) For  $A \geq 1$  and  $n \geq 0$ , monotonicity of  $\psi$  and the recurrence  $\psi(x+1) = \psi(x) + 1/x$  give  $\psi(n+1) \leq \psi(A+n) \leq \psi(\lceil A \rceil + n) \leq \psi(n+1) + H_{\lceil A \rceil - 1}$ . Hence the bracket in (b) lies between

$\log(1/y) - \frac{1}{2}(H_{\lceil A \rceil - 1} + H_{\lceil B \rceil - 1}) \geq 0$  and  $\log(1/y)$ . Moreover  $(A)_n/n! = \prod_{k < n} (A+k)/(k+1) \leq A^n$  for  $A \geq 1$ , and similarly for  $B$ . Summing the geometric series proves (c).  $\square$

Let  $d_{ij}$  be the monomial coefficients of  $P$ , so that  $P(t, t') = \sum_{i,j} d_{ij} t^i t'^j$ , and put  $\eta_0 = 2s - 1$ ,

$$A_{ik} = \eta_0 + i + k, \quad B_{jl} = \eta_0 + j + l, \quad w_{ijkl} = \frac{d_{ij} d_{kl}}{A_{ik} B_{jl}}.$$

We call a quadruple  $(i, j, k, l)$  with  $d_{ij} d_{kl} \neq 0$  a *cross*. In the pair overlap we rename the variable of integration  $v$ , so that  $I_{\mathbb{C}} = \int_{\mathbb{R}} \int_{\mathbb{C}^2} g_{\mathbb{C}}(z, w) g_{\mathbb{C}}(z + e^v, w + e^{-v}) dz dw dv$ .

**Lemma 7.8.** *Let  $s \geq 1$  and  $a > 0$ . Then*

$$I_{\mathbb{C}} = \frac{\pi^2}{a^2} \sum_{(i,j,k,l)} w_{ijkl} \mathbb{E} Q_{A_{ik} B_{jl}}(y), \quad y = a \sqrt{T(1-T)T'(1-T')} \leq a/4,$$

where  $T \sim \text{Beta}(s+i, s+k)$  and  $T' \sim \text{Beta}(s+j, s+l)$  are independent. Each expectation is finite.

*Proof.* Expand  $g_{\mathbb{C}}(z, w) = \sum_{i,j} d_{ij} (1 + a|z|^2)^{-s-i} (1 + a|w|^2)^{-s-j}$  and multiply out  $g_{\mathbb{C}}(z, w) g_{\mathbb{C}}(z + e^v, w + e^{-v})$ . For one cross, the scaling  $z \mapsto z/\sqrt{a}$  and Lemma 7.4 give

$$\int_{\mathbb{C}} (1 + a|z|^2)^{-s-i} (1 + a|z + e^v|^2)^{-s-k} dz = \frac{\pi}{a A_{ik}} \mathbb{E} (1 + aT(1-T)e^{2v})^{-A_{ik}},$$

and similarly in  $w$  with  $e^{-2v}$ ,  $B_{jl}$  and  $T'$ . By Tonelli's theorem, the  $v$ -integral of the product is  $\mathbb{E} \int_{\mathbb{R}} (1 + Y e^{2v})^{-A_{ik}} (1 + Y' e^{-2v})^{-B_{jl}} dv$  with  $Y = aT(1-T)$  and  $Y' = aT'(1-T')$ , and the translation  $v \mapsto v + \frac{1}{4} \log(Y'/Y)$  turns the inner integral into  $Q_{A_{ik} B_{jl}}(\sqrt{Y Y'})$ . By (7.7),  $Q_{AB}(y)$  is at most  $\log(1/y)$  plus a constant, and  $\log(1/(T(1-T)))$  is integrable under every beta law; so each term is finite and the finite signed sum is legitimate.  $\square$

**7.3. Admissibility.** For the Fourier bound let  $m \geq 1$  and write  $P(t, t') = \sum_{i,j=0}^m \beta_{ij} b_i^m(t) b_j^m(t')$  with all  $\beta_{ij} > 0$ . Let  $\Delta_1$  and  $\Delta_2$  be the forward difference operators in the two indices. Fix  $0 < \rho < 1$ , the width of the complex tube in the proof below, and put  $\omega = (\rho + \rho^2)/(1 - \rho^2)$ ,

$$E_{rr'} = \binom{m}{r} \binom{m}{r'} \max_{i,j} |\Delta_1^r \Delta_2^{r'} \beta_{ij}|, \quad q_0 = \frac{1}{\min \beta} \sum_{r+r' > 0} E_{rr'} \omega^{r+r'}.$$

**Lemma 7.9.** *Let  $s > 0$ ,  $sp > 1$  and  $a > 0$ , and suppose  $q_0 < 1$ . Then for all  $\xi_1, \xi_2 \in \mathbb{C}$ ,*

$$|\widehat{g_{\mathbb{C}}^p}(\xi_1, \xi_2)| \leq K_{\mathbb{C}} e^{-\sigma_{\mathbb{C}}(|\xi_1| + |\xi_2|)} A_{\mathbb{C}}, \quad K_{\mathbb{C}} = (1 - \rho^2)^{-2sp} (1 + q_0)^p, \quad \sigma_{\mathbb{C}} = 2\pi\rho/\sqrt{a}.$$

*Proof.* Write  $x = \sqrt{a} z \in \mathbb{R}^2$  and consider complex vectors  $x + iy$  with  $x, y \in \mathbb{R}^2$  and  $|y| \leq \rho$ . Put  $\mathcal{D}(x + iy) = 1 + (x_1 + iy_1)^2 + (x_2 + iy_2)^2$ . Then

$$\text{Re } \mathcal{D} = 1 + |x|^2 - |y|^2 \geq (1 - \rho^2)(1 + |x|^2), \quad |\mathcal{D}^{-1} - (1 + |x|^2)^{-1}| \leq \frac{2|x|\rho + \rho^2}{(1 - \rho^2)(1 + |x|^2)^2} \leq \omega.$$

So the complexified  $t = \mathcal{D}^{-1}$  lies within  $\omega$  of the real  $t_0 = (1 + |x|^2)^{-1} \in (0, 1]$ , and likewise for  $t'$ . By the Bernstein derivative formula,  $\partial_t^r \partial_{t'}^{r'} P/(r! r'!) = \binom{m}{r} \binom{m}{r'} \sum_{i,j} (\Delta_1^r \Delta_2^{r'} \beta)_{ij} b_i^{m-r}(t) b_j^{m-r'}(t')$ , which is at most  $E_{rr'}$  in absolute value on  $[0, 1]^2$ . Taylor expansion of the polynomial  $P$  about  $(t_0, t'_0)$  therefore gives  $|P(t, t') - P(t_0, t'_0)| \leq q_0 \min \beta \leq q_0 P(t_0, t'_0)$ . Thus  $P(t, t')$  lies in the right half-plane, and with principal branches,

$$|\mathcal{D}(x + iy)^{-sp} \mathcal{D}(x' + iy')^{-sp} P(t, t')^p| \leq K_{\mathbb{C}} t_0^{sp} t'_0{}^{sp} P(t_0, t'_0)^p,$$

where  $x' + iy'$  is the variable attached to  $w$ . The same bounds hold for  $|y|, |y'| < \rho'$  with some  $\rho' > \rho$ , since  $q_0$  depends continuously on  $\rho$ , so the left side is holomorphic in each complex coordinate on a neighborhood of the closed tube.

The function  $g_{\mathbb{C}}^p$  is invariant under rotations of  $z$  and of  $w$ . Rotate so that  $\xi_1$  and  $\xi_2$  are positive multiples of the first basis vector, and shift the first real coordinate of  $x$  by  $-i\rho$ , then that of  $x'$  by  $-i\rho$ . Each shift is justified by Cauchy's theorem in one variable: the integrand is holomorphic in the strip and bounded there by a constant times  $(1 + |x|^2)^{-sp}(1 + |x'|^2)^{-sp}$ , which tends to zero at infinity and is integrable because  $sp > 1$ . After the shifts the Fourier factor gains the factor  $e^{-2\pi\rho(|\xi_1|+|\xi_2|)/\sqrt{a}}$ , and the displayed bound gives the lemma.  $\square$

**Proposition 7.10.** *Let  $s \geq 1$ ,  $a > 0$ , and let  $P$  have positive Bernstein coefficients  $\beta_{ij}$  of some degree  $m$  in each variable. Let  $0 < \delta < 1$  with  $a_{\mathbb{R}} = 2p\delta \leq 1$ , let  $f_{\mathbb{R}}$  be the Gaussian of Lemma 7.2, and let  $0 < \rho < 1$  with  $q_0 < 1$ . Then  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  is admissible with Fourier constants*

$$(7.11) \quad M = \max\{2, \sqrt{K_{\mathbb{C}}}\}, \quad \sigma = \min\{1, \sigma_{\mathbb{C}}\}.$$

*Proof.* (P1) The functions  $f_{\mathbb{R}}^p$  and  $g_{\mathbb{C}}^p = (1 + a|z|^2)^{-sp}(1 + a|w|^2)^{-sp}P(t, t')^p$  are smooth, because  $P \geq \min \beta > 0$  on  $[0, 1]^2$ , and they and all their derivatives are bounded.

(P2) The integrals at the compact coordinates are computed in Lemma 7.2. The mass  $A_{\mathbb{C}}$  is finite because  $sp > 1$  (Proposition 7.16 below), and  $0 < I_{\mathbb{C}} < \infty$  because  $g_{\mathbb{C}} \leq \max \beta t^s t'^s$  and Lemma 7.8 applies to the pure Student profile  $t^s t'^s$ .

(P3)  $V_{\mathbb{R}} = a_{\mathbb{R}}|z|^2/p$ , and  $V_{\mathbb{C}} = s \log(1 + a|z|^2) + s \log(1 + a|w|^2) - \log P(t, t')$  is at least  $-\log \max \beta$  and tends to infinity with  $|z|$  or  $|w|$ .

(P4) The endpoint law at a compact coordinate is Gaussian. For the pair,  $-\log P$  is bounded, so it suffices to bound the second moments of  $\log(1 + a|z|^2)$  and  $\log(1 + a|w|^2)$ . Choose  $0 < \tau < s$ ; then  $\tau < 2s - 1$  as well, since  $s \geq 1$ . We have  $\log^2(1 + x) \leq c_{\tau}(1 + x)^{\tau}$  for  $x \geq 0$  and some constant  $c_{\tau}$ , and  $g_{\mathbb{C}} \leq \max \beta t^s t'^s$ . Hence the second moment of  $\log(1 + a|z|^2)$  is at most a constant times the integral of  $(1 + a|z|^2)^{-(s-\tau)}(1 + a|z + e^v|^2)^{-s}(1 + a|w|^2)^{-s}(1 + a|w + e^{-v}|^2)^{-s}$ . By the argument of Lemma 7.8, with exponents  $s - \tau > 0$  and  $s$  and  $A = 2s - \tau - 1 > 0$ , this integral is a multiple of  $\mathbb{E}Q_{A, 2s-1}(y)$ , which is finite by (7.7). The same holds for  $w$ .

(P5) By Lemma 5.16 the Gaussian satisfies  $|\widehat{f_{\mathbb{R}}^p}(\xi)| \leq 2e^{-|\xi|}A_{\mathbb{R}} \leq Me^{-\sigma|\xi|}A_{\mathbb{R}}$ , and by Lemma 7.9 the pair profile satisfies the bound with  $K_{\mathbb{C}}e^{-\sigma_{\mathbb{C}}(|\xi_1|+|\xi_2|)} \leq M^2e^{-\sigma(|\xi_1|+|\xi_2|)}$ .  $\square$

**7.4. A Lower Bound for the Pair Overlap.** For  $n \geq 0$  and integers  $i, k \geq 0$  define

$$K_n(i, k) = \frac{(s+i)_n(s+k)_n}{n!(\eta_0 + i + k + n)_{n+1}}, \quad S_x(j) = \sum_{h=0}^{j-1} \frac{1}{x+h} \quad (S_x(0) = 0),$$

$$r_n(i, k) = -\frac{1}{\eta_0} + \frac{H_n}{2} - \frac{S_{\eta_0}(i+k+n)}{2} + S_{\eta_0}(i+k+2n+1) - \frac{S_s(i+n) + S_s(k+n)}{2},$$

$$R_n(i, k) = K_n(i, k)r_n(i, k), \text{ and}$$

$$G_n = \sum d_{ij}d_{kl}K_n(i, k)K_n(j, l), \quad \tilde{G}_n = \sum d_{ij}d_{kl}[R_n(i, k)K_n(j, l) + K_n(i, k)R_n(j, l)],$$

where the sums run over all crosses. For rational  $s$  and  $d_{ij}$  these are rational numbers. Put

$$(7.12) \quad C_s = -2\psi(s) + \psi(2s) + (2s-1)^{-1} - \gamma, \quad z_0 = a^2/16.$$

**Proposition 7.13.** *Let  $s \geq 1$  and  $a > 0$ , and suppose that for every cross*

$$(7.14) \quad A_{ik}B_{jl}z_0 < 1, \quad 2\log(4/a) \geq H_{\lceil A_{ik} \rceil - 1} + H_{\lceil B_{jl} \rceil - 1}, \quad \log(4/a) > 1/2.$$

*Then for every integer  $m \geq 0$ ,*

$$(7.15) \quad \frac{I_{\mathbb{C}}}{\pi^2 a^{-2}} \geq \sum_{n=0}^m a^{2n} \{G_n[\log(1/a) + C_s] + \tilde{G}_n\} - E_{-,m},$$

$$E_{-,m} = \log(4/a) \sum_{w_{ijkl} < 0} (-w_{ijkl}) \frac{(A_{ik}B_{jl}z_0)^{m+1}}{1 - A_{ik}B_{jl}z_0}.$$

*Proof.* Start from Lemma 7.8. Since  $y \leq a/4 < 1$ , the hypotheses of Lemma 7.6(c) hold at every value of  $y$ , with  $A = A_{ik} \geq 1$  and  $B = B_{jl} \geq 1$ . Split each  $Q_{A_{ik}B_{jl}}(y)$  into the terms  $n \leq m$  of Lemma 7.6(b) and the remainder  $\mathcal{R}_m(y)$ .

Consider the term of index  $n$  for one cross, with  $\alpha = s + i$ ,  $\alpha' = s + k$ ,  $A = A_{ik} = \alpha + \alpha' - 1$ , and similarly for  $T'$ . Its factor  $y^{2n}$  is  $a^{2n}(T(1-T))^n(T'(1-T'))^n$ , and  $\log(1/y) = \log(1/a) - \frac{1}{2}\log(T(1-T)) - \frac{1}{2}\log(T'(1-T'))$ . The beta integral gives  $\mathbb{E}(T(1-T))^n = (\alpha)_n(\alpha')_n/(A+1)_{2n}$ , so

$$\frac{(A)_n}{A n!} \mathbb{E}(T(1-T))^n = \frac{(\alpha)_n(\alpha')_n}{n!(A+n)_{n+1}} = K_n(i, k).$$

Weighting by  $(T(1-T))^n$  turns  $\text{Beta}(\alpha, \alpha')$  into  $\text{Beta}(\alpha+n, \alpha'+n)$ , under which the mean of  $\log(T(1-T))$  is  $\psi(\alpha+n) + \psi(\alpha'+n) - 2\psi(A+2n+1)$ . The recurrence  $\psi(x+j) = \psi(x) + S_x(j)$  and  $\psi(1) = -\gamma$  give

$$\begin{aligned} \psi(n+1) &= -\gamma + H_n, & \psi(\alpha+n) &= \psi(s) + S_s(i+n), & \psi(\alpha'+n) &= \psi(s) + S_s(k+n), \\ \psi(A+n) &= \psi(\eta_0) + S_{\eta_0}(i+k+n), & \psi(A+2n+1) &= \psi(\eta_0) + S_{\eta_0}(i+k+2n+1), \\ \frac{1}{2}C_s &= -\psi(s) + \frac{1}{2}\psi(\eta_0) + \eta_0^{-1} - \frac{1}{2}\gamma, \end{aligned}$$

the last one because  $\psi(2s) = \psi(\eta_0) + 1/\eta_0$ . Substituting,

$$\frac{1}{2}\psi(n+1) - \frac{1}{2}\psi(A+n) - \frac{1}{2}[\psi(\alpha+n) + \psi(\alpha'+n)] + \psi(A+2n+1) = \frac{1}{2}C_s + r_n(i, k).$$

Hence the expectation of the term of index  $n$ , multiplied by  $w_{ijkl}$ , is

$$a^{2n}d_{ij}d_{kl}K_n(i, k)K_n(j, l)[\log(1/a) + C_s + r_n(i, k) + r_n(j, l)].$$

Summing over crosses gives the displayed main terms.

By Lemma 7.6(c),  $0 \leq \mathcal{R}_m(y) \leq \log(1/y)(AB y^2)^{m+1}/(1-AB y^2)$ . The right side increases with  $y$  on  $(0, a/4]$ , because the derivative of  $y^{2m+2}\log(1/y)$  is  $y^{2m+1}((2m+2)\log(1/y) - 1) > 0$  when  $\log(1/y) > 1/2$ . So  $0 \leq \mathcal{R}_m(y) \leq \log(4/a)(AB z_0)^{m+1}/(1-AB z_0)$ . For crosses with  $w_{ijkl} > 0$  we discard the remainder, and for crosses with  $w_{ijkl} < 0$  we subtract its upper bound. This proves (7.15).  $\square$

With  $m = 1$  the bound keeps the exact term of order  $a^2$  and leaves a signed error of order  $a^4 \log(1/a)$ . It is a lower bound for  $I_{\mathbb{C}}$  only.

## 7.5. An Upper Bound for the Pair Mass.

**Proposition 7.16.** *Let  $s > 0$  with  $q = sp - 1 > 0$ , let  $a > 0$ , and let  $P$  be positive on  $[0, 1]^2$ . Then*

$$(7.17) \quad A_{\mathbb{C}} = \frac{\pi^2}{a^2 q^2} \bar{A}_{\mathbb{C}}, \quad \bar{A}_{\mathbb{C}} = \mathbb{E} P(T, T')^p,$$

where  $T$  and  $T'$  are independent with density  $qt^{q-1}$  on  $(0, 1)$ . Partition  $[0, 1]^2$  into rectangles  $\mathcal{Q} = [l, r] \times [l', r']$ , and use on  $\mathcal{Q}$  the coordinates  $x = (T-l)/(r-l)$  and  $x' = (T'-l')/(r'-l')$ . Suppose that the Bernstein coefficients of the restriction of  $P$  to  $\mathcal{Q}$ , in these coordinates, lie in  $[m_{\mathcal{Q}}, M_{\mathcal{Q}}]$  with  $m_{\mathcal{Q}} > 0$ , and put

$$c_{\mathcal{Q}} = \frac{m_{\mathcal{Q}} + M_{\mathcal{Q}}}{2}, \quad \rho_{\mathcal{Q}} = \frac{M_{\mathcal{Q}} - m_{\mathcal{Q}}}{M_{\mathcal{Q}} + m_{\mathcal{Q}}}, \quad P = c_{\mathcal{Q}}(1 + W_{\mathcal{Q}}) \text{ on } \mathcal{Q}.$$

Define

$$(7.18) \quad M_i(l, r) = \int_l^r qv^{q-1} \left( \frac{v-l}{r-l} \right)^i dv = \frac{q}{(r-l)^i} \sum_{j=0}^i \binom{i}{j} (-l)^{i-j} \frac{r^{q+j} - l^{q+j}}{q+j},$$

and, writing  $W_{\mathcal{Q}}^k = \sum_{i,j} c_{ij}^{(k)} x^i x'^j$ ,

$$\Sigma_{\mathcal{Q},N} = c_{\mathcal{Q}}^p \sum_{k=0}^N \binom{p}{k} \sum_{i,j} c_{ij}^{(k)} M_i(l, r) M_j(l', r').$$

Then for every  $N \geq 1$ ,

$$(7.19) \quad \left| \bar{A}_{\mathbb{C}} - \sum_{\mathcal{Q}} \Sigma_{\mathcal{Q},N} \right| \leq \sum_{\mathcal{Q}} c_{\mathcal{Q}}^p \frac{\left| \binom{p}{N+1} \right| \rho_{\mathcal{Q}}^{N+1}}{1 - \rho_{\mathcal{Q}}} M_0(l, r) M_0(l', r').$$

*Proof.* The substitution  $t = (1 + a|z|^2)^{-1}$  gives  $dz = \pi dt/(at^2)$  on  $\mathbb{C}$ . Hence

$$A_{\mathbb{C}} = \frac{\pi^2}{a^2} \int_0^1 \int_0^1 t^{sp-2} t'^{sp-2} P(t, t')^p dt dt',$$

which is (7.17) since  $sp - 2 = q - 1$ . The Bernstein convex-hull property gives  $|W_{\mathcal{Q}}| \leq \rho_{\mathcal{Q}} < 1$  on  $\mathcal{Q}$ . For  $1 < p < 2$  the binomial coefficients satisfy  $\left| \binom{p}{k+1} \right| / \left| \binom{p}{k} \right| = |p - k|/(k + 1) < 1$  for  $k \geq 1$ , so the binomial series of  $(1 + W)^p$  converges uniformly on  $|W| \leq \rho_{\mathcal{Q}}$ , and its tail after index  $N \geq 1$  is at most  $\left| \binom{p}{N+1} \right| \rho_{\mathcal{Q}}^{N+1} / (1 - \rho_{\mathcal{Q}})$ . Integrate against the density on  $\mathcal{Q}$ , whose total mass is  $M_0(l, r) M_0(l', r')$ . The moment formula follows by expanding  $(v - l)^i$ ; the convention  $0^q = 0$  is valid since  $q > 0$ .  $\square$

When  $p, q$ , the partition and the coefficients of  $P$  are rational, the restrictions, the Bernstein coefficients, the powers  $W_{\mathcal{Q}}^k$  and the binomial coefficients are exact rationals, computed by integer polynomial arithmetic after clearing denominators. The endpoint powers  $r^{q+j}$  and the factors  $c_{\mathcal{Q}}^p$  need not be rational; they are enclosed by interval evaluation of  $\exp(y \log x)$ . Exact rational algebra followed by interval evaluation keeps the bound valid despite the cancellation in (7.18). If  $P$  is symmetric and both axes are partitioned alike, a rectangle and its reflection contribute equally, so off-diagonal rectangles may be counted twice and diagonal ones once.

## 7.6. The Pair Functional.

**Corollary 7.20.** *Under the hypotheses of Propositions 7.13 and 7.16, let  $Z_*$  be the right side of (7.15) for some  $m$ , and suppose  $Z_* > 0$ . If  $\bar{A}_{\mathbb{C}}^* \geq \bar{A}_{\mathbb{C}}$ , then*

$$(7.21) \quad J_{\mathbb{C}} \geq \log Z_* + 2(1 + \delta) \log q - 2\delta \log \pi + 2\delta \log a - (1 + \delta) \log \bar{A}_{\mathbb{C}}^*.$$

*Proof.* By Proposition 7.13,  $I_{\mathbb{C}} \geq \pi^2 a^{-2} Z_*$ , and by (7.17),  $A_{\mathbb{C}} \leq \pi^2 a^{-2} q^{-2} \bar{A}_{\mathbb{C}}^*$ . Substitute into  $J_{\mathbb{C}} = \log I_{\mathbb{C}} - (1 + \delta) \log A_{\mathbb{C}}$ .  $\square$

Interval evaluation of the right side of (7.21) gives a certified lower bound for  $J_{\mathbb{C}}$ ; the upper endpoint of such an enclosure is not an upper bound for  $J_{\mathbb{C}}$ . For rational  $s$ , all digamma values in Proposition 7.13 reduce to rational corrections and to  $C_s = 2/\eta_0 + \psi(\eta_0) - 2\psi(s) + \psi(1)$ . The digamma function at rational arguments is enclosed as follows.

**Lemma 7.22.** *For  $x > 0$  and integers  $L \geq 0$  and  $m \geq 1$ , put  $z = x + L$ . Then*

$$\left| \psi(x) - \left( \log z - \frac{1}{2z} - \sum_{k=1}^{m-1} \frac{B_{2k}}{2kz^{2k}} - \sum_{j=0}^{L-1} \frac{1}{x+j} \right) \right| \leq \frac{|B_{2m}|}{2mz^{2m}},$$

where  $B_{2k}$  are the Bernoulli numbers.

*Proof.* By Binet's formula [25, Equation (5.9.15)],

$$\psi(z) = \log z - \frac{1}{2z} - 2 \int_0^\infty \frac{t dt}{(t^2 + z^2)(e^{2\pi t} - 1)}.$$

Write

$$\frac{1}{t^2 + z^2} = \sum_{k=1}^{m-1} (-1)^{k-1} \frac{t^{2k-2}}{z^{2k}} + (-1)^{m-1} \frac{t^{2m-2}}{z^{2m-2}(t^2 + z^2)},$$

and use  $\int_0^\infty t^{2k-1}(e^{2\pi t} - 1)^{-1} dt = |B_{2k}|/(4k)$  and  $(-1)^{k-1}|B_{2k}| = B_{2k}$ . The remaining integral has a fixed sign and is at most  $|B_{2m}|/(2mz^{2m})$  in absolute value, because  $(t^2 + z^2)^{-1} \leq z^{-2}$ . Finally apply the recurrence  $\psi(x+1) = \psi(x) + 1/x$   $L$  times.  $\square$

The certificate evaluates the right side of this lemma with exact rational arguments and Bernoulli numbers,  $L = 64$  and  $m = 16$ , in outward rounded interval arithmetic, without rounding the rational parameters of the profile.

## 8. THE FINAL CERTIFICATE

We apply Proposition 6.15 to the fields of Theorem 2.24, with the ceiling of Theorem 4.1 and the rational data below. Every parameter written as a finite decimal is exact. The inequalities between decimals displayed in this section, and the entries of Tables 6 and 8, are computed and printed by the supplementary program `geom241.py` (Section 8.4); the decimals are rounded outward from interval enclosures and are never substituted into later calculations.

### 8.1. Parameters and Local Data. Set

$$\delta = \frac{4273}{100000} = 0.04273, \quad p = \frac{2}{1 + \delta} = \frac{200000}{104273}, \quad C = \frac{4871285}{10^8} = 0.04871285, \\ \theta_* = \frac{65535}{131072}.$$

At the compact coordinates we use the Gaussian of Lemma 7.2,  $f_{\mathbb{R}}^p = e^{-a_{\mathbb{R}}|z|^2}$  with  $a_{\mathbb{R}} = 2p\delta = 17092/104273 < 1$ . At the pair coordinates we use the profile  $g_{\mathbb{C}}$  of Section 7, with the coefficients (7.1),  $s = 1.14600585$  and  $a = 0.0000128633241758$ . For the Fourier bound we take  $\rho = 10^{-3}$ . For the mass we partition  $[0, 1]^2$  into the 64 squares of side  $1/8$  with rational corners and use the binomial degree  $N = 18$  in Proposition 7.16. For the overlap we use  $m = 1$  in Proposition 7.13.

The selected rational primes are  $\mathcal{R} = \{2, 3, 5, 7, 29\}$ , with the uniform local types of Theorem 2.24. At every place of  $F$  above  $r \in \mathcal{R}$  we use the shell profile with product weights  $w_{ij} = w_{r,i}w_{r,j}$ , in the notation of (6.8) with  $x = x' = (w_{r,i})_i$ , and with  $Q = Q_r = r^{f_r}$  and  $k = k_r$ . Table 6 lists the local data and Table 7 the exact weights. At 7 and 29 the weights of the outer shells are zero, so these profiles have only one and two nontrivial shells, and their gain over the hard window is small (Table 6). By (6.8), the local functional is

$$(8.1) \quad \log \mathcal{F}_{\delta,r} = -\delta k_r \log Q_r + \log((k_r + 1)S_r^2 + 2S_r R_r) - 2(1 + \delta) \log P_r,$$

where  $P_r = \sum_i m_i w_{r,i}^p$ ,  $S_r = \sum_i m_i w_{r,i}^2$  and  $R_r = \sum_{i,i'} \Delta_{ii'} w_{r,i} w_{r,i'}$ , with the shell masses  $m_i$  and the matrix  $\Delta$  of Section 6 for  $Q = Q_r$ .

TABLE 6. Local data at the selected primes. Here  $Q_r = r^{f_r}$ , the profile at  $r$  uses the weights  $w_{r,0}, \dots, w_{r,n_r-1}$ , and the last two columns give  $\log \mathcal{F}_{\delta,r}/(e_r f_r)$  for the shell profile and for the hard window, rounded to ten decimals.

| $r$ | $e_r$ | $f_r$ | $Q_r$   | $k_r$ | $n_r$ | shells       | hard window  |
|-----|-------|-------|---------|-------|-------|--------------|--------------|
| 2   | 8     | 4     | 16      | 7     | 6     | 0.0393401103 | 0.0390666415 |
| 3   | 2     | 2     | 9       | 9     | 6     | 0.3675484302 | 0.3643996093 |
| 5   | 2     | 2     | 25      | 6     | 6     | 0.2817099488 | 0.2801636913 |
| 7   | 1     | 8     | 5764801 | 1     | 2     | 0.0034946608 | 0.0034946569 |
| 29  | 1     | 4     | 707281  | 1     | 3     | 0.0294023321 | 0.0294022443 |

TABLE 7. The exact shell weights  $w_{r,i}$  for  $i \geq 1$ ; all  $w_{r,0} = 1$ . The last column gives the weights rounded to six significant digits, for orientation only; the certificate uses the exact fractions.

| $r$ | $i$ | $w_{r,i}$                    | $\approx w_{r,i}$        |
|-----|-----|------------------------------|--------------------------|
| 2   | 1   | 2854206002066/59041214756005 | $4.83426 \cdot 10^{-2}$  |
| 2   | 2   | 198349814036/95709219081499  | $2.07242 \cdot 10^{-3}$  |
| 2   | 3   | 7762557303/86561984122975    | $8.96763 \cdot 10^{-5}$  |
| 2   | 4   | 109894931/27828186078256     | $3.94905 \cdot 10^{-6}$  |
| 2   | 5   | 17033757/96603874664399      | $1.76326 \cdot 10^{-7}$  |
| 3   | 1   | 5693537714695/63764117760919 | $8.92906 \cdot 10^{-2}$  |
| 3   | 2   | 567838527095/78237129879259  | $7.25792 \cdot 10^{-3}$  |
| 3   | 3   | 43356430379/72943614524859   | $5.94383 \cdot 10^{-4}$  |
| 3   | 4   | 2265261068/45953937834745    | $4.92942 \cdot 10^{-5}$  |
| 3   | 5   | 74149907/18018553422030      | $4.11520 \cdot 10^{-6}$  |
| 5   | 1   | 1343081168627/46386563301209 | $2.89541 \cdot 10^{-2}$  |
| 5   | 2   | 61788974773/82067065665364   | $7.52908 \cdot 10^{-4}$  |
| 5   | 3   | 1389672153/69896165011649    | $1.98820 \cdot 10^{-5}$  |
| 5   | 4   | 32644248/60812698834313      | $5.36800 \cdot 10^{-7}$  |
| 5   | 5   | 511565/34657937034136        | $1.47604 \cdot 10^{-8}$  |
| 7   | 1   | 2309805/69240611498956       | $3.33591 \cdot 10^{-8}$  |
| 29  | 1   | 36498535/95956876917664      | $3.80364 \cdot 10^{-7}$  |
| 29  | 2   | 4/29090285629613             | $1.37503 \cdot 10^{-13}$ |

The weights are positive and nonincreasing, and the coefficients  $\beta_{ij}$  are positive; these are exact rational comparisons. The rational number  $Z(g_r)Q_r^{-k_r}$  is computed twice for each  $r$ : from (6.8), and by writing the profile as the nonnegative combination  $\sum_{i,j}(w_{r,i} - w_{r,i+1})(w_{r,j} - w_{r,j+1})\mathbf{1}_{\mathcal{B}_i \times \varpi^{-k_r}\mathcal{B}_j}$  of indicators of products of balls, with  $w_{r,n_r} = 0$ , and applying (6.3). The two exact values agree.

**8.2. Certified Enclosures.** Let  $q = sp - 1 = 12492817/10427300$ , and let  $T$  and  $T'$  be independent with density  $qt^{q-1}$  on  $(0, 1)$ . Proposition 7.16, with the data above, gives

$$(8.2) \quad 38.7888697256861058 < \bar{A}_{\mathbb{C}} = \mathbb{E} P(T, T')^p < 38.7888697256861071.$$

Here every local Bernstein coefficient is positive, every local ratio  $\rho_Q$  is less than 0.332, and the tail allowances in (7.19) add up to less than  $5.9 \cdot 10^{-16}$ . The threshold conditions (7.14) hold for every cross, and Proposition 7.13 with  $m = 1$  gives

$$(8.3) \quad 348.4186894704059951 < Z_* < 348.4186894704059952,$$

where  $Z_*$  is the right side of (7.15); its signed remainder  $E_{-,1}$  is less than  $6.1 \cdot 10^{-17}$ . The upper endpoint in (8.3) bounds this expression, not the overlap  $I_{\mathbb{C}}$ . Let  $J_{\mathbb{C}}^*$  be the right side of (7.21) with  $\bar{A}_{\mathbb{C}}^*$  the upper endpoint of the enclosure (8.2). By Corollary 7.20,  $J_{\mathbb{C}} \geq J_{\mathbb{C}}^*$ , and

$$1.3556525171042125 < J_{\mathbb{C}}^* < 1.3556525171042127.$$

Formula (7.3) gives

$$-0.2107595968649958 < J_{\mathbb{R}} < -0.2107595968649957.$$

For the finite places, (8.1) and the data of Table 6 give

$$0.7214954821109038790 < \sum_{r \in \mathcal{R}} \frac{\log \mathcal{F}_{\delta,r}}{e_r f_r} < 0.7214954821109038791,$$

against  $J - \delta H < 0.7165268434$  for the hard windows. We also have

$$\begin{aligned} 15.6917787983405342 < H < 15.6917787983405343, \\ 5.6560047235563094 < \ell < 5.6560047235563095, \end{aligned}$$

where  $H = \sum_r k_r \log r / e_r$  and  $\ell = \frac{9}{4} \log 2 + \frac{1}{2} \log 3615$ .

For the Fourier condition, the rational tube quantity of Lemma 7.9 satisfies  $q_0 < 0.026 < 1$ , and

$$\log K_{\mathbb{C}} < 0.0491254364, \quad \sigma_{\mathbb{C}} > 1.7518755886, \quad \mu = 2e^{H/2-\ell} > 17.8683654522.$$

Since  $K_{\mathbb{C}} < 4$  and  $\sigma_{\mathbb{C}} > 1$ , Proposition 7.10 shows that  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  is admissible with Fourier constants  $M = 2$  and  $\sigma = 1$ . The right side of (5.14) is then  $\log 50 + 2 < 5.9121$ , so (6.11), which by (6.17) is the same condition with  $H_f = H$ , holds with a wide margin: although  $\mu$  is not large, only  $\mu \geq \log M + 2 \log 5 + 2$  is needed.

Define the conservative margin

$$\begin{aligned} M_*(\theta) = \sum_{r \in \mathcal{R}} \frac{\log \mathcal{F}_{\delta,r}}{e_r f_r} - (1/2 - \delta)\ell - C + (1 - \delta) \log 2 + (1 - \theta) \log \pi \\ + (1 - 2\theta)J_{\mathbb{R}} + \theta J_{\mathbb{C}}^*. \end{aligned}$$

By (6.17),  $J_{\mathbb{C}} \geq J_{\mathbb{C}}^*$  and  $\theta \geq 0$ , it is a lower bound for (6.16). It is affine in  $\theta$  with slope

$$0.6324418249848039 < -\log \pi - 2J_{\mathbb{R}} + J_{\mathbb{C}}^* < 0.6324418249848040,$$

so on  $[\theta_*, 1/2)$  its minimum is at  $\theta_*$ , and

$$(8.4) \quad M_*(\theta_*) > 0.000176730033534598.$$

Table 8 lists the terms of  $M_*(\theta_*)$ . The proof of Proposition 6.15 only needs some  $\eta > 0$  with  $4\eta$  below the margin. For illustration,  $\eta = 10^{-9}$  gives

$$(8.5) \quad M_*(\theta_*) - 4\eta > 0.000176726033534598 > 0.$$

The margin is small compared with the individual terms. Since  $M_*$  decreases with slope 1 in  $C$  and with slope  $1/2 - \delta$  in  $\ell$ , it would be absorbed by an increase of  $1.77 \cdot 10^{-4}$  in  $C$ , or of  $3.87 \cdot 10^{-4}$  in  $\ell$ . With these profiles and shell weights optimized in the same way for  $\delta = 0.0428$ , the same evaluation gives  $M_*(\theta_*) < -0.0015$ .

TABLE 8. The terms of  $M_*(\theta_*)$ , rounded to ten decimals.

| Term                                             | Value         |
|--------------------------------------------------|---------------|
| $\sum_r \log \mathcal{F}_{\delta,r} / (e_r f_r)$ | 0.7214954821  |
| $-(1/2 - \delta)\ell$                            | -2.5863212799 |
| $-C$                                             | -0.0487128500 |
| $(1 - \delta) \log 2$                            | 0.6635290015  |
| $(1 - \theta_*) \log \pi$                        | 0.5723736765  |
| $(1 - 2\theta_*)J_{\mathbb{R}}$                  | -0.0000032159 |
| $\theta_* J_{\mathbb{C}}^*$                      | 0.6778159157  |
| $M_*(\theta_*)$                                  | 0.0001767300  |

*Remark 8.6.* The certificate uses the bound  $\theta \geq \theta_*$  only through the positivity of  $M_*$  on  $[\theta_*, 1/2)$ . The program `geom241.py` also encloses the value at  $\theta = 0$  and the zero  $\theta_0$  of the affine function  $M_*$ :

$$M_*(0) < -0.316, \quad \theta_0 < 0.49971293.$$

So  $M_*(\theta) > 0$  for  $\theta \geq 0.49971293$ . By the proof of Theorem 2.24,  $b/d = 1/(2N_\ell)$ , where  $N_\ell$  is the number of conjugates of  $\iota_1$  in  $\text{Gal}(K/B)$ , so that  $\theta = 1/2 - 1/(4N_\ell)$ . Hence  $N_\ell \geq 871$  would suffice,

whereas that proof gives  $N_\ell \geq 2^{15}$ . On the other hand, at  $\theta = 0$  the margin (6.16) does not involve  $J_{\mathbb{C}}$  and equals  $M_*(0) < 0$ : with these profiles, mixed signature is essential.

### 8.3. Proof of the Main Theorem.

*Proof of Theorem 1.1.* For each  $n \geq 49$ , choose a Galois extension  $K_n/B$  of degree  $2^n$  as in Theorem 2.24 (with  $m = 2^n$ ), a complex conjugation  $\iota_1$  at a place of  $K_n$  above  $v_1$ , and put  $F_n = K_n^{(\iota_1)}$ . Then  $d_n = [F_n : \mathbb{Q}] = 2^n \rightarrow \infty$ . By that theorem, each  $K_n/F_n$  satisfies (G1)–(G3) with  $\lambda = 2^{9/4}\sqrt{3615}$ , has the uniform local types  $(e_r, f_r)_{r \in \mathcal{R}}$  of Table 6, and has  $\theta_* \leq \theta_n < 1/2$ . By Theorem 4.1,  $d_n^{-1} \log L_{F_n}(1) < C$  for all sufficiently large  $n$ ; we discard the finitely many other fields.

Let  $\mathcal{P}_n$  be the set of places of  $F_n$  above  $\mathcal{R}$ , with the shell profiles of Table 7. Their types form a fixed set of five. By Proposition 7.10 and the enclosures above,  $(f_{\mathbb{R}}, g_{\mathbb{C}})$  is admissible with Fourier constants  $M = 2$ ,  $\sigma = 1$ , and (6.11) holds for every  $n$ . By (6.17), Corollary 7.20 and the slope bound, the margin (6.16) of  $K_n/F_n$  is at least  $M_*(\theta_n) \geq M_*(\theta_*)$ , which is positive by (8.4). Proposition 6.15 therefore gives finite sets  $U_n \subset \mathbb{R}^2$  with  $|U_n| \rightarrow \infty$  and  $u(U_n)/|U_n|^{1+\delta} \rightarrow \infty$ , where  $1 + \delta = 1.04273$ ; these are the sets of Theorem 1.1, indexed by  $n$ . In particular  $u(U_n) \geq |U_n|^{1.04273}$  for all large  $n$ , which gives the last assertion of Theorem 1.1 at the cardinalities  $|U_n|$ .  $\square$

**8.4. Computations.** The program `geom241.py` evaluates the formulas of Sections 6 and 7 for the data above: the local functionals (8.1) from the exponents and weights in the data file `shells241_0.04273.json`, and the quantities of Propositions 7.16 and 7.13, of Lemma 7.9 and of Corollary 7.20 from the profile data above. For these it uses, unchanged, three certified modules of an earlier supplementary archive of the author (Section 1.6), which implement exactly these formulas. It checks every finite hypothesis used above: the positivity and monotonicity of the weights, the two exact computations of each  $Z(g_r)Q_r^{-k_r}$ , the positivity of the  $\beta_{ij}$  and of the local Bernstein coefficients, the conditions (7.14),  $q_0 < 1$ ,  $a_{\mathbb{R}} \leq 1$ ,  $\sigma_{\mathbb{C}} > 1$ ,  $\log K_{\mathbb{C}} < 2 \log 2$ ,  $\mu > 10$  and the positivity of the slope. It then prints the enclosures stated in this section. The exponents and weights were produced by the separate optimization program `shells241.py`; they enter the certificate only as the exact rational data of Table 7.

Rational quantities are computed exactly:  $q_0$ , the coefficients of the powers  $W_{\mathbb{Q}}^k$  and the binomial coefficients in Proposition 7.16,  $G_n$  and  $\tilde{G}_n$ , and the shell sums  $S_r$  and  $R_r$ . The remaining quantities involve the logarithm, the exponential (also through the real powers  $x^y = \exp(y \log x)$  in (7.18), in  $c_{\mathbb{Q}}^p$  and in  $w_{r,i}^p$ ), the square root in  $\sigma_{\mathbb{C}}$ , the constant  $\pi$ , and the digamma function, which enters only through Lemma 7.22. They are evaluated in the interval arithmetic of `mpmath` [40], which computes each arithmetic operation and each of these elementary functions with the lower endpoint rounded down and the upper endpoint rounded up, at a working precision of 80 decimal digits. Hence every computed interval contains the exact value, and the decimals stated above, which are coarser than the working precision by more than fifty orders of magnitude, are rigorous bounds.

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